Amazon L5 Applied Scientist

Onsite interview question bank

01. Loop Format and Hiring Bar: Amazon L5 Applied Scientist (Applied Scientist I I)

- Format notes (what to expect, 2025-2026)
- Questions
- References

02. ML Breadth Round: Classical ML, Statistics, Probability, Optimization, Deep Learning, NLP/CV

- Format notes
- A. Linear models
- B. Trees, ensembles and other classical models
- C. Bias-variance, overfitting, regularization and model selection
- D. Evaluation metrics and class imbalance
- E. Unsupervised learning and dimensionality reduction
- F. Probability and statistics
- G. Optimization
- H. Deep learning fundamentals
- I. Sequence models and classic NLP
- J. Transformers and LLMs (now a breadth staple)
- K. Computer vision fundamentals
- L. Applied breadth: production, ranking and data issues
- Prioritization for an L5 loop (based on report frequency)
- References

03. ML Depth and Science Application: Research and Project Deep Dives, and Applying Science to Amazon Problems

- Format notes
- A. Opening the deep dive: framing, scope and ownership
- B. Problem formulation and data
- C. Modeling choices and alternatives
- D. Loss, optimization and training dynamics
- E. Evaluation and scientific rigor
- F. Deployment, scale and post-launch
- G. Impact, business value and reflection
- H. Research talk or job talk Q&A (when the loop includes a presentation)
- I. Depth in your specialty: NLP and LLMs (common for 2025-2026 L5 loops)
- J. Depth in your specialty: search, ranking and recommendations
- K. Depth in your specialty: vision and multimodal
- L. Depth in your specialty: forecasting, optimization and causal inference
- M. Science application: reported Amazon problems
- N. Science application: likely Amazon problems by org (L5, 2025-2026)
- O. Applying your own research to Amazon
- P. LP probes woven into the depth round
- References

04. Coding Round for Applied Scientists: DSA Questions Reported at Amazon

- Format notes for this round
- A. Arrays, strings and hashing
- B. Intervals and sorting
- C. Two pointers and sliding window
- D. Linked lists and stacks
- E. Trees and BSTs
- F. Graphs and grids
- G. Heaps, top-k and streaming
- H. Binary search
- I. Dynamic programming

J. Randomization and sampling
K. Design-style data structures
L. ML-flavored coding (often in the same round for AS)
M. Data manipulation (AS and Data Scientist-style coding)
N. LP probes typically embedded in the coding round
References

05. ML Coding: Implementing Algorithms and Metrics from Scratch

Format notes for this round
A. Evaluation metrics from scratch
B. Clustering, neighbors and similarity search
C. Linear models and optimization
D. Trees and ensembles
E. Neural networks and backpropagation
F. Attention, transformers and LLM components
G. Probability, sampling and statistics in code
H. Recommendation, search and NLP building blocks
I. Data manipulation and ML pipeline coding
J. Round-level probes (how the interviewer pushes at L5)
Prep priorities (derived from the tags above)
References

06. ML and Science System Design: Recommendations, Search, Ads, Fraud, Forecasting, Alexa/Rufus LLMs, Pricing and Supply Chain

Format notes for this round
A. Generic framing (the questions every design round opens or pivots with)
B. Recommendations and personalization
C. Product search and ranking
D. Advertising (Sponsored Products, DSP, Ads Trust)
E. Fraud, abuse, trust and payments
F. Computer vision and catalog quality
G. Forecasting, supply chain and inventory (SCOT)
H. Pricing, causal inference and experimentation
I. Alexa, Rufus and LLM-powered products
J. Scaling, MLOps and lifecycle (probes across any design)
K. L5-bar probes and LP crossovers inside the design round
L. Team-specific "warm-up" prompts to rehearse (inferred from postings and Amazon Science)
References

07. LLM and GenAI Questions: Transformers, Fine-Tuning, RLHF/DPO, RAG, Evaluation, Hallucination, Efficiency, and Amazon Product Context (Bedrock, Nova, Rufus, Alexa+)

Format notes for this round
A. Transformer fundamentals
B. Inference, decoding and long context
C. Pretraining, fine-tuning and PEFT
D. Alignment: RLHF, DPO and beyond
E. Retrieval-augmented generation (RAG)
F. Agents and tool use
G. Evaluation
H. Hallucination and grounding
I. Efficiency, serving and cost
J. Multimodal and speech (team-dependent)
K. Applying GenAI to Amazon problems (science-application flavor)
L. Depth probes on your own LLM project (depth-round style)
M. GenAI behavioral and LP crossovers
Coverage notes for prep

References

08. Statistics, Causal Inference and A/B Testing: Hypothesis Tests, Power, Variance Reduction, Weblab, Bandits and Forecasting

Format notes for this round
A. Probability and statistical foundations
B. Hypothesis testing and p-values
C. A/B test design, power and sample size
D. Metrics: primary, guardrail and long-term
E. Variance reduction and sensitivity
F. Multiple testing, peeking and sequential testing
G. Pitfalls at Amazon scale: interference, novelty and marketplaces
H. Weblab and the launch decision
I. Causal inference without clean randomization
J. Bandits and adaptive experimentation
K. Forecasting and time series
L. Applied stats case questions (mixed, interview-style)
References

09. Leadership Principles Bank: Behavioral Questions for the L5 Applied Scientist Loop, with L5 Probes and Bar Raiser Favorites

Format notes for the behavioral portion
Questions
Bar raiser favorites and pressure probes
L5 calibration probes (any interviewer, layered on any story)
Generic deepening probes interviewers attach to any LP story
References

10. Bar Raiser Round: Reported Questions, Mixed Technical and Behavioral Probing, and Follow-up Patterns

Format notes
Questions
References

11. Amazon Science Team Context: What Each Hiring Org Asks, and the Recent Amazon Science Work to Know

Format notes for this round
A. Cross-team "why this team" and team-fit questions
B. Search (Stores Foundational AI / Search Science, formerly M5)
C. Advertising (Sponsored Products, DSP, Ads Trust, Ads measurement)
D. Alexa AI, AGI and Amazon Nova (foundation models)
E. AWS AI (Bedrock, SageMaker, Q / coding agents, AI research)
F. Rufus (generative shopping assistant)
G. Personalization and recommendations (homepage, Amazon Music, explanations)
H. SCOT (Supply Chain Optimization Technologies), forecasting and inventory
I. Pricing and causal inference
J. Prime Video (and other Amazon content and devices teams)
K. Payments, trust, fraud and abuse
L. "Have you read our work?" Recent Amazon Science publications as interview questions
M. L5-bar probes that recur across all teams
References

01. Loop Format and Hiring Bar: Amazon L5 Applied Scientist (Applied Scientist II)

> Scope: how the L5 Applied Scientist (AS II) process is structured, who sits in each seat, how the bar raiser and debrief work, how L5 is calibrated against L4 and L6, and the questions each seat tends to ask. Later chapters cover the technical content (ML breadth, depth, coding, ML design, LPs) in full. This chapter covers the format-defining questions and probes.
>
> Research note: everything here comes from Amazon's own pages, candidate self-reports (Glassdoor, Blind, LeetCode, 1point3acres, Medium) and prep-vendor guides (IGotAnOffer, Interview Kickstart, Dataford and others). The live page fetcher could not reach these sites from this environment, so the content was read through search-engine excerpts of each page. Vendor guides are marketing material and are weighted below Amazon's own pages and first-person reports.

Format notes (what to expect, 2025-2026)

Pipeline

Stage	What's reported	Sources
Recruiter call	Logistics, level target, team, and sometimes a light background check on your ML experience.	[22] [24]
Technical phone screen(s)	Amazon says there are one or two screens of about 60 minutes each, run by "a senior leader on our team" (in practice usually a senior or principal scientist, sometimes the hiring manager). Candidates report a mix of ML fundamentals, a project walk-through, 1-2 LP questions, and sometimes one LeetCode easy/medium or a pandas/data-manipulation task. Some screens have no coding at all.	[1] [9] [20] [27] [33]
Onsite "loop" (now usually virtual on Chime)	Amazon's official page says "four 55-minute interviews ... with members of our science community." Candidate reports for AS II more often describe 5 rounds of 55-60 minutes, sometimes spread over 1-3 days. Some teams add a job talk of about 45-50 minutes plus Q&A.	[1] [6] [10] [17] [28]
Debrief	Interviewers submit written feedback and a vote independently. The bar raiser chairs the debrief, and the hiring manager and bar raiser make the final decision.	[3] [4] [21] [30]
Team match (sometimes)	Some candidates pass the loop and are then told they need a team match before an offer letter. Others are "recycled" to another team's opening without re-interviewing. Some are asked to do another loop.	[11] [18] [34]

Typical L5 AS loop composition (as candidates report it)

- **ML/Science Breadth** (about 1 round): fast survey of classical ML, deep learning, statistics, and, increasingly, transformers, LLMs and GenAI. Reported examples include bias-variance, bagging vs boosting, transformer attention, BERT vs GPT, DPO vs RLHF, and how SAM works. [6][7][26]
- **ML/Science Depth** (about 1 round): a deep dive on one project or publication from your resume, where the interviewer keeps drilling down on your choices. [10][22][28]
- **Science Application / ML System Design** (about 1 round, often the hiring manager): a team-specific, open-ended problem. Blind and vendor sources say L5 loops usually include it, while L4 loops may not. [6][12][27]
- **Coding** (1 round, sometimes 2): usually LeetCode easy to medium, sometimes pandas/numpy, ML-from-scratch, or OOD. A Blind commenter says the bar is about SDE I level. [10][12][20][27]
- **Bar Raiser** (1 round): mostly LP/behavioral, run by a trained interviewer from outside the hiring org. It may also include technical probing. Candidates report BRs from non-tech orgs and Principal TPMs. [6][11][13]
- **Leadership Principles in every round:** Amazon's prep page says each interviewer typically asks 2-3 behavioral questions. Candidates report about 15 minutes of each round, and one L5 report put behavioral questions at over 60% of the loop. [1][19][27]

Who owns what

- Each interviewer is pre-assigned a competency (breadth, depth, application, coding) plus about 2-3 Leadership Principles, so the loop covers the full LP set without repeating. Amazon's page confirms that interviewers are assigned competencies "based off of our Leadership Principles". The per-LP split comes from prep guides, not from Amazon. [1][36][23]
- **Hiring manager:** usually owns the Science Application / design round and team fit. With the BR, the HM makes the final hire/no-hire call. [3][37]
- **Bar raiser:** a volunteer interviewer who is nominated, trained, and shadows veteran BRs for several months. The BR comes from outside the hiring team, chairs the debrief, and holds a de facto veto, which former Amazonians say is rarely used explicitly. The BR's job is to judge whether the candidate is better than at least half of current employees at that level. That 50% bar is folklore quoted by one BR, not a published standard. [3][4][5][29][37]

What the L5 bar is (calibration)

- **Level map:** AS I = L4, AS II = L5, Senior AS = L6. A PhD is often credited as about 2-3 years of experience, and new PhDs are frequently interviewed at L5. One example L5 posting asks for a Master's, or a PhD plus 5+ years of practical ML. Levels.fyi data puts typical L5 experience at 5-7 years. [15][31][32]
- **L5 vs L4:** L5 is expected to own projects end to end, including design, to make the technical calls, and to influence peers on the team, with some spillover to sister teams. Vendor debriefs say L5 LP answers face 2-3 layers of follow-up, against 1 layer at L4. [15][27]
- **L5 vs L6:** L6 needs proof of large scope, influence outside the team, mechanisms, and multiple strong examples. Blind commenters say the coding bar is similar at L5 and L6, and that the LP narratives are what differentiate them. [15][16][25]

- **Leveling outcomes:** downleveling from L6 to L5 and from L5 to L4 is common. A BR can push the level down even when the HM is happy (one L5 candidate's BR thought they were "at the cusp of L4 and L5"). One Blind comment says that a weak coding round with strong science can lead to a Research Scientist offer instead. [11][12][16]

Questions

Tag legend:

- **[Reported]**: a candidate or source attributes this question, or a close paraphrase, to Amazon AS (or, where noted, Amazon DS or Senior AS) interviews.
- **[Likely]**: an inference grounded in the format, the LP rubric and the team's domain. It has not been seen verbatim.

A. Recruiter call and technical phone screen

A1. Recruiter / logistics screen

1. Walk me through your background and why Amazon, why Applied Science, and why this team. [Likely]
    - Which Amazon products or science areas have you followed, and why?
    - Are you open to other teams if this one isn't the right fit (team match / recycle)?
2. How many years of hands-on applied ML experience do you have post-degree, and how much of it shipped to production? [Likely]
    - Which of your projects would you consider "L5 scope," and why?
3. Are you interviewing at other companies, and what is your timeline? [Likely]

A2. Technical phone screen (senior scientist or HM, about 60 min)

4. Walk me through the research or project on your resume you're most proud of. [Reported] [33][39]
    - Why that modeling approach over the obvious baseline?
    - What was the business metric, and how did the offline metric map onto it?
5. Explain the bias-variance trade-off and how it shows up in a model you've built. [Reported] [6][20]
    - How did you diagnose which side you were on?
6. How does hyperparameter tuning work, and how do you evaluate models? [Reported] [20]
    - How do you avoid leakage between tuning and evaluation?
7. Explain decision trees, batch normalization and word embeddings. [Reported] [39]
    - Why does batch norm behave differently at train and inference time?
8. Write out the matrix form of OLS and derive the closed-form solution. [Reported] [20]
    - What happens when X^T X is singular, and how does ridge fix it?
9. Load this dataset in pandas, clean it, engineer features, and fit a model. Which breadth questions does each step raise? [Reported] [9][20]
    - How would you handle missing values and class imbalance here?

10. Solve one LeetCode-style easy/medium problem (one candidate reported a graph problem, others arrays or trees). [Reported] [10][20]
    - What is the time and space complexity, and can you do better?
11. Tell me about a time you made a technical decision without complete data. [Reported] [26]
    - What would have changed your mind?
12. (Ads, 2024-25 report) Explain how transformers and attention work. One candidate with strong classical ML answers was rejected after not preparing for this. [Reported] [43]
    - How does a decoder-only LLM generate text at inference time?

B. Job talk / research presentation (team-dependent)

13. Present a recent piece of your work, covering only your own contributions, in about 45-50 minutes with about 10 minutes of Q&A. [Reported] [17][34]
    - Which parts were yours, and which were your collaborators'?
    - What are the impact numbers, and how were they measured?
14. Why did this problem matter to the business or customer, and how did you know? [Likely]
    - If you had to ship a version in four weeks, what would you cut?
15. What baselines did you compare against, and why are they the right ones? [Likely]
    - Which ablation would most likely break your conclusion?
16. Which paper references does this build on, and what is novel versus engineering? [Reported] [17]
17. How would this work transfer to one of our team's problems? [Likely]
    - What data would you need from us that you don't see yet?

C. ML/Science Breadth round

18. Compare bagging and boosting. When would each fail? [Reported] [6][7]
    - Why does boosting overfit noisy labels more readily?
19. Explain L1 and L2 regularization and their effect on the learned model. [Reported] [24]
20. Explain the transformer attention mechanism. [Reported] [7]
    - Why is standard self-attention expensive for long contexts, and how does Longformer-style attention help? [Reported] [7]
21. How does BERT differ from GPT? [Reported] [7]
22. Compare DPO and RLHF. [Reported] [7]
    - What failure modes of PPO-based RLHF does DPO avoid, and which does it introduce?
23. How does KV caching work during autoregressive inference? [Reported] [7]
24. Describe how SAM (Segment Anything Model) works and its key architectural components. [Reported] [6][26]
    - How would you improve CLIP for a product-image retrieval use case? [Reported] [26]
25. Compare gradient descent with Adam. When might a second-order optimizer help? [Reported] [22]
26. State and use Hoeffding's inequality. [Reported] [22]

27. Explain hypothesis testing and design an A/B test for a model change. [Reported] [9] [24]
    - If treatment improves the target metric but a population-level metric drops, what do you do? [Reported] [24]
28. Cover supervised vs unsupervised learning, dimensionality reduction, neural network basics and evaluation metrics, in rapid fire. [Reported] [28]
29. Pick a technique from a field adjacent to your PhD and explain it at grad-course depth. [Likely]
    - Where does your knowledge stop, and how would you find out more?

D. ML/Science Depth round

30. Walk through a project or publication from your resume, end to end. [Reported] [10] [22]
    - Why this loss and this architecture? What did you try that didn't work?
    - What did you personally build, versus what the team built?
31. You proposed an algorithmic fix just now. Go one level deeper: derive it or justify it mathematically. [Reported] [7]
32. Which alternatives did you consider, and what trade-offs drove the final choice? [Reported] [22]
33. What was the business impact, and how did you attribute it to the model rather than to confounders? [Reported] [28]
34. If you restarted this project today with LLMs and foundation models available, what would change? [Likely]
35. What was the hardest unknown problem in that project, and how did you get unstuck? [Likely]
    - How did you know your solution was correct?
36. What did the data look like? How were labels produced, and how noisy were they? [Likely]
    - How did label noise bound your achievable accuracy?
37. Did the model hold up after launch? How did you monitor drift? [Likely]
    - Did the gains hold six months later? [Likely]

E. Science Application / ML system design (often the hiring manager)

38. Design an agentic recommendation system for Amazon, including component architecture, model choices, and quality and safety evaluation. [Reported] [26]
39. Design a pipeline to improve personalized product recommendations on the Amazon homepage. [Reported] [26]
40. Design a GitHub Copilot-style coding agent (Senior AS, Boston report). [Reported] [7]
41. Design an enterprise RAG service. [Reported] [35]
42. Use computer vision to detect defective products in fulfillment-center warehouses. [Reported] [24]
43. Here's a vague business goal from our team. Frame it as an ML problem, choose metrics, and outline v1. [Likely]
    - What's your online/offline metric pair, and how do they disagree?
    - What's the cheapest baseline you would ship first?

44. Evaluate an LLM agent with metrics beyond accuracy: tool use, hallucination, calibration. [Reported] [7]
45. Walk me through how you'd scale this from a prototype to Amazon traffic. [Likely]
    - Where do latency and cost bite first?
46. What would make you kill this project? [Likely]
47. (Product-thinking variant, Seattle 2026 report) Pick an agentic product we could build. Who's the customer, and how do you measure success? [Reported] [6]

F. Coding round

48. String compression (in place, run-length). [Reported] [6][26]
49. Top-K elements in a stream or array, in Python. [Reported] [27]
    - Heap versus quickselect: which would you choose, and when?
50. Design a rental system (object-oriented design). [Reported] [27]
51. A LeetCode medium on trees, followed by ML questions in the same round. [Reported] [7]
52. Implement a simple ML algorithm or training loop from scratch (e.g., logistic regression with gradient descent, k-means). [Reported] [12]
53. A dynamic programming problem, such as an edit-distance-style or knapsack-style problem. [Reported] [7]
54. Compute or minimize a mathematical function in code (e.g., numerically). [Likely]
55. Write code on LiveCode without running it, then test it by hand. [Likely]
    - What edge cases would break this, and what tests would you write?

G. Bar raiser round (LP-heavy, outside interviewer)

56. Tell me about a time you had a conflict with a coworker or manager and how you approached it (Have Backbone; Disagree and Commit). [Reported] [23][41][42]
    - What did you do once the decision went against you?
57. Tell me about a time you had a major conflict with a Principal Engineer. How did you resolve it? [Reported] [25]
58. Tell me about a project failure and how you managed the aftermath. [Reported] [25]
    - What would you do differently? [Reported] [22]
59. Tell me about a time your model underperformed and you had to diagnose the issue. [Reported] [24]
    - What did you learn, and how did you pivot? [Reported] [26]
60. Tell me about a time you made a critical scientific or technical decision without full data. [Reported] [26]
61. (Pushback probe) I think your approach was wrong. Convince me otherwise. Candidates report BRs deliberately telling them they were wrong. [Reported] [41][42]
    - What data would make you concede?
62. How was that measured? [Reported] [41][42]
63. What was your exact individual contribution, stated without "we"? [Reported] [41][42]
64. Tell me about a time you took initiative without waiting for permission. One rejection was reportedly explained as "seemed to wait for permission." [Likely]
65. Tell me about a time you raised the quality bar on a deliverable even though it delayed you (Insist on the Highest Standards). [Likely]

66. Tell me about a time you simplified a complex scientific approach so it could ship (Invent and Simplify). [Likely]
67. Tell me about a time you had to decide quickly with incomplete information and the decision turned out wrong (Bias for Action / Are Right, A Lot). [Likely]
68. Tell me about a time you dove into data or code to find a root cause that others missed (Dive Deep). [Likely]
69. Tell me about the most ambitious science bet you have proposed (Think Big). [Likely]
70. Tell me about a time you did more with less: compute, labels or headcount (Frugality). [Likely]
71. Tell me about a time you mentored someone or raised the bar of your team's science (Hire and Develop the Best). [Likely]
72. Tell me about a time you considered the downstream harm or fairness impact of a model (Success and Scale Bring Broad Responsibility). [Likely]
73. Tell me about a time you made your team's work environment better (Strive to be Earth's Best Employer). [Likely]
74. (TPM- or non-science BR) How do you set ETAs, map dependencies and communicate next steps on a science project? [Reported] [13]
    - When did a science timeline slip, and how did you tell stakeholders?
75. (TPM BR) A few stats/math checks, such as interpreting a p-value or confidence interval. [Reported] [13]

H. LP questions embedded in technical rounds (each interviewer owns about 2-3 LPs)

76. Tell me about a time you started from the customer and worked backwards to the science problem (Customer Obsession). [Likely]
    - Who was the customer, and how did you get their signal?
77. Tell me about a time you took ownership of something outside your formal scope (Ownership). [Likely]
78. Tell me about a time you delivered a model under a hard deadline (Deliver Results). [Likely]
    - What did you descope, and what did that cost?
79. Tell me about a time you learned a new area fast to unblock a project (Learn and Be Curious). [Likely]
80. Tell me about a time you earned the trust of an engineering or product partner who doubted your science (Earn Trust). [Likely]
81. Tell me about a time you were wrong about a technical call. How did you find out? (Are Right, A Lot). [Likely]
82. Tell me about a time you pushed back on a stakeholder's metric or launch decision using data. [Likely]
    - Did you commit after being overruled?

I. Level-calibration probes (L5 vs L4 vs L6 signal)

83. What was the largest scope you have owned end to end, from problem framing to production? [Likely]
    - Who else depended on it, beyond your own team?

84. Tell me about a time your work influenced a sister team's roadmap or technical direction. [Likely]
85. Tell me about a time you set the technical direction for other scientists or engineers, not just executed it. [Likely]
    - How did you get buy-in without authority?
86. What mechanism did you build (review, monitoring, tooling) that kept working without you in the room? [Likely]
87. Give me a second example: a different project where you showed the same thing. L6 rejections often cite a lack of multiple strong examples. [Likely]
88. How do you balance publishing or novelty against shipping on your current team? [Likely]
89. (Depth round, applied to leveling) Where did you have to make a judgment call that your manager didn't make for you? [Likely]
90. (HM) What would you expect to own in your first six months here? [Likely]
    - What would you need from me to get there?

J. Questions to put to the recruiter or HM about format (candidate-side calibration)

91. Is the loop four or five rounds? Is there a job talk, and how long is it? [Likely]
92. Which rounds are breadth, depth, application/design and coding, and which one is the bar raiser? [Likely]
93. Is the coding round DSA (LeetCode), ML-from-scratch, or data manipulation (pandas/SQL)? [Likely]
94. Is the loop calibrated to L5, or is it an L5/L6 "up-level-able" loop? Can an L6 be considered if signal is strong? [Likely]
95. If the loop is positive but this team's headcount closes, does the result carry to other teams (recycle/team match)? [Likely]
96. How soon after the loop does the debrief happen, and when should I expect feedback? [Likely]
97. If I'm not successful, what is the cool-down period, and does it apply org-wide? Reports range from none ("recycle") to 6-24 months. [Likely]
98. Who on the panel is likely to be from outside the team? [Likely]
99. Does the team expect GenAI/LLM depth even if the role is not LLM-titled? [Likely]

K. Debrief-driven follow-up (second loops, extra rounds)

100. (Extra round after a split debrief) Let's revisit coding or design. Can you solve this follow-up problem? [Reported] [14]
101. (New HM after team switch) Tell me about your project again, framed for our domain. [Reported] [12]
102. (Down-level conversation) Would you accept an L4 / AS I offer? What would you need to see to reach L5 quickly? [Likely]

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02. ML Breadth Round: Classical ML, Statistics, Probability, Optimization, Deep Learning, NLP/CV

Format notes

- **Official loop shape.** Amazon's Applied Scientist interview-prep page describes a 60-minute technical phone screen with a senior scientist or leader, then a loop of four 55-minute interviews with members of the science community. Each loop interview covers technical areas plus Leadership Principles (LPs), and each interviewer is assigned different areas [1].
- **What candidates actually see.** Reported loops usually include one dedicated "Science / ML Breadth" interview next to Science Depth (your own project or paper), coding, a team-specific science application or ML design round, and a Bar Raiser. One Applied Scientist II (L5) loop was reported as "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [7]. Other loops had five 60-minute interviews over three days [18], or five rounds including a separate "Scalable ML" round [19]. Interviews are about 55-60 minutes, and LP questions are usually built into each one [1][11][26].
- **Who interviews.** Usually an Applied Scientist, Senior Applied Scientist or Principal Applied Scientist from the hiring org or a partner team [1][15]. The breadth round often starts from one of your projects or from something the team works on, then branches out. A Blind commenter on an L5 virtual onsite said breadth "can go anywhere the interviewer wants to take it" [2].
- **Style.** Expect rapid, quiz-like questions with "why" follow-ups. Blind commenters describe it as a quiz-style series of questions on general ML that tests how deep you can go, and say interviewers care about the intuition behind basic concepts, not just the formula [4]. Candidates report being asked to write things out mathematically: losses, regularization, and recurrent shapes in matrix form [13][16].
- **Scope at L5 (2025-2026).** The round reliably covers classical ML (from linear regression to ensembles), bias-variance and regularization, metrics, and probability and statistics [3][4][7][18]. **Transformers and attention are now treated as core breadth even when the team does not use LLMs.** An L5 candidate reported this [2], and an Ads candidate was rejected at the phone screen after being unprepared for attention, transformers and LLMs [5]. Questions from 2025-2026 go into decoder-only inference, KV caching, long-context attention, retrieval encoders and LoRA [6][10][14].
- **L5-specific bar.** Blind commenters say L5 questions resemble L4 questions, but L5 candidates are vetted more for whether they "raise the bar" [27]. A prep debrief says L5 answers should hold up through two or three layers of follow-up, against one layer at L4 [26]. In practice, the L5 signal is depth on demand: deriving results, naming

trade-offs and failure modes, and tying a technique to a product decision. The official page and prep guides also stress being able to explain *why* algorithms work [1][23].
- **Method note.** This chapter comes from about 45 web searches. Direct page fetches to Glassdoor, Blind, LeetCode, Medium and similar sites were blocked by this environment's egress policy, so every question marked [Reported] rests on search-result extracts of the cited page. Several sources are prep-site aggregators or older posts (2020-2022). Where a question appears only in a prep guide and not in a candidate's account, it is tagged [Likely] with a "see [n]" pointer.

Tag legend: [Reported] means a candidate or source reports it for Amazon Applied Scientist or a closely related Amazon science role (cited). [Likely] means a well-grounded inference from the role, recent loop trends or Amazon products.

A. Linear models

A1. Linear regression

1. What is the closed-form (normal-equation) solution for linear regression, and when would you use gradient descent instead? [Likely] (see [28])
    - What breaks in the closed form when features are highly collinear or when n and d are very large?
    - What is the computational complexity of each approach?
2. What are the assumptions of linear regression, and what happens to your estimates and inference when each is violated? [Reported] [4][5]
    - How would you detect heteroscedasticity or autocorrelated residuals?
3. What is the difference between MLE and MAP for linear regression? [Reported] [15]
    - Under what noise assumption does least squares equal MLE?
    - When does MLE equal MAP?
4. What is the probabilistic interpretation of Bayesian linear regression, and how would you model uncertainty in its predictions? [Reported] [9]
    - How does the posterior predictive variance behave far from the training data?

A2. Logistic regression

5. How does logistic regression work? Write down the model and its loss. [Reported] [9] [21]
    - Why is the decision boundary linear in the original feature space?
    - What does maximizing the likelihood mean for logistic regression? [Reported] [15]
6. What does logistic regression assume about P(y|x)? [Reported] [15]
7. What happens if you put a prior over the weights of logistic regression, and why would you assume one? [Reported] [15]
    - Which prior gives you L1 and which gives L2?
8. What are the differences between L1 and L2 regularization in logistic regression? [Reported] [9][21]
    - Why does L1 produce sparse solutions, geometrically and from a gradient or subgradient view?

9. What is the difference between logistic regression and linear regression, and why not use linear regression for classification? [Reported] [15]
10. Write the loss function for logistic regression and show that it is convex. [Likely] (see [13], which reports "derive loss functions for each model")
    - Derive the gradient. Why does it have such a simple form?
11. How would you classify Amazon product reviews when you do not have a well-labeled dataset? [Likely] (Amazon reviews are a core data source; a similar older Glassdoor report is described in search extracts, but its page could not be verified.)

B. Trees, ensembles and other classical models

12. How is a decision tree built? How are splits chosen? [Reported] [15][19]
    - Compare Gini impurity, entropy or information gain, and variance reduction.
    - How do you stop a tree from overfitting (pruning, depth, min-samples)?
13. How is a random forest different from a single decision tree, and why is the forest better? [Reported] [19][11]
    - The interviewer pressed repeatedly: *why* does averaging decorrelated trees reduce variance? [Reported] [19]
14. What is the difference between bagging and boosting? [Reported] [7][15][26]
    - How does the bias-variance decomposition apply to bagging versus boosting? [Reported] [12]
15. How does gradient boosting (GBDT) work? [Reported] [15]
    - What exactly is being fit at each stage (pseudo-residuals or negative gradients)?
    - Why do the learning rate (shrinkage) and the number of trees interact?
16. What is the computational difference between XGBoost and random forest? [Reported] [21][11]
    - What makes XGBoost fast (histogram splits, sparsity-aware splits, second-order terms)?
    - How does it handle missing values?
17. Compare random forest and gradient-boosted trees in terms of bias and variance. When would you pick each? [Reported] [29]
18. Walk through KNN, Naive Bayes and SVM: the assumptions behind each, its complexity, and when it fails. [Reported] [18]
    - What is the "naive" assumption, and why does Naive Bayes still classify well when it is violated?
    - What do the C parameter and the kernel choice do in an SVM? [Likely] (see [18])
19. What are the failure modes of k-nearest neighbors (scale sensitivity, curse of dimensionality, query cost)? [Likely] (see [28], whose kNN item is labeled an Amazon ML Scientist question)
20. What is your favorite ML algorithm, and what are its drawbacks? [Reported] [12]

C. Bias-variance, overfitting, regularization and model selection

21. Explain the bias-variance trade-off and how it affects model performance. [Reported] [3][7][11]
    - Write the decomposition of expected squared error.
22. How do you detect and reduce overfitting? [Reported] [3][9][11]
    - What do you do about underfitting?
23. You have two models, each with fit and eval functions. How would you tell which one is overfitting, especially on genuinely new test data? [Reported] [19]
    - Why is "train error low, test error high" not a sufficient answer?
24. Explain regularization mathematically. [Reported] [13]
    - Show that L2 corresponds to a Gaussian prior and L1 to a Laplace prior (MAP view). [Likely] (see [15])
25. Is dropout a form of regularization? Why? [Reported] [19]
    - How are weights handled at inference time with dropout? [Reported] (see [21]; also asked in an Amazon intern report, per search extracts)
26. How do you handle data imbalance, collinearity, feature selection and regularization in one model? [Likely] (see [28])
27. How do you approach feature selection? [Reported] [3][20]
    - Compare filter, wrapper and embedded methods.
28. How do you tune hyperparameters, and how do you avoid leaking validation data into model selection? [Reported] [3]
29. Why is standardization of features preferred, and which models need it? [Reported] [15]

D. Evaluation metrics and class imbalance

30. How do you evaluate a model? Which metrics would you choose, and why? [Reported] [3][18]
31. How do you handle severe class imbalance? Would you evaluate with precision, recall and F1, or with ROC-AUC? [Reported] [26]
    - Why can ROC-AUC look optimistic under heavy imbalance, and when is PR-AUC more informative?
    - Compare resampling, class weighting and threshold tuning. Which do you try first?
32. Explain ROC-AUC. What does it measure, and how would you compute it by hand? [Likely] (see [24], which describes a "write a program to calculate AUC" item)
33. How would you train and evaluate a real-time fraud or abuse model with severe imbalance? [Likely] (see [25]; an Ads Trust loop is reported in [18])
34. How would you pick a decision threshold when false positives and false negatives have different business costs? [Likely]
35. Which ranking metrics would you use for search or recommendations (NDCG, MRR, recall@k), and how do they relate to online metrics? [Likely] (Amazon search, ads and recommendation teams; see [32])

E. Unsupervised learning and dimensionality reduction

36. Explain k-means clustering. [Reported] [19]
    - How do you choose k? What objective does it minimize, and does it converge to the global optimum?
    - When would you use a GMM or DBSCAN instead?
37. What methods can reduce dimensionality, and how do they work? [Reported] [18] (see also [28])
    - Distinguish feature selection from feature extraction.
38. Explain PCA. How does covariance relate to PCA? [Reported] [12]
    - Does PCA keep the "most important features"? (Expect a correction: it builds new axes.)
39. What is the difference between eigenvalue decomposition (EVD) and SVD? [Reported] [15]
    - What is the computational complexity of SVD? [Reported] [12]
    - Would you compute PCA by eigendecomposing the covariance matrix or by SVD of the data matrix? Why?
40. What is the difference between Pearson correlation and covariance? [Reported] [12]
41. What is the difference between a VAE and a plain autoencoder? [Reported] [15]
    - Where does the KL term come from in the ELBO?
42. What is mutual information, and how would you use it? [Reported] [15]
43. What are the differences among supervised, unsupervised and reinforcement learning? [Reported] [9][12]

F. Probability and statistics

44. What is Bayes' rule? Explain posterior, likelihood and prior. [Reported] [15]
45. What is maximum likelihood estimation, and how does it differ from Bayesian inference? [Likely] (see [23])
    - Derive the MLE for the probability of heads from k heads in n tosses. [Likely] (a coin-flip MLE question is reported in prep compilations; see [21])
46. A coin-flip probability question worked through on the whiteboard. [Reported] [4] (the exact wording was not captured)
    - How would you get a fair bit from a biased coin? [Likely]
47. A rare disease has prevalence 1 in 1,000, and a test is 98% sensitive with a 1% false-positive rate. Given a positive result, what is P(disease)? [Likely] (attributed to Amazon on prep sites; unverified)
48. State the Central Limit Theorem. Why does it matter for A/B tests? [Reported] [5]
49. What is a p-value? Interpret it in the context of an A/B test. [Reported] [10][20]
    - What is a common misinterpretation of a 95% confidence interval?
50. Walk through hypothesis testing: which statistical tests would you use, and when? [Reported] [5][20]
    - When would you use a t-test, a z-test, chi-squared, or a non-parametric test?
51. How would you design an A/B test for a model launch? [Reported] [20]
    - How do you choose the randomization unit, primary and guardrail metrics, and sample size or power?
    - How would you handle many simultaneous variants (multiple testing)? [Likely] (see [24])

    - How do you measure a recommender when network or interference effects break user-level randomization? [Likely] (see [24])
52. What is KL divergence, and how does it relate to cross-entropy loss in practice? [Reported] [9][20][21]
    - Is KL symmetric? When does forward versus reverse KL matter?
53. What is entropy? Relate it to information gain in decision trees. [Likely]

G. Optimization

54. Describe SGD and Adam, and say when you would use each. [Reported] [3] (see also [24][25])
    - What do momentum and the second-moment estimate each fix?
    - Why might Adam generalize worse than SGD in some settings? What does AdamW change?
55. Which loss functions and optimizers did you use in your project, and why those, given the alternatives? [Reported] [16][3]
56. What changes if you swap cross-entropy for MSE in a classification loss? [Reported] [8]
    - Why do sigmoid or softmax with MSE give weak gradients when the prediction is confidently wrong?
57. What is the role of the learning rate, and how do warmup and decay schedules help? [Likely]
58. What happens to convergence with very large batch sizes, and how do you compensate? [Likely] (see [23])
59. How would you diagnose a training loss that diverges or goes NaN? [Likely] (gradient explosion is reported in [11])
    - What does gradient clipping do?
60. Define convexity. Which common ML objectives are convex and which are not? Why do deep nets still train well? [Likely]

H. Deep learning fundamentals

61. Explain backpropagation for an MLP. [Reported] [4][5]
    - Derive the gradient for one hidden layer with a sigmoid or ReLU.
62. Why use a sigmoid activation, and what are its problems in deep networks? [Reported] [12]
    - Compare ReLU, GELU and tanh. What is the dying-ReLU problem?
63. Explain the vanishing and exploding gradient problems and how to address them. [Reported] [16][26]
64. Explain batch normalization. Why does it help, and how does it behave differently at train and inference time? [Reported] [17]
    - Why do Transformers use LayerNorm instead? What is RMSNorm? [Likely] (a 2024 Glassdoor review mentions "LLM architecture and various normalization, regularization etc methods"; see [6])
65. What is label smoothing, and what is the intuition behind it? [Likely] (see [23])

    - How does it affect calibration?
66. How do you initialize weights, and why does it matter (Xavier or He)? [Likely]
67. What is reinforcement learning, and how does it differ from supervised learning? [Reported] [12]
    - What is a policy gradient, at a high level? [Likely]

I. Sequence models and classic NLP

68. Explain LSTMs and how they differ from a vanilla neural network or RNN. [Reported] [16]
    - Why is an LSTM better than a plain RNN? [Reported] [16]
    - How does backpropagation through time work? [Reported] [16]
69. How does the shape of the input change as data moves through a recurrent network? Write it in matrix form. [Reported] [16]
70. What is the difference between LSTMs and GRUs? [Reported] [15]
    - How do GRU cells address the vanishing gradient problem? How does a BiLSTM work? [Likely] (see [21])
71. Explain the difference between LSTMs and Transformers mathematically. [Reported] [13]
    - What are the compute and parallelism trade-offs? How do they handle long-term dependencies?
72. Which text representations are there? What is the difference between Word2Vec, GloVe and FastText? [Reported] [15]
73. What is Word2Vec, how is it trained, what are the alternatives, and what are its drawbacks? [Reported] [17]
    - Compare skip-gram and CBOW, and explain negative sampling.
74. How is BERT trained? What are its constraints and advantages, and which other members of the BERT family do you know? [Reported] [17]
    - Why might BERT not work in a given use case? [Reported] [17]
75. What are embeddings, and how do you get sentence or token embeddings from BERT? [Reported] [12]
    - CLS token or mean pooling? Why do raw BERT embeddings make poor sentence similarity features?
76. What is statistical language modeling, and what does perplexity measure? [Likely] (see [21]; language modeling is reported in Glassdoor extracts summarized in [7])

J. Transformers and LLMs (now a breadth staple)

77. Explain the Transformer architecture: encoder, decoder, and what each component does. [Reported] [3][8][10]
    - Why scale the dot product by sqrt(d_k)?
    - Why multi-head attention?
78. Why do Transformers need positional encoding? Compare sinusoidal, learned and rotary (RoPE). [Likely] (positional encoding appears in candidate-report extracts for 2025 transformer-heavy screens; see [14])

79. What happens if you set Wq = Wk = Wv in a Transformer? [Reported] [8]
80. How does Transformer inference work for decoder-only models? [Reported] [6]
81. How does KV caching work during autoregressive inference? [Reported] [6]
    - What is the memory cost? How do multi-query and grouped-query attention reduce it? [Likely]
82. What makes standard self-attention expensive for long contexts? [Reported] [6]
    - What is the time and memory complexity in sequence length?
83. Explain Longformer and how its attention mechanism scales to long sequences. [Reported] [6]
    - What are the trade-offs between full attention, sparse or local attention, and long-context architectures? [Reported] [6]
84. What is the difference between a cross-encoder and a bi-encoder for retrieval? [Reported] [6]
    - When would you use each in a retrieve-then-rerank pipeline?
85. Explain LoRA. [Reported] [10]
    - Compare parameter-efficient fine-tuning with full fine-tuning. Which parameters and optimizer states are trained? [Likely] (see [22])
86. Which modern LLM training techniques do you know (pretraining, SFT, preference optimization)? [Reported] [34]
    - Compare RLHF and DPO. What does the DPO beta control? [Likely] (Alexa postings list SFT, DPO, RLHF and RLAIF [31])
87. How do decoding strategies differ (greedy, beam, top-k, top-p, temperature)? [Likely]
88. What is an encoder-decoder model such as T5, and how do you optimize it for inference? [Reported] [11]
89. How would you reduce hallucination in an LLM-powered assistant? Where does retrieval-augmented generation (RAG) help, and where does it not? [Likely] (Amazon's shopping assistant uses LLMs including Amazon Nova, built on Bedrock [30])

K. Computer vision fundamentals

90. How does a CNN work, and how is it different from a feed-forward network? [Reported] [17]
    - What are parameter sharing and translation equivariance?
91. What is a receptive field, and how does it grow with depth, stride and dilation? [Reported] [17]
92. How do you reduce overfitting in CNNs (augmentation, dropout, weight decay, transfer learning)? [Likely] (see [17])
93. Describe how the Segment Anything Model (SAM) works. [Reported] [34]
94. Compare two-stage and one-stage object detectors. What are IoU and non-maximum suppression? [Likely]
95. Explain contrastive multimodal learning such as CLIP, and diffusion models at a high level. [Likely] (see [22])

L. Applied breadth: production, ranking and data issues

96. A model performs well offline but poorly online. What do you check? [Likely] (see [22])
    - How do you detect data drift or feature drift, and what would you monitor?
97. How would you handle noisy labels or non-stationary data? [Likely] (see [22])
98. Implement a rolling average when some dates are missing. What data-manipulation pitfalls arise? [Reported] [3][5] (this sometimes appears inside the breadth or screen round)
99. How does collaborative filtering work, and what are its cold-start problems? [Likely] (Amazon item-to-item collaborative filtering heritage; recommender and learning-to-rank skills are listed in [32]; a recommender design question is reported in [33])
    - Compare two-tower retrieval with a ranking model. Why have separate stages?
100. How would you measure whether a new ranking or recommendation model improved the customer experience, beyond offline metrics? [Likely] (see [24])
101. How would you choose between a simple interpretable model and a deep model for a business-critical launch? [Likely] (this tests L5 judgment and the LP "Dive Deep / Invent and Simplify" [1])

Prioritization for an L5 loop (based on report frequency)

- **Highest frequency in reports:** bias-variance (21), bagging vs boosting (14), regularization L1/L2 (8, 24), overfitting (22), decision tree and random forest (12, 13), logistic regression (5-8), Transformer internals (77-84), loss functions (55, 56), and p-values or A/B tests (49, 51).
- **Rising in 2025-2026:** KV caching, long-context attention, cross-encoder vs bi-encoder, LoRA, and LLM training techniques (80-86).
- **Expect derivations:** loss functions, regularization as MAP, LSTM vs Transformer in math, and RNN shapes in matrix form (10, 24, 69, 71).

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03. ML Depth and Science Application: Research and Project Deep Dives, and Applying Science to Amazon Problems

Format notes

- **Official frame.** Amazon's Applied Scientist interview-prep page describes one or two 60-minute technical phone screens with a senior leader, then a loop of four 55-minute interviews with members of the science community. Each interview covers technical competencies and Leadership Principles (LPs), and the behavioral portion focuses heavily on your past work [1].
- **What "Science Depth" means in practice.** Candidates consistently describe the depth round as a deep dive into one or two projects or papers from the resume, usually led by a Senior Applied Scientist.
    - An AS II (L5) Advertising Trust candidate described "Science Depth" as a resume deep dive plus GenAI fundamentals (Transformers) plus the business impact of past projects [3].
    - Other AS2 reports describe a "deep dive into the specifics of my resume and past projects" [22], "modeling details in my project" [11], and the questions "why went with a particular choice of architecture?" [12].
    - IGotAnOffer summarizes the round as walking through a research project, with follow-ups on technical decisions, the modeling trade-offs considered, and outcomes [2].
    - Some loops replace or supplement this round with a 30-40 minute research talk or job talk followed by Q&A [10][16][26]; senior loops more often include presentations [39]. Blind commenters say the recruiter or coordinator should send a brief naming the topics and LPs for each round, so ask for it [19].
- **What "Science Application" means.** This is the second half of the chapter, and it is reported in two flavors:
    - (a) An open-ended Amazon business or product problem. You propose data sources, labeling, modeling, metrics and risks [2][5][6]. Examples include modeling view, click and purchase probabilities [7]; categorizing 10k products into 100 categories [9]; training retrieval from click logs [9]; trustworthy ad recommendation [3]; and Alexa ASR for accented speech [6].
    - (b) Discussion "based on past project experience," per a Blind comment [18].
    - An L5 candidate (Nov 2024) listed "ML application" as a distinct onsite round alongside depth, breadth, DSA and Bar Raiser [4]. An Alexa guide lists "science application" as one of five loop rounds [5].
- **Interviewer type.** Usually Senior or Principal Applied Scientists, and sometimes the hiring manager (Applied Science Manager), who may run a "tech talk" style mix of design and behavioral questions [3][8].

    - LPs are probed in every round. Reports say about 5-15 minutes of each interview goes to one or two LP questions [5][27].
    - Dive Deep, Ownership and Customer Obsession map most naturally onto the depth round [Likely].
- **L5-specific bar (inferred from sources).** At L5 (Applied Scientist II), the depth interviewer is checking several things:
    - whether you were "the architect or just a passenger" [23][25];
    - whether you can defend every choice (architecture, loss, metric, data split) against alternatives [23][24];
    - whether you understand leakage and "how models cheat" [23][24];
    - whether you can connect the science to measurable customer or business impact [3].
    - A Blind commenter notes that a fresh PhD is typically placed at L5 [21]. That means the bar includes publication-level rigor plus evidence that you can ship [Likely].
    - Interviewers often want "plausible and scientific sounding ideas on the why," even when they do not know the answer themselves [20].
- **Recent trend (2025-2026).** Depth rounds increasingly drift from your project into the fundamentals behind it. Reported topics include:
    - transformer internals, KV caching, long-context attention (Longformer) and bi-encoder vs. cross-encoder retrieval [12];
    - BERT vs. GPT and DPO vs. RLHF [13];
    - beam search decoding [17];
    - knowledge distillation and learning with limited labels [8];
    - how Segment Anything works [12].
    - AGI-team loops are reported to lean heavily on modern LLM training [28].
- **Method note.** This chapter comes from about 38 web searches. Direct page fetches (Glassdoor, Blind, LeetCode, Medium, amazon.jobs, and others) were blocked by this environment's egress policy, and the search budget ran out before the planned 45 queries.
    - Every [Reported] tag rests on search-result extracts of the cited page, not a full read.
    - Several sources are prep-vendor guides (Interview Kickstart, Dataford, OA VO Service). When a question appears only in a vendor guide, it is tagged [Likely] with "see [n]".

Tag legend: [Reported] means a candidate or source reports it for an Amazon Applied Scientist or closely related Amazon science loop (cited). [Likely] means a well-grounded inference from the role, level, team or Amazon products.

A. Opening the deep dive: framing, scope and ownership

1. Walk me through the project or paper on your resume that you are most proud of, end to end. [Reported] [2][3][22]
    - What was the problem statement, and who was the customer? [Reported] (see the Glassdoor depth-round reports summarized at [12])

    - Which parts did you personally design, implement and evaluate, and which were done by others? [Likely] (see [23][25])
    - If I asked your collaborator or manager, how would they describe your contribution?
2. Describe your recent work and projects. [Reported] [2]
    - Which of these would you choose for a 45-minute deep dive, and why that one?
3. Describe one of your proudest projects. What's the most innovative thing you did in it? [Reported] [2]
    - What made it novel compared with the prior art or the existing baseline?
    - Was the novelty necessary, or would a simpler baseline have got most of the gain?
4. Tell me about a project where you were the single-threaded owner of the science. [Likely]
    - What decisions did you make without being told to?
    - What would have happened to the project without you?
5. What was the state of the art or the production baseline when you started, and how far did you move it? [Likely] (see [26])
    - How did you pick baselines, and were they tuned as carefully as your method? [Likely] (see [26])
6. What was the hardest technical obstacle in this project, and how did you get past it? [Likely]
    - What did you try that failed?
    - How long did you spend on dead ends before pivoting? (Bias for Action and Dive Deep)

B. Problem formulation and data

7. How did you turn the business or research question into an ML problem (classification, ranking, regression, generation, RL)? [Likely]
    - Could it have been framed differently, and why did you reject the alternative framings?
8. How large was the training data, and where did it come from? [Reported] (see the Glassdoor depth-round reports summarized at [12])
    - How did you sample it, and is the sample representative of production traffic?
    - How did you down-sample or construct the dataset? [Reported] [7]
9. How did you obtain labels, and how noisy were they? [Likely] (see [2])
    - Did you measure inter-annotator agreement?
    - How did label noise affect your model, and did you correct for it?
10. How would you build this with very little labeled data? [Reported] [8]
    - Would you use weak supervision, semi-supervised learning, active learning, or LLM-generated labels?
    - How would you validate synthetic or LLM-generated labels before training on them?
11. How did you handle class imbalance in your project? [Reported] [27]
    - Why did you choose re-weighting, re-sampling or threshold tuning over the alternatives?
    - How did the imbalance change your choice of evaluation metric? [Reported] [27]

12. Which features mattered most, and how do you know? [Likely] (see [24])
    - How did you guard against features that are not available at inference time?
13. How did you handle missing data, outliers and distribution differences between data sources? [Likely]

C. Modeling choices and alternatives

14. Why did you choose this particular architecture? [Reported] [12]
    - Why X over Y? What concrete evidence made you choose? [Likely] (see [23])
    - What would a simpler model (logistic regression or GBDT) have achieved?
15. What alternatives did you consider, and why did you rule them out? [Reported] (see the Glassdoor depth-round reports summarized at [12])
    - Did you run the alternatives, or reason about them?
16. Why did you use one gradient-boosting library or framework over another? [Likely] (see [23])
17. If you used a pretrained model, why that backbone, and how did you adapt it (full fine-tuning, adapters, LoRA, prompting)? [Likely] (see [28])
    - How did you choose the size of the model given latency and cost limits?
18. Derive or write out the core equations of your method on the whiteboard. [Likely] (see [25])
    - Which assumptions does the derivation rely on, and which of them does your data violate?
19. What are the main hyperparameters in your model, and how sensitive are the results to each? [Likely]
    - How did you tune them, and on which split?
20. If you used knowledge distillation, what exactly was distilled and why? [Reported] [8]
    - What were the teacher-student gap, the temperature and the loss mix?
    - Why distill instead of quantize or prune?

D. Loss, optimization and training dynamics

21. Why did you choose that specific loss function? [Likely] (see [23])
    - How does it align with the metric you actually care about?
    - What happens if you swap it for a ranking loss, focal loss or a calibrated probabilistic loss?
22. Tell me about loss functions and optimizers you used, and why. [Reported] (see the Glassdoor depth-round reports summarized at [12])
    - Why Adam or AdamW over SGD with momentum?
    - What learning-rate schedule did you use, and why?
23. Did you see overfitting or training instability, and how did you diagnose and fix it? [Likely]
    - Training accuracy is high but test accuracy is low: what do you do? [Reported] [2]
24. How was training distributed and how long did it take? What was the compute budget? [Likely]
    - What would you change if you had 10x less compute?

E. Evaluation and scientific rigor

25. How did you evaluate the model, and why did you pick those metrics? [Reported] (see the Glassdoor depth-round reports summarized at [12]; see also [25])
    - Choose between precision, recall, F1 and ROC-AUC for this problem, and justify your choice. [Reported] [27]
    - How does the offline metric relate to the online business metric?
    - Can you write code that computes AUC from scores and labels? [Reported] [14]
26. How did you confirm the model wasn't just memorizing training data? [Likely] (see [23])
27. How did you handle data leakage in that specific pipeline? [Likely] (see [23])
    - How did you split train, validation and test (random, temporal, by user, by product)?
    - Give an example of leakage you found or almost missed.
28. Are your improvements statistically significant? [Likely]
    - How many seeds or runs did you use, and what is the variance?
    - What confidence interval or test did you use?
29. What ablations did you run, and which component contributed most? [Likely] (see [26])
30. Where does your model fail? Show me the error analysis. [Likely]
    - Which slices (new users, long-tail products, low-resource locales) are worst?
    - What would you do next to fix the worst slice?
31. Was the model A/B tested online? [Likely] (see [36])
    - What was the treatment, the primary metric and the guardrails, and how long did the test run?
    - What would you do if offline gains did not translate online?

F. Deployment, scale and post-launch

32. How was the model deployed, and what were the latency, throughput and cost constraints? [Likely] (see [24])
    - What did you trade away to meet latency?
33. How did you monitor the model after launch? [Likely] (see [23])
    - How would you detect data drift vs. concept drift? [Reported] [21]
    - What triggers retraining, and how do you roll back?
34. What happened when the data distribution shifted (seasonality, Prime Day, new marketplace)? [Likely]
35. If you had to scale this 100x (more traffic, more locales, more products), what breaks first? [Likely]

G. Impact, business value and reflection

36. What was the business impact of this project, in numbers? [Reported] [3]
    - How did you attribute the impact to your model rather than to other launches?
    - Who used the output, and how did it change their decisions?

37. What would you do differently if you started this project again today? [Likely] (see [25])
    - Which limitations did you not mention in the paper or launch doc?
38. Tell me about a time your results disagreed with a stakeholder's intuition. How did you resolve it? [Likely] (see [1][27])
39. How did you decide this project was worth doing over other options? [Likely]
40. Tell me about a time you simplified a scientific solution to ship faster. What did you give up? [Likely] (Invent and Simplify)

H. Research talk or job talk Q&A (when the loop includes a presentation)

41. Present your PhD or research work in about 30 minutes, then take ML-focused Q&A. [Reported] [10]
    - Why these baselines, and are they the strongest available? [Likely] (see [26])
    - What is your individual contribution vs. your co-authors'? [Likely] (see [26])
42. Present a project of your choice (a "tech talk"). [Reported] [16][12]
    - How would this research transfer to an Amazon product?
43. The interviewer picks a detail from your presentation and asks "why" repeatedly. [Reported] [20]
    - What would you expect to happen if this assumption were removed?
    - Can you propose a plausible explanation for this unexpected result?
44. What is the next paper you would write on this line of work? [Likely]
45. Which reviewer criticism of your paper was most valid, and how did you address it? [Likely]

I. Depth in your specialty: NLP and LLMs (common for 2025-2026 L5 loops)

46. Explain the transformer architecture in detail. [Reported] [15]
    - Explain the attention mechanism at "root level," including scaled dot-product, multi-head attention and positional encoding. [Reported] [13][15]
    - Walk me through the mathematical formulation of self-attention. [Likely] (see [23])
47. How do BERT and GPT differ (objectives, architecture, use cases)? [Reported] [13]
    - How is BERT trained? [Reported] [9]
    - What are the differences between BERT and RoBERTa? [Reported] (see the Glassdoor depth-round reports summarized at [12])
48. How does decoder-only inference work, and what is the KV cache? [Reported] [12]
    - Why does standard self-attention get expensive on long inputs? [Reported] [12]
    - How does Longformer, or sparse or local attention, handle long context? [Reported] [12]
49. Compare bi-encoders and cross-encoders for retrieval and re-ranking. [Reported] [12]
    - When would you distill a cross-encoder into a bi-encoder? [Likely] (see [34])
50. How does beam search decoding work for LLMs? [Reported] [17]
    - What are the trade-offs vs. sampling (temperature, top-k, top-p)?

51. What is the difference between DPO and RLHF? [Reported] [13]
    - Walk through an SFT training pipeline and its loss. [Reported] [27]
    - What does DPO's beta control? [Likely]
52. How would you fine-tune an LLM for a domain task with limited compute (LoRA or PEFT)? [Likely] (see [25][28])
    - How do you choose the LoRA rank and which layers to adapt?
53. How do you evaluate an LLM-based system (hallucination, faithfulness, helpfulness)? [Likely] (see [32])
    - When would you trust an LLM-as-judge, and how would you calibrate it?
54. Why do LSTMs beat vanilla RNNs, and why did transformers replace both? [Reported] (see the Glassdoor reports summarized at [12])

J. Depth in your specialty: search, ranking and recommendations

55. You have a large collection of user click data for many search queries and product documents. Is click data alone sufficient? How would you train a retrieval or IR model with the data at hand? [Reported] [9]
    - How do you correct for position bias in clicks? [Likely] (see [29][40])
    - What other biases (query intent, presentation, trust) contaminate click labels? [Likely] (see [29])
56. Walk through how you would design, train, validate and deploy a ranking model for product search. [Reported] [40]
    - How would you estimate true relevance from clickstream logs with position bias? [Reported] [40]
57. Why did you choose pointwise, pairwise or listwise learning-to-rank in your project? [Likely]
    - How do NDCG and MRR relate to online engagement?
58. How do you handle cold start for new users or products in your recommender? [Likely]
59. How does matrix factorization for recommendation work, and when would you use it over a two-tower model? [Likely] (see [25])

K. Depth in your specialty: vision and multimodal

60. Describe how SAM (Segment Anything Model) works. [Reported] [12]
    - How would you adapt it to a product-image task with few labels?
61. You have 10k product listings across 100 categories, each with a title, description and photo. How would you categorize them? [Reported] [9]
    - How do you fuse text and image signals?
    - How do you handle a long tail of categories with very few examples?
    - How do you evaluate when the category taxonomy itself is noisy?
62. How does contrastive image-text pretraining (CLIP-style) work, and where would it help Amazon? [Likely] (see [25])

L. Depth in your specialty: forecasting, optimization and causal inference

63. Explain your forecasting model, and how you produce and evaluate uncertainty (quantiles, calibration). [Likely] (see [30][31])
    - Why probabilistic forecasts rather than point forecasts for inventory decisions?
64. What is the newsvendor problem, and what is its solution? [Reported] [38]
    - How does a quantile forecast connect to the newsvendor critical ratio?
65. How would you model an optimization problem from your work: objective, decision variables and constraints? [Reported] [27]
    - When would you use a heuristic instead of an exact solver?
66. How would you estimate the causal effect of a price or promotion change without a clean experiment? [Likely] (see [31])
    - When would you use double ML, instrumental variables, or difference-in-differences?
    - How do you validate a causal estimate you can never observe directly?
67. In your project, how did you separate correlation from causation in the impact claim? [Likely]

M. Science application: reported Amazon problems

68. A user moves from viewing to clicking to purchasing a product. Model the probability of each action given a user and a product. [Reported] [7]
    - Would you build three independent models, or model view, click and purchase jointly? [Reported] [7]
    - How would you construct the dataset, choose features, and down-sample (for example with reservoir sampling)? [Reported] [7]
    - How would you handle the selection bias that purchases are only observed after clicks?
69. Design a system that recommends trustworthy ads to Amazon users while maintaining brand safety and compliance. [Reported] [3]
    - Walk through the full data pipeline, from collection to deployment. [Reported] [3]
    - How do you trade off ad revenue against trust and safety errors?
70. How would you improve Alexa speech recognition for non-native speakers and users with strong regional accents? [Reported] [6]
    - Where would you get diverse accent training data, and how would you handle phonetic variation? [Reported] [6]
    - How would you measure improvement per accent group without hurting the majority?
71. You are given a high-level ML problem. Propose a solution, including data sources, annotation strategy, modeling techniques, evaluation metrics, and trade-offs and risks. [Reported] [2]
72. Answer an ML application question about an Amazon product (team-specific). [Reported] (see the Glassdoor reports summarized at [24])

73. Design and evaluate an ML pipeline to improve personalized product recommendations on the Amazon homepage. [Likely] (see [25])
    - What is the north-star metric, and what are the guardrails?
74. How would you optimize Amazon's recommendation system to improve customer engagement and sales? [Likely] (see [25])
75. Answer a search-results application question (search relevance at Amazon). [Reported] (see the Glassdoor reports summarized at [12])
76. The phone screen or depth round on an LLM team covers LLM decoding and training in the context of an Amazon use case. [Reported] [17][28]

N. Science application: likely Amazon problems by org (L5, 2025-2026)

77. Rufus (shopping assistant): how would you build retrieval-augmented answers to "What do I need for cold weather golf?" over the catalog and reviews? [Likely] (see [32][33])
    - How do you ground answers in product facts and avoid hallucinated specs?
    - How do you evaluate it offline before an A/B test?
78. How would you summarize thousands of customer reviews into a short "customers say" blurb? [Likely]
    - How do you measure faithfulness and coverage of minority opinions?
79. Fraud and abuse: how would you detect fake reviews or abusive sellers with heavily imbalanced and adversarial labels? [Likely]
    - How do you handle adversaries adapting to your model?
80. Supply chain (SCOT): how would you forecast demand for millions of products, including new ones with no history? [Likely] (see [30][31][37])
    - How would forecast errors propagate into buying and placement decisions?
81. Pricing: how would you design an experiment to measure the effect of a pricing algorithm when prices interact across products? [Likely] (see [31])
    - What about interference between treated and control products?
82. Last mile and geospatial: how would you predict delivery time or improve address geocoding accuracy? [Likely] (see [35])
    - What loss would you use if late deliveries cost more than early ones?
83. Ads: how would you predict click-through rate for sponsored products, and how does calibration interact with the auction? [Likely] (see [3])
    - Revenue is flat, CPC is up and conversions are down after a pacing change. How would you diagnose it?
84. Product categorization and catalog quality: how would you detect and fix wrong attributes (size, color, brand) on listings at catalog scale? [Likely]
85. Alexa or devices: how would you make an on-device model 10x smaller with minimal quality loss? [Likely] (see [34])
86. AWS AI services: how would you evaluate whether a new foundation-model fine-tuning method is worth exposing to customers? [Likely]
87. How would you build a query-understanding model (intent, product type, attributes) for a new marketplace and language with little data? [Likely] (see [36])

O. Applying your own research to Amazon

88. How would you apply your research or project to a problem at Amazon? [Reported] [18]
    - Which team would benefit most, and what metric would move?
    - What would have to change in your method to work at Amazon's scale and latency?
89. If you joined this team, what would you work on in your first six months, and how would you measure success? [Likely]
90. Take your method and tell me the first experiment you would run with our data. [Likely]
    - What result would make you abandon the approach?
91. How would you explain your project's value to a non-scientist product manager in two minutes? [Likely] (Earn Trust)
92. Write the one-paragraph "PR/FAQ" headline for a product built on your research. [Likely]

P. LP probes woven into the depth round

93. Tell me about a time you went deep into the data and found something that changed the direction of the project. [Likely] (Dive Deep; see [1][5])
94. Tell me about a time you had to make a decision with incomplete data. [Likely] (Bias for Action)
95. Tell me about a time a model you shipped failed or underperformed in production. What did you do? [Likely] (Ownership)
96. Tell me about a time you disagreed with a senior scientist's technical approach. [Likely] (Have Backbone; Disagree and Commit)
97. Tell me about a time you raised the scientific bar on your team (reviews, standards, reproducibility). [Likely] (Insist on the Highest Standards)
98. Tell me about a time you learned a new area quickly to solve a problem. [Likely] (Learn and Be Curious)
99. Tell me about the most ambiguous problem you worked on, and how you scoped it. [Likely]
100. Tell me about a time you traded scientific purity for customer impact, or refused to. [Likely] (Customer Obsession)

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04. Coding Round for Applied Scientists: DSA Questions Reported at Amazon

> Scope: the live-coding (data structures and algorithms) round in the Amazon L5 Applied Scientist (Applied Scientist II) onsite or virtual loop, 2025-2026. Questions only, with no solutions. Tags: **[Reported]** means a candidate report or prep source ties the question to Amazon (citation [n]). Where the report comes from an SDE, intern or Data Scientist loop and not an AS loop, the line says so. **[Likely]** means a reasoned guess based on the role, team or Amazon's products. No source reported it.
>
> Research note: page fetching was blocked in this environment, so sources were read through search-result extracts and not as full pages. Treat the details of each report as what the candidate said happened, not as verified fact.

Format notes for this round

- **Where coding appears.** Amazon's official AS prep page says a technical phone screen comes first, and some teams add a second one [1]. Candidate reports describe the L5 loop as 4 to 6 back-to-back 60-minute interviews: ML breadth, ML depth, live coding, science application or ML system design, and a Bar Raiser [5][6][18][41]. An L5 poster said their recruiter described **one coding interview, usually a LeetCode medium**, with no coding in the other rounds [18]. One Seattle 2026 report listed "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [9].
- **Difficulty.** Most reports say easy to medium, and often an "easier medium" [18][19][22]. Some reports say medium/hard, with "clean production level, optimal code" and edge cases expected [20]. Others got one medium and one hard, with brute force explained before the optimal solution [7], or a LeetCode Medium/Hard such as median of a data stream [12]. Some AS candidates say they got almost no LeetCode at all (one poster: "roughly 4/5 of the AS I asked") [18][22]. **The bar depends on the team and on who interviews you.** A Reddit commenter said an SDE interviewer usually gives medium DSA, while an AS interviewer often gives ML or data-science coding [5].
- **Environment.** Amazon's prep pages ask for syntactically correct code with no pseudo code, written in its Livecode shared editor with no IDE, and they say to check edge cases and bad input [1][2]. Third-party guides add that you usually cannot run the code, so you are expected to trace through it by hand [47]. Python is the norm for AS candidates, but any language is accepted [4].
- **What is assessed at L5.** Problem framing and clarifying questions, choice of data structure, stating time and space complexity, moving from brute force to optimal, testing and edge cases, and readable, maintainable code [2][4][46][47]. Expect 1 or 2 Leadership Principle (LP) questions to open or close the coding round. Reports

describe 20 to 30 minutes of LPs inside technical rounds [45][48]. The ML-coding variant (for example AUC, k-means, IoU or k-NN from scratch) bans libraries to test whether you really understand the method [6][43][45][50].

- **Prep signals from candidates.** Candidates say Blind 75 or 150 at medium level is usually enough, and that you should know arrays, hash maps, trees, graphs, heaps and basic DP well [21][30][34]. L5 phone screens mix coding, statistics and LPs [31]. Alexa, AGI and LLM teams lean toward ML-implementation coding [35][55]. Amazon SDE-tagged frequency lists (islands, LRU, top-k, intervals) are useful backup practice, because AS coding interviewers are often SDEs [3][49][51][52].
- **Common follow-ups.** Scaling to data that does not fit on one machine (MapReduce) [26], streaming versions, and "now make it production-grade" [20].

A. Arrays, strings and hashing

1. [Reported] Given an array of integers and a target, return the indices of the two numbers that add up to the target. [17]
    - What changes if the array is sorted, or if there are many queries against the same array?
    - How would you return all unique pairs without duplicates?
2. [Reported] Find the maximum subarray sum in an integer array. [20]
    - How would you also return the start and end indices?
    - What if the array is circular?
3. [Reported] Compress a string in place by run-length encoding (for example "aabccc" becomes "a2bc3"), and analyze time and space complexity. [9][41]
    - How do you handle counts of 10 or more?
    - What if the compressed string would be longer than the original?
4. [Reported] For each position in a list, return the product of all other elements. A fraud-detection team screen asked this without allowing division. [27]
    - How do you handle zeros?
    - Can you do it in O(1) extra space, not counting the output?
5. [Reported] Find the longest substring without repeating characters. [41][4]
    - How would you return the substring itself, not just its length?
6. [Reported] Divide two integers without using multiplication, division or modulo. [4]
    - How do you handle overflow and negative numbers?
7. [Reported] In a sorted array where every element appears twice except one, find the single element. [4]
    - Can you beat O(n)?
8. [Reported] For each element, find the nearest smaller element to its left. This was reported in an Amazon loop, but the role is not specified. [53]
    - Which data structure gives you amortized O(n)?
9. [Reported] Merge two sorted arrays. This appears as a sample AS coding question in a third-party guide. [44]
    - How would you merge in place if the first array has spare capacity at the end?
10. [Reported] Implement a data structure from scratch (one 2022 report says the round was 4 or 5 small "implement a data structure" prompts). [5]
    - Which operations must be O(1), and what trade-off does that force?

11. [Reported] A hash-map counting problem, followed by: what if the data is too big for one machine? [26]
    - How would you split it into map and reduce stages?
    - How do you handle a skewed key that overloads one reducer?
12. [Likely] Group anagrams from a list of product search queries.
    - How does your key choice change if the input is Unicode?
13. [Likely] Given customer reviews as strings, return the k most frequent words, ignoring stop words.
    - How do you break ties?
    - What if the vocabulary is far larger than k?
14. [Likely] Check whether two strings are one edit apart (insert, delete or replace).
    - How would you generalize this to a full edit-distance threshold k?

B. Intervals and sorting

15. [Reported] Merge overlapping intervals. Reported in AS loops (Sunnyvale, 2020, and a June 2026 AS technical screen). [5][28][39]
    - Should touching intervals such as [1,3] and [3,5] merge? How does your code change if they should not?
    - How would you insert one new interval into an already merged list?
16. [Reported] Merge overlapping time ranges for room reservations given as epoch-second pairs. This was tagged Amazon technical screen. [39]
    - How do you handle intervals where start equals end?
17. [Likely] Given delivery time windows for drivers, find the minimum number of vans needed so that no two overlapping windows share a van.
    - Can you do it with a sweep line and no heap?
18. [Likely] Given Prime Video viewing sessions, find the peak number of concurrent viewers and when it happened.
19. [Likely] Sort a large file of (customer_id, timestamp) records that does not fit in memory.
    - How many passes does external merge sort need, given memory size M?

C. Two pointers and sliding window

20. [Reported] A sliding-window variant with "rigorous follow-ups". [22]
    - What invariant does your window maintain?
21. [Reported] Find the k closest elements to a target x in a sorted array. [36]
    - Can you do it in O(log n + k)?
22. [Likely] Find the shortest substring of a string that contains all characters of a pattern (minimum window substring).
23. [Likely] Given daily sales, find the longest run of days whose total is at most a budget B, assuming all values are non-negative.
    - What breaks if negative values are allowed?
24. [Likely] Trapping rain water. This is Amazon SDE-tagged. [54]
    - Can you do it in O(1) extra space?

D. Linked lists and stacks

25. [Reported] Swap nodes in pairs (pairwise swap) in a linked list. A Research/Applied Scientist phone screen asked for this as a medium problem in about 25 to 30 minutes. [23][24]
    - Can you write both the recursive and iterative versions?
    - How would you generalize it to reverse every k nodes?
26. [Reported] Asteroid collision. This was reported in an Applied Scientist intern DSA round (Bengaluru, 2026) and is also common in SDE loops. [7][56]
    - Which edge cases involve equal sizes or asteroids all moving the same way?
27. [Likely] Evaluate an arithmetic expression string with +, -, *, / and parentheses.
28. [Likely] Merge k sorted lists (for example, merging ranked result lists from k retrieval shards). This is Amazon SDE-tagged. [54]
    - What is the complexity with a heap compared with pairwise merging?
29. [Likely] Detect a cycle in a linked list and return the node where the cycle starts.

E. Trees and BSTs

30. [Reported] A standard LeetCode medium on trees, followed by Transformer and long-context questions (AS, 2025). [5]
31. [Reported] An easy tree-traversal question. [41]
    - Can you do level-order, and iterative in-order without recursion?
32. [Reported] Basic DS questions covering trees, an LRU cache and counting islands. [14]
33. [Likely] Serialize and deserialize a binary tree. In an ML context, this could be a decision tree learned by a model.
    - How does your format handle null children compactly?
34. [Likely] Find the lowest common ancestor of two nodes in a binary tree, and then in a BST.
    - What if one node might not be in the tree?
35. [Likely] Validate a binary search tree.
36. [Likely] Given a product category tree (Electronics > Audio > Headphones), return the path from the root to a category and the depth of the deepest leaf.
37. [Likely] Implement prediction for a trained decision tree stored as an array of nodes (feature, threshold, left, right, leaf value).
    - How would you batch-predict efficiently for a million rows?
38. [Likely] Return the right-side view of a binary tree.

F. Graphs and grids

39. [Reported] Count the number of islands in a 2D grid. Reported in AS loops in 2020, 2025 and 2026, and in a London 2026 screen. [5][7][14][29]
    - Should diagonals count as connected?
    - What if the grid is streamed row by row and does not fit in memory?
    - Can you solve it with union-find?

40. [Reported] A BFS problem on a grid (for example, the minimum number of steps from start to target). [32]
    - How do you reconstruct the path?
41. [Reported] Shortest path with Dijkstra's algorithm where the grid or graph has obstacles. [38]
    - Why do negative edge weights break Dijkstra?
    - When would A* be the better choice?
42. [Reported] Explain how you would implement Dijkstra: which data structures, and what runtime? [13]
    - Which data structure gives O((V+E) log V), and which gives O(V^2)?
43. [Reported] A DFS-based coding exercise in a Senior AS onsite (Boston, 2025). [11]
44. [Reported] Word ladder (BFS on implicit graph). This is Amazon SDE-tagged and frequently reported. [47]
    - How do you build neighbors efficiently using wildcard patterns?
45. [Likely] Course schedule: given prerequisite pairs, return a valid order or report that there is a cycle. This could be scheduling ML pipeline steps in a DAG.
46. [Likely] Clone a graph, for example a knowledge-graph subgraph used by Alexa.
47. [Likely] Given warehouse aisles as a grid with blocked cells, find the shortest route for a robot from a start to multiple pick locations. This is related to Amazon Robotics.
    - How does the problem change with 2 pick locations compared with k?
48. [Likely] Rotting oranges and multi-source BFS: how many minutes until every reachable cell is infected?
49. [Likely] Find the connected components of users who share a payment method or address, which is a fraud-detection framing.
    - How would union-find with path compression help at scale?

G. Heaps, top-k and streaming

50. [Reported] Find the median of a data stream. Reported as a LeetCode Medium/Hard in an AS loop (Luxembourg, 2023). [12]
    - What if most values are small integers in a known range?
    - How would you support a sliding-window median?
51. [Reported] Top K frequent elements. This is a high-frequency Amazon-tagged problem. [47][54]
    - Can you compare heap and bucket sort?
    - How would you approximate this with Count-Min Sketch when there are billions of items?
52. [Reported] K closest points to the origin. This is Amazon SDE-tagged. [54]
    - Can you use quickselect, and what is its worst case?
53. [Likely] Find the kth largest element in an unsorted array, such as the kth highest model score in a batch.
54. [Likely] Return the top k products by click-through rate in a sliding one-hour window over a click stream.
    - What state must you keep per product?
55. [Likely] Merge k sorted model-score streams from different shards into a global top-N ranking.

H. Binary search

56. [Reported] A binary search coding question in a Senior AS loop (New York, 2026). [11]
57. [Reported] Search a 2D matrix whose rows and columns are both sorted, and find the optimal approach. [41]
    - Can you compare the staircase search with binary search on each row?
58. [Reported] Implement binary search and state its time complexity. [44]
    - Can you write it to find the first and the last occurrence of a value?
59. [Likely] Search in a rotated sorted array.
60. [Likely] Given a monotonic model-score threshold function, find the smallest threshold that keeps precision at or above a target. This is binary search on the answer.
61. [Likely] Koko eating bananas or ship packages within D days. This is binary search on capacity, in a package-delivery framing.

I. Dynamic programming

62. [Reported] A minimum-coins variant solved with 1-D DP (AS intern, 2025). [32]
    - Can you return the actual coins used, not just the count?
    - How do you count the number of ways instead?
63. [Reported] A dynamic programming problem from the Blind 150 list, asked by the hiring manager in a phone screen. [5][17]
64. [Reported] Two string DP and hash problems, from a PracHub Amazon listing. [40]
65. [Likely] Edit distance between two strings, for example to correct misspelled Alexa transcriptions.
    - Can you reduce memory to O(min(m, n))?
66. [Likely] Longest increasing subsequence.
    - Can you do it in O(n log n)?
67. [Likely] Word break: can a query string be split into dictionary words? This is a framing for search-query segmentation.
    - How does a trie help?
68. [Likely] Decode ways, or count paths in a grid with obstacles.
69. [Likely] Viterbi-style DP: given emission and transition scores, find the highest-scoring tag sequence.

J. Randomization and sampling

70. [Reported] Random pick with weight: build a weighted random number generator from a probability array. Reported in AS and Senior AS screens in 2026. [11][41][43]
    - How do you handle weights that change frequently?
    - What does the alias method give you, and what does it cost?
71. [Likely] Reservoir sampling: choose k items uniformly from a stream of unknown length.
    - What changes for weighted reservoir sampling?
72. [Likely] Shuffle an array uniformly at random (Fisher-Yates), and explain why the naive swap-with-any-index version is biased.

73. [Likely] Implement rand7() using rand5().

K. Design-style data structures

74. [Reported] Implement an LRU cache. Reported in Amazon loops, including a basic-DS AS-adjacent report. [14][47]
    - Can every operation be O(1)?
    - How would you make it thread-safe?
    - What about LFU, or TTL-based eviction?
75. [Likely] Design a key-value store that keeps a time-stamped history of values, for features stored at a point in time.
76. [Likely] Implement a trie with insert, search and startsWith, used for query autocomplete. Tries appear in AS candidates' prep lists. [33]
    - How would you return the top-3 completions by query frequency?
77. [Likely] Design a hit counter that returns the number of events in the past 5 minutes.
78. [Likely] Implement a min-stack or a max-queue that supports O(1) min or max.

L. ML-flavored coding (often in the same round for AS)

79. [Reported] Write a program to calculate AUC (Applied Scientist II). [6][16]
    - What do you do with tied scores?
    - Can you do it in O(n log n) without building the full ROC curve?
80. [Reported] Write a function that computes Intersection over Union (IoU) for two bounding boxes. [43]
    - What edge cases come from boxes that do not overlap or have zero area?
    - How would you vectorize it for N x M boxes?
81. [Reported] Implement k-means from scratch in Python. [41][4]
    - How do you initialize the centroids (k-means++)?
    - When should it stop, and what happens with empty clusters?
82. [Reported] Implement k-nearest neighbors from scratch. [50]
    - How do you scale features, choose k, and deal with high dimensionality?
83. [Reported] Compute the Chamfer distance between two point clouds efficiently, using an exact nearest-neighbor structure such as a KD-tree instead of computing all pairwise distances. [42]
    - How does a KD-tree perform as dimensionality increases?
84. [Reported] Code a traditional ML classifier from scratch. [17]
85. [Reported] Write gradient descent without libraries. [45]
    - How would you change it to mini-batch SGD with momentum?
86. [Likely] Implement logistic regression training with gradient descent and the loss function, using NumPy only.
87. [Likely] Implement non-maximum suppression for object detection, for example for Amazon Go or Lab126 vision teams.
88. [Likely] Implement a numerically stable softmax and cross-entropy.
89. [Likely] Implement scaled dot-product attention with a causal mask (for AGI or LLM teams).
    - What does a KV cache change at inference time?

90. [Likely] Implement beam search decoding over a toy next-token probability function.
91. [Likely] Implement byte-pair encoding (BPE) merges on a small corpus.
92. [Likely] Compute precision@k and NDCG@k for a ranked list of search results.

M. Data manipulation (AS and Data Scientist-style coding)

93. [Reported] Compute a rolling average when some dates are missing. A Blind commenter advised preparing "leetcode medium with focus on data manipulation". [18]
    - Should missing days count as zeros or be skipped, and how do you defend that choice?
94. [Reported] A simple Python question plus a medium SQL question. [20]
95. [Reported] Simple data-wrangling questions in Python alongside arrays, hash maps, trees and linked lists. [6]
96. [Likely] Given a log of (user_id, session_start, session_end), compute each user's total active time, merging overlapping sessions.
97. [Likely] Deduplicate near-identical product titles using normalized string keys.

N. LP probes typically embedded in the coding round

98. [Reported] Tell me about a time you had to dive deep into data or code to find a root cause. LP questions open or close most technical rounds. [45][48]
99. [Likely] You found a better algorithm after you had already shipped. What did you do (Insist on the Highest Standards)?
100. [Likely] Tell me about a time you simplified a complex piece of code or a pipeline (Invent and Simplify).

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05. ML Coding: Implementing Algorithms and Metrics from Scratch

> Scope: the "ML coding" variant of the coding round in the Amazon L5 Applied Scientist (Applied Scientist II) onsite or virtual loop, 2025-2026. This round asks you to implement an ML algorithm, metric or training loop in plain Python or NumPy, usually with no scikit-learn or PyTorch. Questions only, with no solutions.
>
> Tags: **[Reported]** means a candidate report or an Amazon-specific prep source ties this coding task to Amazon (citation [n]). Where the source is a vendor guide or prep blog and not a first-person report, the reference list says so. **[Likely]** means an inference from the role, Amazon's teams and products, or a concept that was reported in a breadth or depth round, turned into a coding task. A [Likely] item was not reported as a coding question.
>
> Research note (read this): in this run the web-search budget was already used up and direct page fetches failed with DNS errors. This chapter therefore uses the sources collected during the research for chapters 01-04 of this bank. Those sources were mostly read through search-result extracts, not full pages. Every [Reported] tag below matches something those sources recorded. No new reports were added without a source. Treat what each source describes as the candidate's account of what happened.

Format notes for this round

- **Where it sits in the loop.** Amazon's official AS prep page describes the onsite as four 55-minute interviews with members of the science community. Candidates for AS II usually report 5 rounds: ML breadth, ML depth, coding, science application or system design, and a Bar Raiser [1][3][11]. Most L5 loops have **one coding round**. Depending on the team and the interviewer, it can be LeetCode-style DSA, data manipulation in pandas or NumPy, or ML from scratch [11][12][13]. One Reddit-sourced comment, summarized on Glassdoor, says SDE interviewers tend to give medium DSA, while AS interviewers often give ML or data-science coding [6].
- **Which teams lean toward ML coding.** Reports from Alexa, AGI and LLM teams lean toward implementing ML from scratch [24][25]. A Blind loop summary lists "implement a simple ML algorithm or training loop from scratch (e.g., logistic regression with gradient descent, k-means)" as one coding format [12]. Vendor guides say libraries are banned so the interviewer can test whether you truly understand the method [19][21].
- **Environment.** You code in Amazon's shared editor (Livecode), with no IDE and usually no way to run the code. Amazon expects syntactically correct code, not pseudo code, and expects you to check edge cases [1][2][30]. Python is the norm for AS candidates.

- **Bar at L5.** You are expected to write correct, readable, vectorized code. Interviewers also check that you state time and space complexity, handle edge cases such as ties, empty clusters, division by zero and numerical overflow, and can test your code by tracing a small example [14][19][20][30]. Expect follow-ups that scale the solution up (batching, streaming, distributed) or tie it to an Amazon use case. Each technical round typically also spends 1-2 questions, about 5-15 minutes (some reports say longer), on Leadership Principles (LPs) [1][21].
- **Typical pacing (inferred).** About 5 minutes to clarify the task and the input/output contract, 25-35 minutes to implement, 5-10 minutes on tests and complexity, then extensions. Not every report confirms this split.

A. Evaluation metrics from scratch

A1. Classification metrics

1. Write a program that computes ROC-AUC from a list of scores and binary labels. [Reported] [3][4][5][17]
    - How do you handle tied scores?
    - Can you do it in O(n log n) using ranks (Mann-Whitney U) instead of building the full ROC curve?
    - What do you return if all labels are the same class?
    - How would you compute AUC over a dataset too large for one machine?
2. Compute precision, recall and F1 from predicted and true labels, without libraries. [Likely] (metric choice was reported in breadth and depth rounds [21])
    - How do you extend it to multi-class with macro, micro and weighted averaging?
    - What should the function return when there are no predicted positives?
3. Build a confusion matrix for K classes from two integer arrays, vectorized in NumPy. [Likely]
4. Given scores and labels, find the threshold that maximizes F1, or that maximizes recall while precision stays at or above a target. [Likely]
    - Can you do it in a single pass after sorting?
5. Compute the precision-recall curve and the average precision (PR-AUC). [Likely]
    - Why can ROC-AUC look optimistic under heavy class imbalance, for example fraud or abuse detection?
6. Implement binary and multi-class log loss (cross-entropy) with numerically safe clipping. [Likely] (the cross-entropy versus MSE question was reported in breadth [7])
7. Compute expected calibration error (ECE) with equal-width bins, and the data for a reliability diagram. [Likely] (ad click-through-rate (CTR) prediction and auction teams)
    - Equal-width or equal-mass bins: which would you use, and why?

A2. Ranking and retrieval metrics

8. Compute precision@k, recall@k and NDCG@k for a ranked list of search results with graded relevance. [Likely] (Amazon search, ads and recommendations)
    - What is IDCG, and how do you handle queries with no relevant items?
    - How do you average across queries?

9. Compute mean reciprocal rank (MRR) and mean average precision (MAP) for a batch of queries. [Likely]
10. Given impressions with click logs, compute CTR per position, and suggest a simple correction for position bias. [Likely] (position bias in clickstream relevance was reported in a depth round)

A3. Vision, NLP and forecasting metrics

11. Write a function that computes Intersection over Union (IoU) for two bounding boxes. [Reported] [19]
    - What edge cases come up when boxes do not overlap or have zero area?
    - How would you vectorize it for N x M boxes?
12. Implement non-maximum suppression (NMS) using your IoU function. [Likely] (vision teams such as Lab126, Amazon Go and robotics)
    - What is the complexity, and how would you speed it up?
13. Compute mean average precision (mAP) at an IoU threshold for an object detector over a dataset. [Likely]
14. Implement BLEU (n-gram precision with brevity penalty) or ROUGE-L for a candidate and a reference. [Likely] (Alexa, translation and LLM teams)
15. Compute perplexity from per-token log-probabilities, handling padding masks. [Likely]
16. Compute word error rate (WER) with edit distance between a hypothesis and a reference transcript. [Likely] (Alexa ASR; accented-speech ASR was reported as a science-application problem)
    - How would you also return the counts of substitutions, insertions and deletions?
17. Implement forecasting metrics: MAPE, sMAPE, and weighted quantile (pinball) loss at P50 and P90. [Likely] (Amazon supply-chain forecasting teams)
    - What goes wrong with MAPE when actuals are zero?

B. Clustering, neighbors and similarity search

18. Implement k-means from scratch in Python or NumPy. [Reported] [12][16][17]
    - How do you initialize centroids? Can you implement k-means++?
    - When should it stop, and what do you do with an empty cluster?
    - What is the per-iteration complexity? Can you vectorize the distance computation?
    - How would you choose k?
    - How would you run it on 1 billion product embeddings (mini-batch, distributed)?
19. Implement k-means++ initialization on its own. [Likely] (follow-up to the reported k-means task)
20. Implement k-nearest neighbors classification from scratch. [Reported] [22] (prep blog labeled as an Amazon ML Scientist question)
    - How do you choose k and break ties in voting?
    - Why must features be scaled, and how do you avoid leaking test statistics into the scaling?
    - What is the query cost, and how does it change in high dimensions?
21. Extend kNN to distance-weighted regression. [Likely]

22. Compute the Chamfer distance between two point clouds efficiently, using an exact nearest-neighbor structure such as a KD-tree instead of computing all pairwise distances. [Reported] [18] (Amazon-framed practice question from a prep site)
    - How does a KD-tree perform as dimensionality grows?
23. Build a KD-tree and run a nearest-neighbor query. [Likely] (follow-up to the Chamfer task)
24. Given a query embedding and a matrix of N item embeddings, return the top-k items by cosine similarity, using a heap rather than a full sort. [Likely] (semantic search, Rufus retrieval [33])
    - What is the complexity of the full sort versus the heap?
    - What changes for 100 million items (approximate nearest neighbor: IVF, HNSW, product quantization)?
25. Implement random-hyperplane locality-sensitive hashing for near-duplicate product detection. [Likely]
26. Compute pairwise Euclidean distances between two matrices without Python loops, using the expansion ||a||^2 - 2ab + ||b||^2. [Likely] (core subroutine of the reported k-means and kNN tasks)
    - What numerical issue can produce small negative values, and how do you fix it?

C. Linear models and optimization

27. Implement a simple ML training loop from scratch, for example logistic regression with gradient descent. [Reported] [12]
    - Derive the gradient of the log loss before you code it.
    - How do you add L2 (and L1) regularization? Should the bias be regularized?
    - How do you avoid overflow in the sigmoid?
    - How do you check convergence?
28. Write gradient descent without libraries. [Reported] [21]
    - How would you change it to mini-batch SGD with momentum?
29. Code a traditional ML classifier from scratch (the candidate chose the model). [Reported] [8]
    - Why did you choose that model, and what are its assumptions?
30. Implement linear regression with the closed-form normal equation and with gradient descent, and compare them. [Likely] (closed form versus GD and OLS were reported in breadth [13][23])
    - When is the matrix X^T X singular, and what do you do about it (ridge, pseudo-inverse)?
31. Implement multinomial (softmax) regression for K classes with a vectorized gradient. [Likely]
32. Implement the Adam optimizer update for a parameter vector. [Likely] (Adam versus SGD was reported in breadth)
    - Why is bias correction needed in the early steps?
33. Implement a numerical gradient check to verify an analytic gradient. [Likely]
34. Implement L1-regularized regression with proximal gradient (soft-thresholding) or coordinate descent. [Likely] (L1 versus L2 sparsity was reported in breadth [10][16])
35. Implement a Newton-Raphson (IRLS) step for logistic regression. [Likely]

    - When does Newton's method beat gradient descent, and what does it cost per step?
36. Implement stratified k-fold cross-validation and use it to tune the regularization strength of your logistic regression. [Likely]

D. Trees and ensembles

37. Implement the split search for a decision-tree node: find the feature and threshold that maximize information gain or Gini reduction. [Likely] (how splits are chosen was reported in breadth [36])
    - What is the complexity per node, and how does pre-sorting help?
    - How do you handle categorical features and missing values?
38. Implement a full decision-tree classifier with recursive building, max-depth and min-samples stopping rules, and predict. [Likely]
39. Implement prediction for a trained tree stored as arrays of nodes (feature, threshold, left, right, value). [Likely]
40. Implement a random forest on top of your tree: bootstrap sampling, feature subsampling and majority voting. [Likely] (bagging versus boosting is the most-reported breadth topic)
    - How would you compute out-of-bag error?
41. Implement gradient boosting for regression with depth-1 stumps and squared loss. [Likely]
    - What changes for log loss, and what are the "pseudo-residuals"?
42. Compute permutation feature importance for any trained model. [Likely]

E. Neural networks and backpropagation

43. Implement forward and backward passes for a 2-layer MLP in NumPy, and train it on a toy dataset. [Likely] (backpropagation for an MLP was reported in breadth [35])
    - What are the shapes of every gradient?
    - How would you initialize the weights, and why?
44. Implement a numerically stable softmax and softmax-plus-cross-entropy, with its gradient with respect to the logits. [Likely]
    - Why does the combined gradient simplify to p - y?
45. Implement ReLU, sigmoid, tanh and GELU with their derivatives. [Likely]
46. Implement dropout (inverted dropout) for training and inference. [Likely]
47. Implement the batch normalization forward pass with running statistics, and explain or implement the backward pass. [Likely] (batch norm was reported in breadth)
    - How does it behave differently at train and inference time?
48. Implement a 2D convolution (single channel, then multi-channel) with stride and padding, using loops and then im2col. [Likely] (vision teams)
49. Implement a single LSTM or GRU cell forward step in matrix form. [Likely] (LSTM math and recurrent shapes in matrix form were reported in breadth [26][32])
    - How does backpropagation through time work, and where do vanishing gradients come from?

50. Build a minimal scalar autograd engine (a Value class with +, *, tanh and backward via topological sort). [Likely]
51. Implement a simple training loop with early stopping on validation loss and a learning-rate schedule (warmup plus cosine). [Likely]

F. Attention, transformers and LLM components

52. Implement scaled dot-product attention in NumPy. [Likely] (attention "at root level" and the math of self-attention were reported in depth and breadth rounds [27][29])
    - Why divide by sqrt(d_k)?
    - What changes if Wq = Wk = Wv? (reported as a breadth question [7])
53. Add a causal mask and a padding mask to your attention function. [Likely]
    - Where exactly is the mask applied, and why use -inf instead of 0?
54. Implement multi-head self-attention, including the reshape, split heads and output projection. [Likely]
    - What is the time and memory complexity in sequence length?
55. Implement sinusoidal positional encoding, or rotary embeddings (RoPE). [Likely]
56. Implement layer normalization and RMSNorm. [Likely]
    - Pre-norm or post-norm: what is the difference in training stability?
57. Implement a decoder step with a KV cache: given cached K and V and a new token, return the output and the updated cache. [Likely] (KV caching and decoder-only inference were reported in a 2026 breadth round [6])
    - How much memory does the cache use for a given model size and context length?
    - How would multi-query or grouped-query attention change your code?
58. Implement a sliding-window (local) attention mask, as in Longformer. [Likely] (Longformer and long-context attention were reported [6])
59. Implement beam search decoding over a toy next-token probability function. [Likely] (beam search decoding was reported as a conceptual question for an LLM-team AS II [28])
    - How do you handle end-of-sequence tokens and length normalization?
60. Implement temperature, top-k and top-p (nucleus) sampling from a logits vector. [Likely]
61. Implement byte-pair encoding (BPE): learn merges on a small corpus, then tokenize a new word. [Likely]
62. Implement a LoRA linear layer forward pass and explain which parameters are trained. [Likely] (LoRA was reported in AWS AS breadth [4])
    - How do you merge the adapter weights for inference?
63. Implement the SFT loss for a causal LM, masking prompt tokens so only response tokens count. [Likely] (SFT pipelines were reported in a depth round [21])
64. Implement the DPO loss given log-probabilities from the policy and reference models for chosen and rejected responses. [Likely] (Alexa and AGI job descriptions mention SFT, DPO and RLHF [37])
65. Implement InfoNCE (in-batch negatives) contrastive loss for a bi-encoder retriever. [Likely] (cross-encoder versus bi-encoder was reported in a 2026 round [6])

G. Probability, sampling and statistics in code

66. Build a weighted random number generator from an array of probabilities or weights (random pick with weight). [Reported] [9] (Senior AS screen, 2026)
    - What is the complexity per sample with prefix sums plus binary search? Can you do O(1) with the alias method?
67. Implement reservoir sampling to pick k items uniformly from a stream of unknown length. [Likely]
    - How would you make it weighted?
68. Compute a bootstrap confidence interval for a metric difference between two models, such as AUC. [Likely]
69. Implement a two-sample t-test (Welch) or a two-proportion z-test for an A/B test, from scratch. [Likely] (A/B testing and p-values were reported in breadth)
    - How would you compute the required sample size for a given minimum detectable effect?
70. Implement Gaussian or multinomial Naive Bayes for text classification, with Laplace smoothing and log-space computation. [Likely] (Naive Bayes was reported in breadth [38])
71. Implement PCA via SVD, or via eigendecomposition of the covariance matrix, and project data onto the top k components. [Likely] (PCA and covariance were reported in breadth)
    - How would you implement power iteration for the top component only?
72. Implement EM for a 1-D Gaussian mixture model with two components. [Likely]
73. Implement Thompson sampling or UCB for a multi-armed bandit over ad creatives or widgets, and simulate regret. [Likely] (ads and personalization)
74. Estimate pi, or an expected value, by Monte Carlo simulation, and report its standard error. [Likely]

H. Recommendation, search and NLP building blocks

75. Implement matrix factorization for implicit or explicit ratings with SGD or ALS. [Likely] (Amazon recommender-systems roles list collaborative filtering [34])
    - How do you handle cold-start users and items?
76. Implement item-to-item collaborative filtering: compute cosine similarity between item vectors from a sparse user-item matrix, and recommend the top N. [Likely] (Amazon's well-known item-to-item recommendations)
    - How would you scale this computation to millions of items?
77. Implement TF-IDF and BM25 scoring for a query against a small document collection. [Likely] (product search)
78. Implement a pointwise or pairwise learning-to-rank loss (for example, RankNet) and its gradient. [Likely]
79. Implement n-gram language-model counting with add-k smoothing, and score a sentence. [Likely]
80. Implement a simple word2vec skip-gram negative-sampling update for one (center, context) pair. [Likely]
81. Deduplicate near-identical product titles using normalized string keys or character shingles plus Jaccard similarity. [Likely]

I. Data manipulation and ML pipeline coding

82. Compute a rolling average when some dates are missing. [Reported] [11]
    - Should missing days count as zero or be skipped? Defend your choice.
83. Load a dataset in pandas, clean it, engineer features, and fit a model, explaining each step. [Reported] [13][31]
    - Which steps can leak information from the test set?
84. Answer simple data-wrangling questions in Python (group-by, joins, aggregations) alongside standard DSA. [Reported] [3]
85. Return the top-K most frequent elements in a stream or array, in Python. [Reported] [21]
86. Implement a train, validation and test split that is stratified by label and grouped by user, so no user appears in two splits. [Likely]
87. Implement a time-based split and an expanding-window backtest for a demand-forecasting model. [Likely] (supply-chain forecasting)
88. Implement standard scaling and one-hot encoding as fit/transform classes that handle unseen categories. [Likely]
89. Implement mini-batch iteration over a dataset that does not fit in memory, with shuffling. [Likely]
90. Implement random oversampling, or class-weighted loss, for an imbalanced dataset, and show the effect on your metric function. [Likely] (class imbalance was reported in depth rounds [21])

J. Round-level probes (how the interviewer pushes at L5)

91. Before writing code: what are the inputs, outputs, data types and sizes? What should happen on empty or degenerate input? [Likely] (Amazon's prep pages stress edge cases and bad input [1][2])
92. What is the time and space complexity of your implementation, and where is the bottleneck? [Likely]
93. You cannot run the code. Walk me through it on a tiny example by hand. [Likely] (third-party guides note that code is usually not executed [30])
94. How would you test this function? Which unit tests and invariants would you write? [Likely]
95. How would this change if the data were 1000x larger, or arrived as a stream? [Likely]
96. Now apply it to our team's problem. For example, where would k-means or kNN fit in product categorization or Rufus retrieval? [Likely]
97. An extra round after a split debrief: let's revisit coding with a follow-up problem. [Reported] [15]
98. Tell me about a time you dove deep into data or code to find a root cause (an LP question opening or closing the coding round). [Reported] [21]
99. Tell me about a time you had to deliver a model under a tight deadline. What corners did you refuse to cut? [Likely] (LPs "Deliver Results" and "Insist on the Highest Standards")
100. Tell me about a time you simplified a complex ML approach into something you could ship. [Likely] (LP "Invent and Simplify")

Prep priorities (derived from the tags above)

- **Most-reported ML coding tasks:** AUC from scores and labels (1), k-means (18), logistic regression or a gradient-descent training loop (27, 28), kNN (20), IoU (11), and a generic classifier from scratch (29).
- **Most likely extensions for 2025-2026 L5 loops on LLM, Alexa or AGI teams:** attention with masks and a KV cache, sampling and beam search, and LLM losses (52-65). These are [Likely] as coding tasks, but the concepts were reported often in breadth and depth rounds.
- **Do not neglect:** data-manipulation coding (82-90). It shows up in screens and loops where the interviewer is a scientist, not an SDE.

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06. ML and Science System Design: Recommendations, Search, Ads, Fraud, Forecasting, Alexa/Rufus LLMs, Pricing and Supply Chain

> Scope: the "Science Application" / "ML System Design (team-specific)" round in the Amazon L5 Applied Scientist (Applied Scientist II) onsite loop, 2025-2026. This chapter is questions only, with no answers.
>
> Research note (please read): no web searches or page fetches worked for this chapter. The run's shared web-search budget was already used up, and direct fetches to Glassdoor, Blind, Medium, Dataford, IGotAnOffer, amazon.jobs and similar sites were refused by this environment's egress proxy. Every [Reported] tag below therefore comes from the sourced findings already gathered for chapters 01-04 of this guide. Those findings were themselves built from search-result extracts of the cited pages, not full reads. Each [Reported] question cites the original page. A question is tagged [Reported] only if one of those chapters attributes it to an Amazon science interview. Everything else is [Likely]. A follow-up session with a fresh search budget could add more first-person design-round reports, especially from Blind and 1point3acres.

Format notes for this round

- **Where it sits in the loop.** L5 AS loops are usually 4-6 interviews of 55-60 minutes each. One 2025-2026 Applied Scientist II report gave the loop as "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [2]. Amazon's official prep page describes the loop as about four 55-minute interviews with members of the science community. It says interviewers are assigned competencies based on the Leadership Principles and typically ask 2-3 behavioral questions each [1]. Other candidates and guides list "ML application" or "science application" as a distinct onsite round [17][18].
- **L5 versus L4.** Blind commenters say L5 loops usually include the design or application round and L4 loops may not [5]. The L5 expectation is end-to-end ownership: framing a vague business problem, choosing data, labels and models, defining offline and online metrics, and launching and monitoring the system, with sound trade-off calls along the way [5][16].
- **Who runs it.** It is often the hiring manager or a senior scientist from the hiring team, and the problem is drawn from that team's domain [2][8]. Examples include Ads trust [9], Alexa speech [10], product search and recommendations [3][12][13], and LLM agents and RAG [3][14].
- **What is assessed.** IGotAnOffer describes the round as being given a high-level ML problem and proposing the data sources, annotation strategy, modeling technique,

evaluation metrics, and trade-offs and risks [7]. In candidate reports, interviewers drill into the data pipeline "from collection to deployment" [9], dataset construction and sampling [11], and bias in implicit feedback such as click position bias [12][13]. In 2025-2026, GenAI design prompts appear more often: agentic recommenders, RAG and LLM-agent evaluation [3][6][14].

- **Leadership Principles.** Expect 1-2 LP questions inside this round, often Customer Obsession, Dive Deep, Invent and Simplify, or Think Big. Reports put LPs at roughly 5-15 minutes of each technical interview [1][16].
- **How it differs from SDE system design.** The weight is on the science (problem framing, labels, modeling, metrics, experimentation) more than on distributed-systems boxes. Candidates still report questions about scale, latency and deployment [9][22]. One campus loop had a separate "Scalable ML" round [21].

Tag legend: **[Reported]** means a cited source attributes the question, or a close paraphrase, to an Amazon science interview (AS unless noted). **[Likely]** means a grounded inference from the team's products, job postings or Amazon Science publications. It has not been seen verbatim.

A. Generic framing (the questions every design round opens or pivots with)

1. Here is a high-level ML problem from our team. Propose a solution, covering data sources, annotation strategy, modeling techniques, evaluation metrics, and trade-offs and risks. [Reported] [7]
    - Which parts of the problem actually need ML, and which could be rules or heuristics?
    - What is your v1 that could ship in one quarter, and what is v2?
2. Walk me through the full data pipeline for your design, from collection to deployment. [Reported] [9]
    - Where do labels come from, and how noisy are they?
    - How do you keep training and serving features consistent?
3. Answer an ML application question about one of our team's Amazon products. [Reported] [8]
    - Who is the customer, and how would this change their experience? (Customer Obsession)
4. Pick an agentic product we could build. Who is the customer, and how do you measure success? (Seattle 2026 report) [Reported] [2]
    - What would the press release headline say (Working Backwards)?
    - Which single metric would you defend to a VP?
5. How would you take this from a prototype to Amazon-scale traffic? [Likely]
    - Where do latency and cost hit first: feature fetch, model inference, or candidate generation?
    - What do you precompute offline, and what must be computed in real time?
6. What offline metric and online metric would you pair, and what do you do when they disagree? [Likely]

    - How do you design the A/B test, including the unit of randomization, duration, power and guardrails?
7. What would make you kill or pivot this project after launch? [Likely]

B. Recommendations and personalization

8. Design a pipeline to improve personalized product recommendations on the Amazon homepage. [Reported] [6]
    - What is the north-star metric, and what are the guardrails?
    - How do you split the work into candidate generation, ranking and re-ranking?
    - How do you trade relevance against diversity and novelty on one page of widgets?
9. Design an agentic recommendation system for Amazon, covering component architecture, model choices, and quality and safety evaluation. [Reported] [6]
    - Which decisions should the LLM agent make, and which belong to a classical ranker?
    - How do you evaluate multi-turn recommendation quality before an A/B test?
    - How do you stop the agent from recommending unsafe, unavailable or irrelevant products?
10. Design a recommender system (reported as the design round in an AS loop). [Reported] [15]
    - Two-tower retrieval, matrix factorization, or sequence models: which, and why?
    - How do you handle cold start for new users and new ASINs?
11. A user goes from viewing to clicking to purchasing a product. Model the probability of each action given a user and a product. [Reported] [11]
    - Would you build three independent models, or model view, click and purchase jointly? [Reported] [11]
    - How would you build the dataset, choose features, and down-sample (for example with reservoir sampling)? [Reported] [11]
    - How do you handle the selection bias that purchases are only observed after clicks?
12. How would you optimize Amazon's recommendation system to improve customer engagement and sales? [Likely] (see [6])
    - How do you avoid a feedback loop that only recommends already-popular products?
13. Design "Frequently bought together" / "Customers who bought this also bought". [Likely]
    - How do you separate substitutes from complements?
    - How do you deal with spurious co-purchases from promotions or bundles?
14. Design recommendations for Prime Video or Amazon Music, where consumption is sequential and the feedback is implicit. [Likely]
    - How do you define a positive label for a show that was watched for only 3 minutes?
15. How would you personalize email or push notifications without fatiguing customers? [Likely]
    - How would you frame this as uplift modeling rather than response modeling?

16. How would you detect and fix popularity bias or exposure unfairness toward small sellers in recommendations? [Likely]

C. Product search and ranking

17. You have a large collection of user click data for many search queries and product documents. Is click data alone sufficient? How would you train a retrieval or IR model with the data at hand? [Reported] [12]
    - How do you correct for position bias in clicks? [Likely] (see [23])
    - What other biases (presentation, trust, query intent) contaminate click labels?
18. Walk through how you would design, train, validate and deploy a ranking model for product search. [Reported] [13]
    - How would you estimate true relevance from clickstream logs with position bias? [Reported] [13]
    - Pointwise, pairwise or listwise learning-to-rank: which would you choose, and why?
19. Answer a search-relevance application question for Amazon search. [Reported] [3]
    - How do you define "relevant" for a query like "running shoes" versus "nike pegasus 40 size 10"?
20. Compare bi-encoders and cross-encoders for retrieval and re-ranking. Where would each sit in a product-search stack? [Reported] [3]
    - What is your latency budget per stage, and how do you distill the cross-encoder?
21. Design query understanding for Amazon search (intent, product type and attribute extraction). [Likely] (see [27])
    - How would you launch it in a new marketplace and language with little labeled data?
22. Design a system to detect and reduce zero-result or low-quality-result searches. [Likely]
    - When should you relax the query, and when should you show a "did you mean"?
23. How would you combine relevance with business objectives (conversion, margin, sponsored placement) in the final ranking? [Likely]
    - How do you stop business boosts from eroding relevance over time?
24. Design an evaluation framework for search relevance using human annotation and LLM judges. [Likely]
    - How do you measure annotator agreement and calibrate an LLM judge against humans?
25. How would you build semantic or vector search over a catalog of hundreds of millions of products? [Likely]
    - How do you handle index freshness for new listings and price or stock changes?

D. Advertising (Sponsored Products, DSP, Ads Trust)

26. Design a system that recommends trustworthy ads to Amazon users while maintaining brand safety and compliance. [Reported] [9]
    - Walk through the full data pipeline from collection to deployment. [Reported] [9]
    - How do you trade ad revenue against trust and safety errors?

    - How do you get labels for rare policy violations?
27. Design click-through-rate prediction for Sponsored Products ads. [Likely] (see [20])
    - Why does calibration matter in a second-price or first-price auction?
    - How do you handle delayed conversions in conversion-rate modeling?
28. Revenue is flat, cost-per-click is up and conversions are down after a budget-pacing change. How would you diagnose it? [Likely]
29. Design an ad-relevance or ad-moderation classifier for images and text in sponsored creatives. [Likely] (see [9])
    - How do you handle adversarial advertisers who iterate on rejected creatives?
30. How would you measure the incremental impact (lift) of an ad campaign for an advertiser? [Likely]
    - Ghost ads, geo experiments or observational methods: what are the trade-offs?
31. Design a bidding or budget-pacing algorithm for advertisers on Amazon DSP. [Likely]
    - How would you frame it as constrained optimization, or as a bandit problem?
32. Design keyword or audience targeting recommendations for advertisers. [Likely]

E. Fraud, abuse, trust and payments

33. Design a fraud detection system for Amazon payments transactions. [Likely] (see [28])
    - How do you deal with extreme class imbalance and delayed chargeback labels?
    - How do you pick the operating threshold given the costs of false positives and false negatives?
    - How do you handle adversaries adapting to your model?
34. How would you detect fake reviews or abusive sellers with heavily imbalanced and adversarial labels? [Likely]
    - Would you use graph-based signals across accounts, devices and payment instruments?
    - How do you evaluate when your labels come from your own past enforcement?
35. Design a system to detect counterfeit products or listings. [Likely]
    - How do you combine image, text, seller history and brand-registry signals?
36. Design account-takeover detection at login time under a strict latency budget. [Likely]
    - What do you do with the "uncertain" band: friction, step-up auth, or human review?
37. Design returns-abuse detection. How do you avoid punishing legitimate customers? [Likely]
38. How would you prioritize a human-investigator queue when you can review only 1% of flagged cases? [Likely]
    - How do the investigators' decisions feed back into training without biasing it?
39. Your fraud model's precision drops sharply over two weeks. Walk me through the investigation. [Likely]

F. Computer vision and catalog quality

40. Use computer vision to detect defective products in fulfillment-center warehouses. [Reported] [8]
    - How do you collect and label defect images when defects are rare?

    - Edge or cloud inference, and what is the throughput requirement per station?
41. You have 10k product listings across 100 categories, each with a title, description and photo. How would you categorize them? [Reported] [12]
    - How do you fuse text and image signals?
    - How do you handle a long tail of categories with very few examples?
    - How do you evaluate when the taxonomy itself is noisy?
42. How would you detect and fix wrong attributes (size, color, brand) on listings at catalog scale? [Likely]
    - How would you use an LLM for attribute extraction, and how do you verify its outputs?
43. Design duplicate-listing detection (the same product listed under multiple ASINs). [Likely]
    - How do you scale pairwise matching to hundreds of millions of items?

G. Forecasting, supply chain and inventory (SCOT)

44. How would you forecast demand for millions of products, including new ones with no history? [Likely] (see [24][25][29])
    - Why use probabilistic or quantile forecasts rather than point forecasts for buying decisions?
    - Would you use a global model or per-item models, and why?
45. What is the newsvendor problem, and how would you use it to set inventory levels? (Amazon supply chain interviews) [Reported] [26]
    - How does a quantile forecast map to the newsvendor critical ratio?
46. How would you model an optimization problem from your work, including objective, decision variables and constraints? [Reported] [16]
    - When would you use a heuristic instead of an exact solver?
47. Design an inventory placement system that decides which fulfillment centers should stock each product. [Likely] (see [25])
    - How do forecast errors propagate into placement and transportation costs?
48. How would you forecast demand during Prime Day or other promotions, when history is sparse and distorted? [Likely]
    - How do you correct for censored demand when items were out of stock?
49. How do you evaluate a forecasting system against a business outcome (in-stock rate, holding cost) rather than MAPE alone? [Likely]
    - How would you run an experiment on a buying policy when products share warehouses and trucks?
50. Design a system to predict delivery time (the promise date shown on the detail page). [Likely] (see [19])
    - What loss would you use if a late delivery costs more than an early one?
    - How do you calibrate promises across regions and carriers?
51. How would you improve address geocoding or delivery-location accuracy for last-mile delivery? [Likely] (see [19])
52. Design a capacity or labor-planning forecast for fulfillment centers. [Likely]

H. Pricing, causal inference and experimentation

53. Design a system that recommends competitive prices for millions of products. [Likely] (see [29])
    - How do you estimate price elasticity when prices are set by your own algorithm (endogeneity)?
    - What guardrails stop the system from causing price spirals or customer-trust problems?
54. How would you design an experiment to measure the effect of a pricing algorithm when prices interact across products? [Likely]
    - How do you handle interference between treated and control products, for example with switchback or cluster designs?
55. How would you estimate the causal effect of a promotion without a clean experiment? [Likely]
    - When would you use double ML, instrumental variables, or difference-in-differences?
56. How would you design an A/B test for a model launch? [Reported] [4]
    - What do you do if the test is underpowered or the metric is heavy-tailed?
    - How would you use variance reduction such as CUPED?
57. A new ranking model wins offline but is neutral online. What could explain it, and what do you do next? [Likely]
58. How would you decide whether to discount a product, and by how much, for a deal event? [Likely]
    - How do you account for cannibalization and pull-forward demand?

I. Alexa, Rufus and LLM-powered products

59. How would you improve Alexa speech recognition for non-native speakers and users with strong regional accents? [Reported] [10]
    - Where would you get diverse accent training data, and how would you handle phonetic variation? [Reported] [10]
    - How would you measure improvement per accent group without regressing the majority?
60. Design an enterprise RAG (retrieval-augmented generation) service. [Reported] [14]
    - How do you chunk, embed and index heterogeneous documents?
    - How do you enforce document-level access control in retrieval?
    - How do you evaluate retrieval quality separately from generation quality?
61. Evaluate an LLM agent with metrics beyond accuracy, such as tool-use correctness, hallucination and calibration. [Reported] [3]
    - How do you build an offline evaluation set for multi-step agent tasks?
    - When would you trust an LLM-as-judge?
62. Design a GitHub Copilot-style coding agent (reported in a Senior AS loop, Boston). [Reported] [3]
    - What context does the agent retrieve, and how do you rank it?
    - How do you measure whether it helps developers (acceptance rate versus downstream bugs)?
63. On an LLM team, discuss LLM decoding and training choices in the context of an Amazon use case. [Reported] [22]

    - Greedy, beam or sampling decoding for a customer-facing assistant: which would you use, and why?
64. Design Rufus: answer "What do I need for cold weather golf?" using the catalog, reviews and Q&A. [Likely] (see [27][31][32])
    - How do you ground answers in product facts and avoid hallucinated specs?
    - How do you evaluate it offline before an A/B test, and what is the online success metric?
    - How do you mix organic and sponsored products in a generative answer?
65. How would you summarize thousands of customer reviews into a short "Customers say" blurb? [Likely]
    - How do you measure faithfulness and coverage of minority opinions?
    - How do you stop manipulated reviews from steering the summary?
66. Design an Alexa skill or intent router that maps an utterance to the right skill or API (natural-language understanding). [Likely]
    - How do you handle new skills with no training data?
    - What is your latency budget end to end, from wake word to response?
67. How would you serve an LLM-based assistant to millions of users at acceptable cost and latency? [Likely]
    - Would you use distillation, quantization, caching, speculative decoding, or routing between small and large models?
68. How would you fine-tune and align an assistant for a new language or marketplace (SFT, DPO, RLHF or RLAIF)? [Likely] (see [27])
    - How do you get preference data, and how do you detect reward hacking?
69. How would you make an on-device model 10x smaller with minimal quality loss (Echo or Kindle devices)? [Likely]
70. How would you detect and mitigate unsafe, biased or policy-violating responses from a shopping assistant? [Likely]
    - Which red-teaming process and which guardrail models would you use?
71. Design a conversational customer-service agent that can issue refunds or replacements. [Likely]
    - Which actions require human-in-the-loop, and how do you decide?
    - How do you evaluate resolution rate without incentivizing over-refunding?

J. Scaling, MLOps and lifecycle (probes across any design)

72. How do you monitor this model in production, and how do you detect data or concept drift? [Likely]
    - What triggers a retrain, and how do you validate it before rollout?
73. How would you run inference at scale, in real time versus batch, and on what infrastructure (for example, SageMaker endpoints)? [Likely]
74. How do you handle feature-store consistency and training/serving skew? [Likely]
75. How would you roll back safely if the new model misbehaves on a subset of traffic? [Likely]
76. Your labels arrive weeks late (returns, chargebacks, long-term retention). How do you train and evaluate under label delay? [Likely]
77. How would you reduce the cost of this system by 50% with minimal quality loss? [Likely] (Frugality)

78. Which parts of your design would you build, and which would you reuse from existing Amazon services or other teams? [Likely]
79. How would you handle personally identifiable information and privacy constraints in the training data? [Likely]

K. L5-bar probes and LP crossovers inside the design round

80. Of everything you proposed, what is the single riskiest assumption, and how would you de-risk it in two weeks? [Likely]
81. A product manager wants to launch in a month, and your offline evaluation is incomplete. What do you do? [Likely] (Bias for Action versus Insist on the Highest Standards)
82. Your design improves conversion but increases returns. How do you decide whether to ship it? [Likely]
83. Tell me about a time you designed an ML system end to end. What would you change now? [Likely] (Dive Deep, Ownership)
84. Which metric would you refuse to optimize directly, and why? [Likely]
85. How would you explain the trade-offs in your design to a non-scientist stakeholder? [Likely] (Earn Trust)
86. If you had 10x the data, or one-tenth of it, how would the design change? [Likely]
87. How would you hand this system over to an engineering team for long-term ownership? [Likely]

L. Team-specific "warm-up" prompts to rehearse (inferred from postings and Amazon Science)

88. (Search/IR internship and AS postings) Design a learning-to-rank system that uses collaborative-filtering signals. [Likely] (see [30])
89. (SCOT Buying Science) Design a system that decides how much of each product to buy from which vendor and when. [Likely] (see [29])
90. (Payments) Design a model to predict payment-method failure and pick the best fallback for the customer. [Likely] (see [28])
91. (Last mile) Design a model that predicts whether a package will be delivered successfully on the first attempt. [Likely] (see [19])
92. (Alexa International) Design a multilingual natural-language-understanding model for a marketplace with code-switching (for example Hinglish). [Likely] (see [10][17])
93. (Search relevance) Design a system that estimates relevance from click logs, accounting for query intent. [Likely] (see [23])
94. (Ads Trust) Design a policy-violation detection system for ads with an annotation budget of 5,000 labels per month. [Likely] (see [9])
95. (Rufus) Design an offline evaluation harness for a shopping assistant, covering factuality, helpfulness, safety and commercial impact. [Likely] (see [27][31][32])

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07. LLM and GenAI Questions: Transformers, Fine-Tuning, RLHF/DPO, RAG, Evaluation, Hallucination, Efficiency, and Amazon Product Context (Bedrock, Nova, Rufus, Alexa+)

> Scope: the LLM and GenAI questions an L5 Applied Scientist (Applied Scientist II) candidate should expect across the 2025-2026 onsite loop. This chapter is questions only, with no answers.
>
> Research note (please read): the shared web-search budget for this run was used up before this chapter started, and direct page fetches to Glassdoor, Blind, LeetCode, 1point3acres, Medium, amazon.jobs, amazon.science, IGotAnOffer and similar sites failed in this environment (DNS refused by the egress proxy). This chapter therefore draws on three kinds of evidence:
>
> - Candidate-report findings already gathered for chapters 01-06 of this guide. Those came from search-result extracts of the cited pages, not full reads. Each [Reported] tag cites the original page.
> - Three GitHub-hosted interview aggregators that were fetched and read directly for this chapter [22][23][24][25]. They compile Amazon candidate reports from sources they cite (mostly trueinterview.io, CC BY 4.0) or do not name. They are labeled as aggregators wherever used.
> - Amazon job postings and product pages, used only to ground [Likely] questions.
>
> A follow-up session with a fresh search budget could add more first-person reports, especially from Blind and 1point3acres threads about Rufus, AGI/Nova and Alexa teams.

Format notes for this round

- **There is usually no separate "LLM round."** LLM and GenAI questions come up inside the standard loop seats: ML/Science Breadth, Science Depth (your own project), Science Application / ML System Design, and sometimes ML Coding. Amazon's official page describes a loop of about four 55-minute interviews with members of the science community. Each interviewer is assigned technical competencies and Leadership Principles and typically asks 2-3 behavioral questions [1]. Candidate reports for AS II more often describe five rounds of 55-60 minutes, for

example "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [2]. Aggregated Amazon science reports also describe a "modern deep learning" round on some loops, and one or two ML-theory rounds on GenAI teams [22][23].

- **Transformers are now baseline breadth, even off-LLM teams.** An L5 virtual-onsite thread on Blind treats transformers as core breadth [13]. An Amazon Ads candidate reported being rejected after not preparing for attention, transformers and LLMs [14]. 2025-2026 reports cover decoder-only inference, KV caching, Longformer-style long-context attention, bi-encoders vs cross-encoders, LoRA, BERT vs GPT, DPO vs RLHF, beam search, and "modern LLM training techniques" [3][4][7][8][10][11].
- **Team-dependent depth.** AGI/Nova-team loops are reported to lean hard on modern LLM training and ML coding. One aggregator reports a "debug a broken Transformer implementation" coding problem, and says candidates compare the ML-coding style to OpenAI's [10][15][17][25]. LLM-team and Rufus/Alexa loops tie decoding, training and evaluation to an Amazon use case [8][27][28]. Science-application prompts in 2025-2026 include agentic products, agentic recommenders, enterprise RAG and LLM-agent evaluation [2][3][16][21].
- **Interviewer type.** Usually Applied Scientists, Senior Applied Scientists or Principal Applied Scientists. The application round is often run by the hiring manager [1][2].
- **The L5 bar.** L5 answers are expected to hold up through two or three layers of follow-up, against one layer at L4 [19]. Blind commenters say L5 candidates are vetted harder on whether they "raise the bar" [34]. For LLM topics that means:
  - deriving the mechanics (attention, the DPO loss, KV-cache memory);
  - naming failure modes and trade-offs;
  - choosing between prompting, RAG, fine-tuning and preference tuning for a concrete Amazon problem;
  - defining how you would evaluate it offline and online, including the cost and latency budget.
- **Leadership Principles still apply.** Expect 1-2 LP questions inside each technical round [1][19]. A GenAI-flavored behavioral prompt is now reported in Amazon screens: how you use AI tools, and a time one gave you a wrong answer [24].

Tag legend. [Reported] means a cited source attributes the question, or a close paraphrase, to an Amazon science interview. That is Applied Scientist unless noted. Items from the Research Scientist or MLE aggregators are marked "(Amazon RS aggregate)" or "(Amazon MLE aggregate)". [Likely] means a grounded inference from the role, Amazon's products or job postings, or adjacent reported questions. It has not been seen verbatim.

A. Transformer fundamentals

A1. Architecture and attention

1. Explain the Transformer architecture in detail: encoder, decoder, and what each component does. [Reported] [4][9][11]
   - Why scale the dot product by sqrt(d_k)? What happens to the softmax if you don't?
   - Why multiple heads instead of one wide head?

    - Where are the residual connections and layer norms, and why does pre-norm vs post-norm matter for training stability?
2. Explain the attention mechanism "at root level," including scaled dot-product attention, multi-head attention and positional encoding. [Reported] [5][7]
    - Write the attention equation with shapes for Q, K and V.
    - How does the causal mask enter the equation?
3. What happens if you set Wq = Wk = Wv in a Transformer? [Reported] [6]
    - What does that do to the attention matrix's symmetry, and what expressiveness do you lose?
4. Explain the difference between LSTMs and Transformers mathematically. [Reported] [12]
    - How does each handle long-range dependencies, and what are the parallelism and compute trade-offs?
    - Why did transformers replace RNNs for language modeling? [Reported] (Amazon MLE aggregate) [23]
5. Why do Transformers need positional information? Compare sinusoidal, learned, RoPE and ALiBi encodings. [Reported] (Amazon RS aggregate: RoPE formula, RoPE and ALiBi) [22]
    - How does RoPE behave when you extrapolate past the training context length?
6. Compare multi-head, multi-query and grouped-query attention. [Reported] (Amazon MLE aggregate, phone screen) [23]
    - How does each change KV-cache memory and quality?
7. Transformer internals: how does the feed-forward block contribute compared with attention, and where do most of the parameters and FLOPs go? [Reported] (Amazon RS aggregate, "modern deep learning" round) [22]
    - Where do the parameter count and FLOPs come from as d_model, the number of layers and the sequence length change?
8. Why does AdamW decouple weight decay, and why does that matter for training transformers? [Reported] (Amazon RS aggregate) [22]
9. You are handed a Transformer implementation that trains but performs badly. Debug it. [Reported] (Amazon AGI, aggregator) [25]
    - What bugs would you check for first (mask direction, softmax axis, scaling, positional-encoding offsets, layer-norm placement)?

A2. Model families

10. How do BERT and GPT differ in objectives, architecture and use cases? [Reported] [7]
    - How is BERT trained (MLM and NSP), and why did RoBERTa drop NSP?
    - When would an encoder-only model still beat a decoder-only LLM for an Amazon task, such as product classification?
11. What is an encoder-decoder model such as T5, and how would you optimize it for inference? [Reported] [5]
12. Compare architectures: dense vs mixture-of-experts, and encoder-only vs decoder-only vs encoder-decoder. [Reported] (Amazon MLE aggregate, "LLM depth" onsite) [23]
    - How do you load-balance experts in MoE, and what goes wrong without it?
13. What do scaling laws say about how to split a fixed compute budget between parameters and tokens? [Reported] (Amazon MLE aggregate) [23]

    - How would you use a scaling-law fit to decide whether to train a larger Nova-sized model or serve a smaller one? [Likely]
14. Explain the history of embedding models (word2vec, then BERT pooling, then contrastive sentence encoders, then LLM-based embedders). How would you compare two sizes of the same embedding family? [Reported] (Amazon RS aggregate, breadth) [22]

B. Inference, decoding and long context

15. How does inference work for decoder-only models? [Reported] [3]
    - Walk through prefill vs decode. Which phase is compute-bound and which is memory-bandwidth-bound?
16. What is the KV cache, and how does it speed up autoregressive generation? [Reported] [3][22][23]
    - Estimate the KV-cache memory for a given model (layers, heads, head dim, sequence length, batch).
    - How do GQA/MQA, paged attention and cache quantization reduce it? [Likely]
17. What makes standard self-attention expensive for long contexts? [Reported] [3]
    - What are the time and memory complexity in sequence length, and what does FlashAttention change (and not change)? [Likely]
18. Explain Longformer and how its attention scales to long sequences. [Reported] [3]
    - Compare full attention, sparse or local attention, and other long-context approaches. [Reported] [3]
    - For a multi-turn shopping or search agent, when would you summarize or retrieve rather than extend the context window? [Reported] (Amazon RS aggregate: multi-turn search agent with long-context handling) [22]
19. How does beam search decoding work for LLMs? [Reported] [8]
    - When does beam search hurt open-ended generation, and why?
    - Compare greedy, beam, temperature, top-k and top-p sampling for a customer-facing assistant. [Reported] (LLM team: decoding choices for an Amazon use case) [8]
20. How would you add constrained or structured decoding so an assistant always returns valid JSON for a tool call? [Likely]
    - What does it cost in latency and quality?
21. What is speculative decoding, and when does it give a real speed-up? [Likely]
    - How do you pick the draft model, and how does the acceptance rate depend on the domain?

C. Pretraining, fine-tuning and PEFT

22. Which modern LLM training techniques do you know (pretraining, SFT, preference optimization)? [Reported] [10]
    - For an AGI-team role, expect several follow-ups in a row on each stage. [Reported] [10]
23. Pretraining vs fine-tuning: what changes in data, objective and learning rate? [Reported] (Amazon MLE aggregate) [23]

    - What is catastrophic forgetting, and how do you detect and limit it?
24. Walk me through an SFT training pipeline and its loss. [Reported] [19]
    - Why mask the prompt tokens in the loss?
    - How do you build and clean an instruction dataset for a domain like shopping or Alexa intents?
    - How do you mix general and domain data, and how would you choose the ratio?
25. Explain LoRA. [Reported] [4][22][23]
    - Which matrices do you adapt, and how do you choose the rank and alpha?
    - What is the memory cost of an adapter compared with full fine-tuning, counting optimizer states? [Reported] (Amazon MLE aggregate) [23]
    - Compare LoRA variants (QLoRA, DoRA, adapters, prefix tuning). [Reported] (Amazon RS aggregate) [22]
26. You need a domain-specialized model with limited compute. Do you use prompting, RAG, LoRA, full fine-tuning or continued pretraining? [Likely]
    - What evidence would make you move from one option to the next?
    - How does the answer change if the base model is only reachable as a hosted endpoint, for example through Bedrock? [Likely]
27. If you used knowledge distillation from an LLM, what exactly was distilled, and why? [Reported] [20]
    - Why distill instead of quantizing or pruning?
    - How would you distill an LLM into a small classifier using unlabeled data? [Likely] (see Amazon Science work on LLM distillation [31])
28. How would you generate synthetic training data with an LLM without collapsing diversity or leaking evaluation data? [Likely]
29. Your fine-tuning run's loss spikes at step 40k. How do you debug it? [Likely]
    - Which checks would you run on the data, the learning-rate schedule, mixed precision and gradient norms?

D. Alignment: RLHF, DPO and beyond

30. What is the difference between DPO and RLHF? [Reported] [7]
    - Derive how DPO removes the explicit reward model. What role does the reference model play?
    - What does DPO's beta control, and what happens at very small or very large beta?
31. Explain RLHF and PPO, including the clipped objective. [Reported] (Amazon RS aggregate) [22]
    - Why include a KL penalty to the SFT policy, and what failure mode does it prevent?
32. Compare PPO, GRPO and GSPO. Why does GRPO drop the critic? [Reported] (Amazon RS and MLE aggregates) [22][23]
    - When would you prefer verifiable rewards (RLVR) over a learned reward model?
33. RLHF in practice: what goes wrong at scale (reward hacking, length bias, sycophancy, mode collapse)? [Reported] (Amazon MLE aggregate) [23]
    - How would you detect reward hacking from logs?
34. How do you design a reward signal or preference dataset for a shopping assistant? [Reported] (Amazon RS aggregate: reward design) [22]

    - Who labels, how do you measure agreement, and how do you handle annotator disagreement?
    - Where could RLAIF replace human labels, and where can it not? [Likely] (Alexa postings list SFT, DPO, RLHF and RLAIF [27])
35. Your DPO-tuned model gives longer but less accurate answers. Diagnose it. [Likely]
    - How do you control for length in the preference data and in evaluation?
36. How would you align a multilingual Alexa model when preference data is scarce in some locales? [Likely] (Alexa International Tech posting [27]; the Alexa International loop guide [32])

E. Retrieval-augmented generation (RAG)

37. Design an enterprise RAG service. [Reported] [16]
    - How do you chunk, embed and index heterogeneous documents?
    - How do you enforce per-user, document-level access control at retrieval time?
    - How do you evaluate retrieval quality separately from generation quality?
38. What is the difference between a cross-encoder and a bi-encoder for retrieval? [Reported] [3]
    - When would you use each in a retrieve-then-rerank pipeline, and with what latency budget?
    - When would you distill a cross-encoder into a bi-encoder? [Likely] [31]
39. Explain RAG, and where it helps and does not help with hallucination. [Reported] (Amazon RS aggregate: LoRA, RAG and agents in breadth) [22]
    - What failure modes remain once the right passage is retrieved (ignoring it, misreading it, or mixing it with parametric knowledge)?
40. Compare dense, sparse (BM25) and hybrid retrieval for product search queries. [Likely]
    - How do you handle exact attributes such as model numbers, sizes and brand names?
41. Contrastive embedding training: write out the InfoNCE loss and explain how to keep it numerically stable. [Reported] (Amazon RS aggregate, coding round) [22]
    - How do hard negatives change what the encoder learns, and how do you mine them without false negatives?
42. Rufus-style question: build retrieval-augmented answers to "What do I need for cold weather golf?" over the catalog, reviews and community Q&A. [Likely] [28][29][30]
    - How do you ground claims in product facts and avoid made-up specifications?
    - How do you handle stale or conflicting sources (listing vs reviews)?
    - How do you show citations, and how do you check that a citation supports the claim?
43. How would you decide between a longer context window and better retrieval for an assistant? [Likely]
44. How would you evaluate a RAG system offline (context recall and precision, faithfulness, answer relevance)? [Likely]
    - How do you build the golden set, and how often do you refresh it?

F. Agents and tool use

45. Pick an agentic product we could build. Who is the customer, and how do you measure success? [Reported] [2]
46. Design an agentic AI solution for a hypothetical product. [Reported] (Amazon RS aggregate, science application) [22]
47. Design an agentic recommendation system for Amazon, covering the component architecture, model choices, and quality and safety evaluation. [Reported] [21]
48. Design a GitHub Copilot-style coding agent. [Reported] (Senior AS loop) [3]
    - What context does the agent retrieve, and how do you rank it?
    - How do you measure whether it actually helps (acceptance rate vs downstream defects)?
49. Evaluate an LLM agent with metrics beyond accuracy: tool-use correctness, hallucination and calibration. [Reported] [3]
    - How do you build an offline evaluation set for multi-step agent tasks?
    - How do you assign credit when a five-step trajectory fails at step four?
50. Design a multi-turn search agent that handles long context. [Reported] (Amazon RS aggregate) [22]
51. Design an AI chatbot with intent, dialogue, retrieval and generation layers. [Reported] (Amazon MLE aggregate, phone-screen design) [23]
    - Which layers would you keep as classic NLU or rules, and which would you hand to the LLM?
52. Alexa+-style question: an LLM-based assistant has to call real device and service APIs (timers, smart home, ordering). How do you make tool calls reliable and safe? [Likely]
    - How do you handle ambiguous requests: ask a clarifying question, or act?
    - Which actions require confirmation, and how do you decide?
53. Design a browser-operating agent (in the spirit of Nova Act) that completes multi-step web tasks. How do you make it reliable enough to ship? [Likely] (Nova Act named as an active Amazon AGI product [25])
54. How do you defend an agent against prompt injection from retrieved web pages, product listings or reviews? [Likely]

G. Evaluation

55. How would you evaluate an LLM or LLM-based system? [Reported] (Amazon MLE aggregate, "project plus foundations" onsite) [23]
    - What do you measure offline, and which online guardrails do you set? [Reported] (Amazon MLE aggregate) [23]
56. How do you evaluate an LLM-based system for hallucination, faithfulness and helpfulness? [Likely] (follows from [3][16])
    - When would you trust an LLM-as-judge, and how would you calibrate it against human labels?
    - What biases do LLM judges have (position, length, self-preference), and how do you control for them?
57. Design an evaluation framework that checks whether brand names are spelled correctly in generated images (OCR, embedding checks, human protocol). [Reported] (Amazon RS aggregate) [22]
58. How would you decide an LLM-powered assistant is ready to launch to millions of customers? [Likely]

    - Which launch gates would you use for quality, safety, latency and cost?
59. Offline metrics improved but the A/B test is flat. What happened? [Likely]
    - How do you design an online experiment for a conversational assistant, where sessions interact and the outcome (purchase) is delayed?
60. How do you detect benchmark contamination when evaluating a new foundation model? [Likely]
61. How would you measure an assistant's calibration and teach it to say "I don't know"? [Likely] (calibration reported as an agent-evaluation metric [3])
62. How would you run a red-teaming and safety evaluation for a shopping or voice assistant (toxicity, unsafe advice, PII leakage, jailbreaks)? [Likely]
63. How would you evaluate a multilingual assistant fairly across locales when you have far fewer human raters in some languages? [Likely] [27][32]

H. Hallucination and grounding

64. How would you reduce hallucination in an LLM-powered assistant? [Likely] (Amazon's shopping assistant runs on LLMs including Amazon Nova, built on Bedrock [29])
    - Which causes are data-side, which are decoding-side, and which are retrieval-side?
    - How do you trade off abstaining against helpfulness?
65. An assistant states a wrong battery life for a product. Walk through your root-cause analysis. [Likely] [28][30]
    - How do you tell whether the cause was retrieval, a stale catalog attribute, or the model?
66. How would you build an automatic, claim-level factuality checker for generated answers? [Likely]
    - How do you validate the checker itself?
67. How would you fine-tune an LLM to be a shopping assistant that cites its sources? [Likely]
    - How would you reward grounded claims and penalize unsupported ones in preference tuning?
68. How would you summarize thousands of customer reviews into a short "Customers say" blurb? [Likely]
    - How do you measure faithfulness and coverage of minority opinions?

I. Efficiency, serving and cost

69. What are the trade-offs of putting LLMs into production? [Reported] (Amazon MLE aggregate, "LLM depth" onsite) [23]
70. Quantization: compare int8 and int4, and weight-only vs activation quantization (GPTQ, AWQ). What is the quality cost? [Likely]
    - Which layers or tasks are most sensitive?
71. A customer's Bedrock-hosted workload costs too much. How would you cut inference cost sharply without unacceptable quality loss? [Likely] (Bedrock-shaped prompt, synthesized in an Amazon prep list [26])

    - Compare model routing, caching, distillation, quantization, batching and prompt compression.
    - How do you prove the quality loss is acceptable?
72. Your LLM endpoint's p99 latency doubled after a deploy, but the model weights did not change. Walk me through debugging it. [Likely] [26]
73. Explain continuous batching and why it improves throughput for LLM serving. [Likely]
74. Compare data, tensor and pipeline parallelism for training a large model across hundreds of accelerators. [Reported] (Amazon RS and MLE aggregates: pipeline parallelism) [22][23]
    - Where do the communication bottlenecks sit for each?
    - How does ZeRO/FSDP sharding change memory per device? [Likely]
75. Is an MLP layer compute-bound or memory-bound at a given batch size? Do the roofline analysis. [Reported] (Amazon MLE aggregate, infrastructure loop) [23]
76. You have one latency budget for Rufus answers (say, first token in under 1 second). How do you split it across retrieval, reranking, generation and safety filters? [Likely] [28]
77. When would you route a request to a small model and when to a large one, and how would you learn the router? [Likely]
78. Design a training-data pipeline for LLM pretraining: tokenization, deduplication and quality filtering. [Reported] (Amazon MLE aggregate) [23]
    - How do you detect near-duplicates at web scale?
    - How do you measure the downstream effect of a quality filter?

J. Multimodal and speech (team-dependent)

79. Compare a dual-encoder (CLIP-style) model with a vision-language model that uses a resampler or adapter. [Reported] (Amazon RS aggregate) [22]
    - Which would you use for product image search, and which for visual question answering on a listing?
80. How does the Segment Anything Model work? [Reported] [3][10]
81. How do diffusion noise schedules affect sample quality and training? [Reported] (Amazon RS aggregate) [22]
82. What causes codebook collapse in VQ-VAE, and how do you fix it? [Reported] (Amazon RS aggregate) [22]
83. Image-text embeddings and LLM inference engines (online-assessment topics on some science pipelines). [Reported] (Amazon RS aggregate) [22]
84. For a speech-to-speech assistant (in the spirit of Nova Sonic or Alexa+), what changes when you move from an ASR, then LLM, then TTS cascade to an end-to-end speech model? [Likely] [25]
    - How do you evaluate turn-taking, barge-in and latency?
85. How would you improve Alexa speech recognition for non-native speakers and users with strong regional accents, and could an LLM rescoring stage help? [Reported for the base question] [32]; LLM follow-up [Likely]

K. Applying GenAI to Amazon problems (science-application flavor)

86. On an LLM team, discuss LLM decoding and training choices in the context of an Amazon use case. [Reported] [8]
87. Here is a high-level problem from our team. Would you solve it with an LLM at all? Propose a solution: data sources, annotation strategy, modeling, evaluation, trade-offs and risks. [Reported for the frame] [33]; the LLM-or-not framing is [Likely]
    - What is the cheapest baseline that might be good enough?
88. Use an LLM to categorize millions of products into a large taxonomy. How do you combine LLM labeling, a distilled classifier and human audit? [Likely] (product categorization is reported in classical form in earlier chapters; LLM distillation context [31])
89. Use an LLM to generate or enrich product listings (titles, bullet points, attributes) from seller input. How do you prevent made-up attributes? [Likely]
90. Use LLMs as relevance judges for product search. How do you validate them against human judges and behavioral signals? [Likely]
91. How would you personalize Rufus answers with a customer's purchase history while respecting privacy? [Likely] [29]
    - How would you measure whether personalization helps rather than creeps people out?
92. Sponsored products in a generative answer: how do you blend ads into an LLM response while keeping trust, and how do you measure the impact? [Likely]
93. Design an LLM-based system to detect policy-violating content or fraud in reviews or ads. [Likely] (Advertising Trust context [18]; content moderation listed as an Amazon AI-assisted task [24])
    - How do you deal with adversaries who adapt to the classifier?
94. Design a system that turns customer-support transcripts into structured issue reports. How do you evaluate extraction quality? [Likely]
95. A team wants to replace an existing BERT classifier with an LLM prompt. What do you need to see before you agree? [Likely]
    - How do you compare cost per 1,000 requests, latency and accuracy on the long tail?

L. Depth probes on your own LLM project (depth-round style)

96. Walk me through your most significant LLM or NLP project, including the GenAI fundamentals behind it and its business impact. [Reported] [18]
97. Why did you choose this base model and size? What would you do with 10x more compute? [Reported] (Amazon RS aggregate: backbone choice, use of extra compute) [22]
98. How did you build the evaluation set, and how do you know it reflects real customer traffic? [Likely]
99. What did your error analysis show, and which fix had the biggest effect? [Likely]
    - What did you try that did not work?

100. If you fine-tuned, how did you rule out leakage between training data and evaluation prompts? [Likely]
101. Where would your approach break if deployed to a different marketplace or language? [Likely]
102. How did you decide the project was "done" and ready to ship (Deliver Results, Insist on the Highest Standards)? [Likely]

M. GenAI behavioral and LP crossovers

103. How do you use AI tools in your work, and when did one give you a wrong answer? How did you catch it? [Reported] (Amazon general aggregate, phone screen) [24]
104. Tell me about a time you pushed back on using an LLM where a simpler model was better (Frugality, Invent and Simplify). [Likely]
105. Tell me about a time an LLM feature you shipped behaved badly in production. What did you do in the first 24 hours (Ownership, Dive Deep)? [Likely]
106. Tell me about a time you had to explain LLM risks (hallucination, safety) to a non-scientist stakeholder (Earn Trust). [Likely]

Coverage notes for prep

- **Highest-yield set across reports:** transformer and attention mechanics; KV cache and decoder inference; long context; BERT vs GPT; LoRA; SFT and DPO vs RLHF; beam search and sampling; bi-encoder vs cross-encoder; RAG design; and agent and LLM evaluation (questions 1-2, 10, 15-19, 22-25, 30, 37-38, 45-49, 55) [3][4][7][8][10][16][19].
- **Team-specific tilt:**
  - AGI/Nova teams: training, RL, parallelism and ML coding [15][17][22][25].
  - Rufus/Stores: RAG, grounding, ranking and evaluation [28][29][30].
  - Alexa: multilingual alignment, tool use and speech [27][32].
  - AWS/Bedrock-adjacent science: PEFT, serving cost and customization [4][26].
- **Aggregator caveat:** questions marked "(Amazon RS aggregate)" or "(Amazon MLE aggregate)" come from compilations of Amazon Research Scientist and ML Engineer reports [22][23]. Those loops overlap heavily with AS loops but are not identical.

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08. Statistics, Causal Inference and A/B Testing: Hypothesis Tests, Power, Variance Reduction, Weblab, Bandits and Forecasting

> Scope: the statistics and experimentation content an L5 Applied Scientist (Applied Scientist II) candidate meets in the Amazon onsite loop. This chapter covers hypothesis testing, power and sample size, metric design, variance reduction, experiment pitfalls at Amazon scale, causal inference without clean randomization, bandits, and forecasting. Questions only; no answers.

> Sourcing note: the live web search and fetch budget for this run was exhausted, and outbound fetches were blocked by network policy. Every [Reported] tag below cites a source that earlier chapters of this bank (01-06) verified and summarized. Questions without a direct candidate report are tagged [Likely], with the reasoning given. No question is presented as reported unless a cited source reports it.

Format notes for this round

- **There is no dedicated "stats round" in most L5 AS loops.** Amazon's official prep page describes the onsite as four 55-minute interviews with members of the science community. Each interviewer is assigned competencies and typically asks 2-3 Leadership Principle questions [1]. Candidates more often report 5 rounds of 55-60 minutes for AS II: ML breadth, science depth, science application or ML design, coding, and a bar raiser [2][3][15].
- **Where statistics shows up:**
  - **ML/Science Breadth round:** p-values, statistical tests, CLT, and "design an A/B test" are reported as part of the breadth survey [4][8][9].
  - **Science Application / ML design round** (often the hiring manager, and usually present at L5 [20]): candidates are asked how they would A/B test the launch, choose metrics and handle conflicting metrics [5][6][10].
  - **Phone screen:** an L5 phone screen mixed coding, statistics and LPs [12], and a 2025 first-round AS interview covered ML and statistics alongside LLM theory [21].
  - **Bar raiser:** a Principal TPM bar raiser asked a few stats checks, such as interpreting a p-value or a confidence interval [7].
  - **Team-specific depth:** supply-chain, pricing and forecasting teams (SCOT) ask forecasting and inventory questions such as the newsvendor problem [16][17][18].
- **The L5 bar.** L5 candidates are expected to own a problem end to end. That includes framing a vague business goal, defining offline and online metrics, designing the launch experiment and making the ship or no-ship call when metrics disagree [6][20].

In stats questions, interviewers probe past definitions into judgement. Expect follow-ups like "the test is underpowered, now what?", "the primary metric is up but a guardrail is down", and "you can't randomize, now what?" [5][6][10].

- **Experimentation at Amazon (context).** Amazon's internal A/B testing platform is widely known as Weblab. Teams treat online experiments as the launch gate for ranking, recommendation, ads, pricing and Alexa/Rufus changes. Interviewers outside retail may phrase this generically as "how would you A/B test this?" Questions that mention Weblab are tagged [Likely], because no source in this bank reports a candidate being asked about Weblab by name.

A. Probability and statistical foundations

1. State the Central Limit Theorem. Why does it matter for A/B tests? [Reported] [8]
    - When does the CLT approximation break down for a revenue-per-visitor metric with a heavy right tail?
    - How large must n be before you trust a normal approximation for a skewed metric?
2. What is Bayes' rule? Explain the prior, likelihood and posterior. [Reported] [13]
    - Walk through a base-rate example: a fraud model flags 1% of orders, fraud prevalence is 0.1%, and recall is 90%. What is the precision?
3. Work through a coin-flip probability question on the whiteboard. [Reported] [14] (the exact wording was not captured)
    - Derive the MLE for p after k heads in n tosses. What is its variance?
    - How would you build a confidence interval for p when k = 0?
4. State Hoeffding's inequality and use it. [Reported] [11]
    - How many samples do you need to estimate a click-through rate to within plus or minus 0.1% with 95% confidence?
    - How does a Hoeffding bound compare with a CLT-based interval in practice?
5. What is the difference between a frequentist confidence interval and a Bayesian credible interval? [Likely] (follows from the reported CI interpretation probe [7])
6. What is the law of large numbers, and how does it differ from the CLT? [Likely]
7. Explain the delta method. When do you need it for an A/B test metric? [Likely] (ratio metrics like CTR or revenue per session are standard in Amazon retail and ads)
    - Derive the variance of a ratio metric such as clicks / impressions when the randomization unit is the customer.
8. What is the bootstrap? When would you use it instead of an analytic standard error? [Likely]
    - How do you bootstrap correctly when observations are clustered by customer?
9. Compare the Poisson, negative binomial and zero-inflated distributions for modeling orders per customer. [Likely] (retail count data)

B. Hypothesis testing and p-values

10. What is a p-value? Interpret it in the context of an A/B test. [Reported] [4][9]
    - What does a p-value of 0.03 NOT tell you?
    - Explain it to a product manager with no statistics background.

11. Interpret a p-value or a confidence interval. [Reported] [7] (asked by a Principal TPM bar raiser)
    - What is a common misinterpretation of a 95% confidence interval?
12. Walk through hypothesis testing end to end: which statistical tests would you use, and when? [Reported] [8][9]
    - When would you use a t-test, a z-test, a chi-squared test or a non-parametric test?
    - Welch's t-test or Student's t-test: which is the safer default, and why?
13. Explain hypothesis testing, then design an A/B test for a model change. [Reported] [5] [6]
14. Define Type I and Type II errors. Which is more costly for a checkout-flow change compared with a recommendation widget? [Likely]
15. Compare one-sided and two-sided tests. When, if ever, is a one-sided test justified for a launch decision? [Likely]
16. How would you test whether conversion rates differ between two arms? Compare a two-proportion z-test, a chi-squared test and Fisher's exact test. [Likely] (conversion is the core retail metric)
17. When would you use Mann-Whitney U or a permutation test instead of a t-test? [Likely]
    - Does Mann-Whitney test a difference in means? What does it actually test?
18. You observe a statistically significant 0.05% lift on a site with hundreds of millions of sessions. How do you think about statistical versus practical significance? [Likely] (Amazon traffic scale)
19. What is a paired test? When could an Amazon experiment use a paired design? [Likely]
20. How do you test for equivalence or non-inferiority, for example to show that a cheaper model is "no worse"? [Likely] (cost-reduction launches such as model distillation)
    - How do you choose the non-inferiority margin?

C. A/B test design, power and sample size

21. How would you design an A/B test for a model launch? [Reported] [9][10]
    - How do you choose the randomization unit, primary and guardrail metrics, and sample size or power?
    - What do you do if the test is underpowered or the metric is heavy-tailed?
22. Treatment improves the target metric, but a population-level metric drops. What do you do? [Reported] [6]
    - How would you decide whether the drop is real, a segment effect, or noise?
    - Who do you bring into the decision, and what do you write in the launch doc?
23. Derive the sample-size formula for a two-sample test of means. Which inputs drive it the most? [Likely]
    - How does required sample size scale with the minimum detectable effect (MDE)?
24. How do you choose the MDE for an experiment? [Likely]
    - Who should own that number, science or the business?
25. Your power analysis says you need 8 weeks, but the team wants an answer in 2. What are your options? [Likely]

26. How do you choose the randomization unit (customer, session, page view, product, seller, marketplace)? [Likely] (the Amazon marketplace has customers, sellers and products as distinct units)
    - What goes wrong if the analysis unit differs from the randomization unit?
27. How long should an experiment run? How do you account for day-of-week effects and paydays? [Likely]
28. What is an A/A test, and what would you learn from one? [Likely]
    - Your A/A test shows a significant difference. Walk through your debugging.
29. What is a sample ratio mismatch (SRM)? How do you detect and diagnose it? [Likely]
    - The split is 50.4/49.6 on 20 million users. Is that a problem?
30. How do you handle triggering and dilution, for example when only 3% of users ever see the changed widget? [Likely] (retail page widgets)
    - How do you translate a triggered-population effect into a site-wide impact estimate?
31. How would you run an experiment for a change that affects only Prime members, or only a single marketplace? [Likely]

D. Metrics: primary, guardrail and long-term

32. Choose an overall evaluation criterion for a change to product search ranking. [Likely] (search is a core AS team)
    - Why might clicks be a poor primary metric? What about revenue?
33. What guardrail metrics would you add for a recommendation change? [Likely]
    - Consider latency, returns, customer-service contacts, and seller-side metrics.
34. How do you define a metric that captures long-term customer value when the experiment runs for only two weeks? [Likely]
    - How would you build and validate a surrogate or proxy metric?
35. Your offline metric (NDCG or AUC) improved, but the online A/B test is neutral. What could explain it? [Likely] (ranking and recommendation teams launch through online tests)
36. How do you evaluate a metric's sensitivity, or directionality, before using it as a launch criterion? [Likely]
37. How would you measure a change whose effect shows up as fewer returns 30 days later? [Likely] (delayed outcomes in retail)
38. What is Simpson's paradox? Give an Amazon-flavoured example where it could reverse an experiment readout. [Likely]
39. A metric moved by +2% in the US and -1% in the UK. How do you decide whether to launch globally? [Likely]

E. Variance reduction and sensitivity

40. What is CUPED, and why does it reduce variance? [Likely] (CUPED appears as a probe on the reported "design an A/B test" question in this bank's design chapter)
    - Which pre-period covariate would you choose, and what happens for new customers with no pre-period data?
    - How does CUPED relate to regression adjustment (ANCOVA)?

41. Compare stratified sampling, post-stratification and regression adjustment as variance reduction methods. [Likely]
42. How do you handle heavy-tailed metrics like revenue per customer? [Likely] (the heavy-tail probe is in [10])
    - Compare winsorization, capping, log transforms and switching to a quantile metric. What do you trade off with each?
    - Does capping introduce bias, and when is that acceptable?
43. When would you use interleaving instead of an A/B test for a ranking change? [Likely] (search and ads ranking teams)
    - What can interleaving tell you, and what can it not?
44. How can you use ML predictions as control variates for variance reduction? [Likely]
45. Your experiment is underpowered and you cannot get more traffic. List five ways to increase sensitivity. [Likely]

F. Multiple testing, peeking and sequential testing

46. Your dashboard tracks 40 metrics. Three are significant at 0.05. What do you conclude? [Likely]
    - Compare Bonferroni, Holm and Benjamini-Hochberg. When is each appropriate?
47. How would you handle many simultaneous variants (multiple testing)? [Likely] (follow-up to a reported A/B design question; see [6])
48. A PM checks results every day and wants to stop when $p < 0.05$. What is wrong with that? [Likely]
    - How would you design a sequential test (alpha spending, group sequential, mSPRT or always-valid p-values) to allow safe early stopping?
49. You slice results by 20 segments and one shows a large effect. How do you decide whether to believe it? [Likely]
    - How would you estimate heterogeneous treatment effects in a principled way?
50. Many teams run experiments on the same pages at the same time. How do you handle interactions between concurrent experiments? [Likely] (Amazon runs many concurrent experiments across teams)

G. Pitfalls at Amazon scale: interference, novelty and marketplaces

51. How do you measure a recommender when network or interference effects break user-level randomization? [Likely] (follow-up to a reported design question; see [6])
52. How would you design an experiment to measure the effect of a pricing algorithm when prices interact across products? [Likely] (pricing science teams; see [18])
    - When would you use switchback, cluster or marketplace-level randomization?
53. A new seller-facing feature changes how sellers price, which affects buyers in control. How do you get an unbiased estimate? [Likely] (two-sided marketplace)
54. How would you experiment on a fulfillment or inventory-placement policy when products share warehouses and trucks? [Likely] (SCOT; see [18][19])
55. What are novelty and primacy effects? How do you detect them in a two-week test? [Likely]

56. The test wins in week 1 and is flat in week 3. What do you report? [Likely]
57. How do you A/B test an ads model when advertiser budgets are shared between treatment and control? [Likely] (ads teams; see [8])
    - What is budget cannibalization, and how do budget-split designs address it?
58. How would you A/B test a change to Alexa or Rufus, where the conversation spans many turns and outcomes are sparse? [Likely] (LLM product teams)
    - What would your primary metric be, and how would you combine it with human or LLM-judge evaluation?
59. When is an A/B test the wrong tool, and what would you do instead? [Likely]

H. Weblab and the launch decision

60. Walk me through how you would set up, monitor and close out an experiment on Amazon's internal experimentation platform (Weblab) for a model change. [Likely] (Weblab is Amazon's internal A/B platform; no source here reports it being named in an interview)
    - What would you check on day 1, and what before the launch decision?
61. How would you ramp a risky change (1%, 5%, 50%), and what triggers a rollback? [Likely]
62. What should go in a launch recommendation, a narrative-style write-up for leadership, after an experiment? [Likely] (Amazon's written narrative culture)
    - How do you present a result with a wide confidence interval that includes zero?
63. The result is significant on the primary metric, but the effect is smaller than the cost of serving the new model. How do you decide? [Likely]
64. Leadership wants to launch despite a non-significant result. How do you respond? [Likely] (also a Leadership Principle probe for "Have Backbone; Disagree and Commit")
65. After launch, how do you check that the measured lift persists? [Likely]
    - What is a holdback (long-term holdout) group, and how big should it be?
66. How would you estimate the cumulative impact of 30 launched experiments in a year? Why might the sum of individual lifts overstate it? [Likely]
    - What are the winner's curse and regression to the mean, and how do you correct for them?

I. Causal inference without clean randomization

67. How would you estimate the causal effect of a promotion or price change without a clean experiment? [Likely] (pricing and promotion teams; see [18])
    - When would you use difference-in-differences, synthetic control, instrumental variables or double ML?
68. Explain difference-in-differences. What is the parallel-trends assumption, and how would you check it? [Likely]
    - What goes wrong with staggered adoption, for example a feature rolled out marketplace by marketplace?
69. Explain synthetic control. When would you use it to measure a launch in a single country? [Likely]

70. What is an instrumental variable? Give a plausible instrument for estimating price elasticity on Amazon. [Likely]
    - How do you defend the exclusion restriction?
71. How do you estimate price elasticity when prices are set by your own algorithm (endogeneity)? [Likely] (pricing system design; see [18])
72. Explain propensity score matching and inverse propensity weighting. What assumptions do they need? [Likely]
    - How do you check overlap (positivity)?
    - What does doubly robust estimation buy you?
73. What is regression discontinuity? Give an example from a free-shipping threshold. [Likely] (free-shipping thresholds are a familiar Amazon retail mechanic)
74. How would you measure the effect of joining Prime on customer spending? [Likely]
    - Why is a naive comparison of Prime and non-Prime members biased, and in which direction?
75. What is double or debiased machine learning, and why use cross-fitting? [Likely]
76. Draw a causal DAG for "recommendation exposure leads to purchase". Name the confounders, colliders and mediators. [Likely]
77. How would you measure the incremental impact (lift) of an ad campaign for an advertiser? [Likely] (ads teams; see [8])
    - What are the trade-offs between ghost ads, geo experiments and observational methods?
78. What is uplift modeling? How does it differ from response modeling? [Likely]
    - How would you evaluate an uplift model (Qini or uplift curves)?
79. How do you validate a causal estimate when you can never observe the counterfactual? [Likely]

J. Bandits and adaptive experimentation

80. When would you use a multi-armed bandit instead of an A/B test? [Likely] (ads creative and widget selection)
    - What do you give up in inference when you use a bandit?
81. Explain epsilon-greedy, UCB and Thompson sampling. How do their exploration strategies differ? [Likely]
    - Write the Beta-Bernoulli posterior update for Thompson sampling.
82. Define regret. What regret bounds do UCB and Thompson sampling achieve? [Likely] (Hoeffding bounds are reported [11] and underpin UCB)
83. How would you design a contextual bandit for recommending widgets on the homepage? [Likely]
    - How do you handle delayed rewards, such as purchases that happen hours later?
    - How do you handle non-stationarity, for example during Prime Day?
84. How do you evaluate a new policy offline from logged bandit data? [Likely]
    - Compare inverse propensity scoring, self-normalized IPS and doubly robust estimators. When does each have high variance?
    - What must the logging policy record for off-policy evaluation to be possible?
85. A bandit is optimizing CTR on deal-page creatives and converges to clickbait. How do you fix the reward? [Likely]

86. How would you frame budget pacing or bidding as a bandit problem or as constrained optimization? [Likely] (ads teams; see [8])

K. Forecasting and time series

87. What is the newsvendor problem, and what is its solution? [Reported] [16]
    - How does a quantile forecast map to the newsvendor critical ratio?
88. How would you forecast demand for millions of products, including new ones with no history? [Likely] (SCOT forecasting teams; see [17][18][19])
    - What does a global model trained across products buy you over per-product models?
89. Why use probabilistic or quantile forecasts rather than point forecasts for inventory decisions? [Likely] (see [17])
    - What is quantile (pinball) loss, and how do you evaluate calibration of P50 and P90 forecasts?
90. Compare MAPE, sMAPE, WAPE, RMSE and quantile loss. Which would you use for intermittent demand? [Likely]
    - Why does MAPE behave badly for low-volume products?
91. How do you backtest a forecasting model correctly? [Likely]
    - What is an expanding-window or rolling-origin evaluation, and how does leakage creep in?
92. How do you forecast demand during Prime Day or other promotions, when history is sparse and distorted? [Likely] (see [17])
    - How do you separate a promotion lift from the baseline?
93. Demand is censored when products are out of stock. How do you correct for it? [Likely] (in-stock is a core SCOT metric)
94. Compare ARIMA, exponential smoothing, and neural forecasters such as DeepAR- or transformer-style models. When does each win? [Likely] (Amazon publishes neural forecasting research; see [17])
95. Explain stationarity, autocorrelation and seasonality. How do you test for them? [Likely]
96. How would you forecast a hierarchy (product to category to total) so the forecasts add up? [Likely]
97. How do you evaluate a forecasting system against a business outcome (in-stock rate, holding cost) rather than accuracy alone? [Likely] (see [18])
    - How would you A/B test a new forecast when buying decisions in treatment and control share inventory?
98. How would you detect anomalies or structural breaks in a demand time series, such as the start of a pandemic? [Likely]

L. Applied stats case questions (mixed, interview-style)

99. Orders dropped 5% week over week. Walk me through how you would diagnose it. [Likely]
100. A product page change increased add-to-cart but not purchases. What hypotheses would you test, and how? [Likely]

101. You are asked to estimate how many customers would buy a new Echo device at a given price. What data and statistical approach would you use? [Likely]
102. Customer reviews have an average rating of 4.6 from 5 reviews and 4.3 from 2,000 reviews. How should ranking treat them? [Likely] (product reviews ranking) - How would you apply Bayesian shrinkage, or a lower confidence bound, to the rating?
103. How would you estimate the false-positive rate of a fraud model when you only see labels for orders you blocked? [Likely] (selective labels in fraud and trust teams)
104. Two models have AUCs of 0.812 and 0.815 on the same test set. Is the difference meaningful? [Likely] - How would you get a confidence interval for the AUC difference (DeLong or bootstrap)?
105. How would you design a survey or human-labelling study to estimate the defect rate of search results, with a confidence interval? [Likely] - How many labelled samples do you need, and how do you handle disagreement between labelers?

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09. Leadership Principles Bank: Behavioral Questions for the L5 Applied Scientist Loop, with L5 Probes and Bar Raiser Favorites

> Scope: behavioral (Leadership Principle, "LP") questions as asked of Amazon Applied Scientists, grouped by all 16 LPs, plus the follow-up probes used to test the L5 (Applied Scientist II) bar, and the bar raiser's favorite questions and pressure tactics. Questions only, no model answers.
>
> Research note (read this first): this chapter was built under tight limits. The session's web-search budget had already been used by earlier chapters, and this environment's egress policy blocked direct fetches of Glassdoor, Blind, LeetCode, Medium, IGotAnOffer, interviewing.io and amazon.jobs. Two kinds of evidence were used: (a) Sourced claims and reported questions already recorded, with citations, in chapters 01-08 of this bank. Those were gathered from search-engine excerpts of the cited pages and are carried over here with the same sources. (b) Pages read in full for this chapter: GitHub-hosted material, including a candidate's notes that repost 1point3acres Amazon AS interview reports [21][22] and a curated Amazon AI-role question guide [20]. [Reported] means a source attributes the question, or a close paraphrase, to Amazon interviews (AS where possible). [Likely] means an inference from the LP text, the AS role and Amazon's products. It has not been seen verbatim. Treat vendor guides as weaker than first-person reports.

Format notes for the behavioral portion

- **No single "LP round"; LPs are in every round.** Amazon's AS prep page says interviewers are assigned competencies "based off of our Leadership Principles" and each typically asks 2-3 behavioral questions [1]. Candidates report 1-2 LP questions opening or closing every technical round. Estimates of LP time per round range from 5-15 minutes [15] to "at least 30 minutes" of each hour-long interview [21]. One L5 candidate put behavioral content at over 60% of the loop [23].
- **The bar raiser (BR) round is mostly or entirely LP.** One 1point3acres AS report said the BR round was "pure LPs" [21]. The BR comes from outside the hiring org, has been trained and shadowed into the role, chairs the debrief, and holds a de facto veto [3][19]. Glassdoor AS and AS II reports describe the BR round as LP-heavy [4][5]. AS candidates have reported BRs who were non-science, including a Principal TPM who also asked about ETAs, dependencies and a few stats checks [7].
- **Assigned LPs.** Each interviewer usually owns 2-3 LPs, so the loop covers the set without overlap [1][16]. Some interviewers name the LP before asking [21]. Candidates recommend asking the recruiter in the prep call which LPs each

interviewer is assigned [21]. A candidate's compiled notes say an Amazon recruiter advised against reusing one story with more than two interviewers [22].

- **Fresh-PhD adaptation.** Interviewers prompt candidates without industry projects to draw on research experience. Reported examples: "what did you do when you didn't like a project your advisor assigned?" and "have you worked in a research area you weren't familiar with?" [21]
- **What the L5 bar looks like in LP answers.** L5 means owning a project end to end, making the technical calls, and influencing peers, with some spill-over to sister teams [8]. A vendor debrief says L5 stories face 2-3 layers of follow-up, against about 1 layer at L4 [15]. Blind commenters say LP narratives, not coding, are what separate L5 from L6, and that L6 needs multiple strong examples and influence beyond the team [8][9]. A BR can pull the level down even when the hiring manager is positive. One AS candidate's BR thought they were "at the cusp of L4 and L5" [6].
- **Grading habits reported for BRs.** BRs push on metrics ("how was that measured?"), on individual contribution ("I", not "we"), and sometimes say outright that your approach was wrong to see how you hold your position [17][18][20]. Weak or rehearsed stories "collapse" under several "why" layers [20].
- **The 16 LPs** are published on amazon.jobs [2]. The last two, Strive to be Earth's Best Employer and Success and Scale Bring Broad Responsibility, were added in 2021 and are asked less often in reports, but they are in scope.

Questions

Tag legend: [Reported] = a source attributes it to Amazon interviews (cite). [Likely] = a grounded inference. Indented bullets are follow-up probes, phrased as interviewers ask them.

1. Customer Obsession

1. Tell me about a time you started from the customer and worked backwards to define the science problem. [Likely]
    - Who exactly was the customer: the end shopper, an internal team, or a seller?
    - How did you get the customer signal: data, anecdotes, or a PR/FAQ?
    - What metric did you choose as the customer proxy, and where did it diverge from what customers actually felt?
2. Tell me about a time you used data to anticipate a customer need nobody had asked for. [Likely]
    - How did you validate that the need was real before you built it?
3. Tell me about a time you traded scientific elegance or novelty for customer impact, or refused to. [Likely]
    - Looking back, was that the right call for the customer?
4. Tell me about a time a model's offline metric improved but the customer experience got worse. [Likely]
    - How did you find out, and what did you change in how you evaluate?
5. Tell me about a time you dealt with a difficult or demanding internal customer (a product, ops or business team consuming your model). [Reported] [22]
    - What did they want that you wouldn't give them, and why?

6. Tell me about a time you had to say no to a stakeholder request because it would hurt end customers. [Likely]
    - Did the stakeholder agree in the end? If not, what happened?

2. Ownership

7. Tell me about a time you went above and beyond your job responsibilities. [Reported] [21]
    - Why was it your job to do it, and who else could have?
8. Tell me about a time you went well beyond the scope of a project and still delivered. [Reported] [21]
    - What did you stop doing to make room for it?
9. Tell me about a time you took on something significant outside your area of responsibility. [Reported] [22]
    - How did the owning team react?
10. Tell me about a time you worked on a project with unclear responsibilities. [Reported] [22]
    - How did you decide what you would own?
11. Tell me about a time a model you shipped failed or underperformed in production. What did you do? [Reported] [12] (model underperformance reported in BR-style questioning; production framing [Likely])
    - Who found it first, you or someone else?
    - What did you put in place so it couldn't happen silently again?
12. Tell me about a time you sacrificed a short-term result for long-term value, for example paying down technical or science debt instead of shipping another model. [Likely]
    - How did you justify it to your manager?
13. Tell me about a time you took initiative without waiting for permission. [Likely] (chapter 01 notes a rejection reportedly explained as "seemed to wait for permission")
    - What would have happened if you had waited?
14. Tell me about something you own today that will keep working after you leave the team. [Likely]
    - What mechanism, such as a runbook, monitor or review, makes it independent of you?

3. Invent and Simplify

15. Tell me about a time you gave a simple solution to a complex problem, or found a new way to do something. [Reported] [22]
    - What did the complex alternative look like, and why did you reject it?
16. Tell me about the most innovative project you have worked on. [Reported] [20] (Exponent-style "invented a solution to a problem that didn't have an obvious answer"; quoted in a prep source, so weak)
    - What, specifically, was your invention versus prior art?
17. Tell me about a time you simplified a complex ML approach into something that could ship. [Likely]
    - What did you give up in accuracy, and how did you decide it was worth it?

18. Tell me about a time you started from a heuristic or simple baseline and it beat the sophisticated model. [Likely]
19. How do you solve a problem when the capability you need isn't in your programming tools or libraries? [Reported] [21] (AS NLP phone screen)
    - Give me a specific time you built that missing piece yourself.
20. Tell me about a time you reused an idea from outside your field or from another team rather than inventing from scratch. [Likely]
    - How did you know it would transfer?

4. Are Right, A Lot

21. Tell me about a time you made a critical scientific or technical decision without full data. [Reported] [14]
    - What did you learn, and how did you pivot? [Reported] [14]
22. Tell me about a time you realized your initial modeling hypothesis was wrong. [Likely]
    - What evidence made you change your mind, and how fast did you act on it?
23. Tell me about a time you were wrong about a technical call. How did you find out? [Likely]
24. Tell me about a time your judgment went against the data, or the data went against your intuition. Which did you trust? [Likely]
25. Tell me about a time you sought out a view that disagreed with yours before making a decision. [Likely]
    - Whose view did you seek, and did it change the decision?
26. Tell me about a time you had to choose between two model architectures or approaches with no clear winner. [Likely]
    - What was the decision rule, and in hindsight was it right?

5. Learn and Be Curious

27. Tell me about a time you learned a new area quickly to solve a problem. [Likely] (fresh-PhD variant reported: "worked in a research area you weren't familiar with" [21])
    - How did you know you had learned enough to make decisions?
28. Tell me about a time you worked in a research or technical area you were not familiar with. [Reported] [21]
29. What is something you taught yourself in the last year (for example LLM fine-tuning, RL from human feedback, or a new framework), and how have you applied it? [Likely]
    - Where did you get stuck?
30. Tell me about a time curiosity about an anomaly in the data led to a bigger finding. [Likely]
31. How do you stay current with the literature, and tell me about a paper that changed how you work. [Likely]

6. Hire and Develop the Best

32. Tell me about a time you mentored a junior scientist, engineer or intern. [Likely]
    - How did you measure their growth?
    - What did you do when they struggled?

33. Tell me about a time you raised the science bar on your team through reviews, standards or reproducibility. [Likely]
34. Tell me about a time you gave someone difficult feedback. [Likely]
35. Tell me about a time you were involved in hiring or interviewing. What bar did you hold? [Likely]
    - Did you ever vote no on someone others liked?
36. Tell me about a time you helped a teammate move to a role or team where they could grow, even though your team lost them. [Likely]

7. Insist on the Highest Standards

37. Tell me about a time you raised the quality bar on a deliverable even though it delayed you. [Likely]
    - Who disagreed with the delay, and how did you handle it?
38. Tell me about a time you had to deliver a model under a tight deadline. What corners did you refuse to cut? [Likely]
39. Tell me about a time you found a bug or leak in someone else's experiment, such as data leakage or a bad A/B setup. [Likely]
    - How did you tell them?
40. Tell me about a time you were not satisfied with the quality of your own shipped work. What did you do about it? [Likely]
41. Tell me about a time insisting on high standards turned out to be the wrong call. [Likely]
    - How do you now decide when "good enough" is good enough?

8. Think Big

42. Tell me about the most ambitious science bet you have proposed. [Likely]
    - How did you sell it, and how far did it get?
43. Tell me about a time you turned a one-off model into a platform or capability other teams use. [Likely]
44. Where do you think your field will be in 3-5 years, and what would you build at Amazon toward that? [Likely] (prep sources describe a BR "bold vision" prompt; not seen verbatim in first-person reports)
45. Tell me about a time you challenged a team to aim for a much bigger goal than planned. [Likely]
    - What was the inspiring direction, in one sentence?
46. Tell me about a time your vision for a project was larger than your manager's. What happened? [Likely]

9. Bias for Action

47. Tell me about a time you had to make a decision with incomplete information. [Reported] [14] [20]
    - Was it a one-way or a two-way door?
    - What was your backup plan if you were wrong?
48. Tell me about a time you had to take a risk, or had very little time to decide. [Reported] [22]
49. Tell me about a time you took a calculated risk and it failed. [Reported] [22]

    - What did you learn, and would you take the same risk again?
50. Tell me about a time you did something without asking for your manager's approval. [Reported] [22]
    - How did your manager react afterwards?
51. Tell me about a time you shipped a quick baseline or experiment to unblock a decision instead of waiting for the full model. [Likely]
52. Tell me about a time you moved too fast. What broke? [Likely]

10. Frugality

53. Tell me about a time you did more with less: compute, labels or headcount. [Likely]
    - What specific constraint forced you to be creative?
54. Tell me about a time you reduced the training or inference cost of a model significantly. [Likely]
    - What did it cost in quality?
55. Tell me about a time you dealt with limited or no labeled data. [Likely]
    - Why not just buy more annotation?
56. Tell me about a time you chose not to use a large or expensive model (for example an LLM) because a cheaper one was sufficient. [Likely]

11. Earn Trust

57. Tell me about a time you failed, or the biggest mistake you made. [Reported] [20][22]
    - What signal existed before the failure that you missed? [Reported] [20]
    - How did you tell your customers or leadership? [Reported] [20]
58. Tell me about a time you received negative feedback. [Reported] [22]
    - What did you change, and how would the person who gave it know?
59. Tell me about a time you earned the trust of an engineering or product partner who doubted your science. [Likely]
60. Tell me about a time you had to deliver bad news, such as a model not working or a launch slipping. [Likely]
    - When did you know, and when did you tell them?
61. Tell me about a time you explained a complex model trade-off to a non-scientist stakeholder. [Likely]
62. Tell me about a time you were vocally self-critical about your own team's work. [Likely]

12. Dive Deep

63. Tell me about a time you had to dive deep into data or code to find a root cause. [Reported] [15] (asked opening or closing an AS coding round)
    - What was the first hypothesis, and how did you rule it out?
    - At what level of detail did you personally look at the raw data?
64. Tell me about a time your model underperformed and you had to diagnose the issue. [Reported] [12]
65. Tell me about a time a metric and anecdotes disagreed. Which did you believe? [Likely] (taken from the LP text)
66. Tell me about a time you found a data quality or pipeline problem that was silently degrading a model. [Likely]

    - How long had it been broken, and why didn't monitoring catch it?
67. Tell me about a time an A/B test result looked too good to be true. [Likely]
68. Walk me through one number in your resume. How exactly was it computed? [Likely] (BR "how was that measured?" probe reported [17][18])

13. Have Backbone; Disagree and Commit

69. Tell me about a time you had a conflict with a coworker or manager, and how you approached it. [Reported] [10][17][18]
    - What did you do once the decision went against you?
70. Tell me about a time you had a major conflict with a Principal Engineer (or senior scientist). How did you resolve it? [Reported] [13]
71. Tell me about a time you disagreed with your team's technical direction. What did you do? [Reported] [20]
    - What would you have done if you had been overruled and then proven right? [Reported] [20]
    - Tell me about a time you were the one who was wrong. [Reported] [20]
72. Tell me about a time you disagreed with a teammate or manager. [Reported] [22]
73. Tell me about a time you pushed back on a stakeholder's metric or launch decision using data. [Likely]
    - Did you commit after being overruled? How would we know you really committed?
74. Tell me about a time you disagreed with a senior scientist's technical approach. [Likely]
75. Tell me about a time you committed to a decision you disagreed with. How did it turn out? [Likely]
76. Tell me about a time you stopped, or tried to stop, a model launch. [Likely]

14. Deliver Results

77. Tell me about a time you had a tight deadline. [Reported] [21] (AWS AS phone screen)
    - If you did it again, what would you do better? [Reported] [21]
78. Tell me about a time you missed a deadline. [Reported] [22]
    - When did you realize it would slip, and who did you tell?
79. Tell me about your most challenging or proudest project, or a time you solved a complex problem. [Reported] [22]
    - What was the business result, in numbers?
80. Tell me about a time you delivered a model under a hard deadline. What did you descope? [Likely]
    - What did that descoping cost?
81. Tell me about a project failure and how you managed the aftermath. [Reported] [13]
    - What would you do differently? [Reported] [11]
82. Tell me about a time you hit a major setback mid-project, such as data being unavailable or a dependency slipping, and still shipped. [Likely]
83. Tell me about a time you had to choose between shipping something imperfect on time and slipping for quality. [Likely]

15. Strive to be Earth's Best Employer

84. Tell me about a time you made your team's work environment better. [Likely]
85. Tell me about a time you noticed a teammate struggling (workload, morale, inclusion) and acted. [Likely]
86. A senior manager assigned you work when you already had a full load. How did you handle it? [Likely] (seen in consolidated MLE behavioral banks; not verified as Amazon-specific)
87. Tell me about a time you made a process or on-call rotation less painful for others. [Likely]

16. Success and Scale Bring Broad Responsibility

88. Tell me about a time you considered the downstream harm or fairness impact of a model. [Likely]
    - What did you measure, and what did you change?
89. Tell me about a time you identified a secondary effect of your model on sellers, customers or another team that others had missed. [Likely]
90. Tell me about a time you handled privacy, safety or responsible-AI constraints on a model, such as hallucination in a customer-facing LLM feature. [Likely]
91. Tell me about a time you decided not to launch because the risk to customers outweighed the gain. [Likely]

Bar raiser favorites and pressure probes

These are the questions and tactics most often attributed to the BR seat. The BR may come from any org, so expect plain-language questions, not science ones.

92. Tell me about your most significant failure. What happened, and what did you change afterwards? [Reported] [20] (described as "a Bar Raiser staple")
93. Tell me about a time you had a conflict with a coworker or manager (Have Backbone; Disagree and Commit). [Reported] [10][17][18]
94. (Pushback) "I think your approach was wrong. Convince me otherwise." [Reported] [17][18]
    - What data would make you concede?
95. How was that measured? [Reported] [17][18]
96. What was your exact individual contribution? Say it without "we." [Reported] [17][18]
97. (TPM or non-science BR) How do you set ETAs, map dependencies and communicate next steps on a science project? [Reported] [7]
    - When did a science timeline slip, and how did you tell stakeholders?
98. (TPM BR) Interpret this p-value or confidence interval for me. [Reported] [7]
99. Give me a second, different example where you showed the same principle. [Likely] (L6 rejections often cite a lack of multiple strong examples [8][9])
100. Why are you leaving your current role, and why Amazon now? [Likely]
101. What is the hardest feedback you have received from a manager, and what did you do with it? [Likely]
102. Tell me about a time you were not proud of how you behaved in a disagreement. [Likely]

L5 calibration probes (any interviewer, layered on any story)

These follow-ups test whether a story is L5 scope rather than L4 execution. Expect 2-3 layers per story [15].

103. What was the largest scope you have owned end to end, from problem framing to production? [Likely]
    - Who else depended on it, beyond your own team?
    - What were you accountable for that your manager was not?
104. Tell me about a time your work changed a sister team's roadmap or technical direction. [Likely]
105. Tell me about a time you set the technical direction for other scientists or engineers, not just carried it out. [Likely]
    - How did you get buy-in without authority?
106. Where did you make a judgment call that your manager did not make for you? [Likely]
107. What mechanism did you build (a review, monitor or tool) that kept working without you in the room? [Likely]
108. If I called your skip-level manager, how would they describe your contribution to this project? [Likely]
109. What were the alternatives you rejected, and what data did you use to reject them? [Likely]
110. What would you do differently if you ran this project again today? [Reported] [11][21]

Generic deepening probes interviewers attach to any LP story

These are not standalone questions. Expect a few of them after each story. [Likely, except where cited]

- What was the situation in one sentence, and what was your specific task?
- What did you personally do, as opposed to your team? [Reported] [17][18]
- What was the measurable result, and how was it measured? [Reported] [17][18]
- What was the hardest moment, and what options did you consider?
- Who disagreed with you, and what was their best argument?
- What did you learn, and where have you applied it since? [Reported] [14]
- How long ago was this? Do you have a more recent example?
- What would have happened if you had not acted?

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10. Bar Raiser Round: Reported Questions, Mixed Technical and Behavioral Probing, and Follow-up Patterns

> Scope: the one interview in the L5 Applied Scientist (Applied Scientist II) onsite run by the Bar Raiser (BR). It covers who the BR is, what they assess at L5, the questions candidates report, the questions the format makes likely, and the follow-up patterns BRs use to test whether a story holds up.
>
> Research note (please read): no live web research worked for this chapter. The run's shared web-search budget was already used up, and direct page fetches (igotanoffer.com, medium.com, aboutamazon.com, teamblind.com) failed with DNS errors in this environment. Every [Reported] tag below therefore comes from findings already gathered and cited in chapters 01-08 of this guide. Those findings were themselves built from search-result extracts of the cited pages, not full reads. Each [Reported] item cites the original page, and anything those chapters did not attribute to an Amazon interview is tagged [Likely]. A follow-up session with a fresh search budget could add more first-person BR reports, especially from Blind, 1point3acres and Glassdoor individual reviews.

Format notes

Aspect	What is known	Sources
Length and slot	One of the onsite loop interviews. Amazon's official AS page describes the loop as four 55-minute interviews with members of the science community. Candidates more often report 5 rounds of 55-60 minutes, one of them labeled "Bar Raiser".	[1][8] [9] [13]
Who the BR is	A trained interviewer from outside the hiring team. BRs are nominated, trained, and shadow veteran BRs before they run debriefs. Science candidates report BRs from non-tech orgs, and one AS II candidate was told the final interview would be a Principal TPM.	[2][3] [5][6] [10]
Power	The BR chairs the debrief and, with the hiring manager, makes the hire/no-hire call. The BR holds a de facto veto that former Amazonians say is rarely used explicitly.	[2][3] [7] [12]
What is assessed	Mostly Leadership Principles (LPs), with technical probing possible. The BR asks whether the hire raises the bar for the role and level. The "better than 50% of current employees at that level" line is BR folklore, not a published Amazon standard.	[4][5] [6][8]
L5-specific bar	L5 means owning a project end to end, making the technical calls, and influencing peers and sometimes sister teams. A vendor-published debrief says L5 LP answers face 2-3 layers of follow-up, against 1 layer at L4. Blind commenters say LP narratives, not coding, separate L5 from L6.	[14] [15] [18]

Aspect	What is known	Sources
Leveling	A BR can push the level down even when the HM is positive. One L5 candidate's BR saw them "at the cusp of L4 and L5". Downleveling from L6 to L5 and from L5 to L4 is commonly reported.	[11] [15]
Behavioral weight overall	Amazon says each interviewer typically asks 2-3 behavioral questions. One L5 report put behavioral content above 60% of the loop. Vendor debriefs put about 15 minutes of each technical round on LPs. The BR round is where LPs get the most time.	[1] [16] [18]
Split decisions	When the debrief splits, candidates report an extra round, for example one more coding round for an AGI-team AS II.	[17]

Observed probing patterns (what makes the BR round different)

- **Layered "why" and "how do you know":** 2-3 levels of follow-up on one story, ending in metrics, mechanisms and your personal decisions. [18][19][21][22]
- **"I" not "we":** BRs separate your individual contribution from the team's. [21][22]
- **Deliberate pushback:** candidates report BRs telling them their approach was wrong to see whether they defend it with data or fold. [21][22]
- **Metric interrogation:** "How was that measured?", including baselines, attribution and counterfactuals. [21][22]
- **Lay-audience translation:** a non-science BR (for example a Principal TPM) asks for project-management mechanics and basic stats interpretation. [10]
- **Story reuse checks:** because each interviewer owns about 2-3 LPs, a BR may return to an LP that another interviewer covered weakly, or ask for a second, different example. This comes from prep guides and is [Likely], not confirmed. [23][26]

Questions

Tag legend:

- **[Reported]**: a candidate or source attributes this question, or a close paraphrase, to an Amazon BR round or Amazon AS/science LP questioning. Citations point to the References.
- **[Likely]**: an inference grounded in the BR format, the 16 LPs, the L5 bar and Amazon's science products. It has not been seen verbatim.

Indented bullets are follow-up probes. In a BR round, expect at least two of them per story.

A. Opening and scope calibration (sets the level the BR will argue in the debrief)

1. Walk me through your career and the one project that best shows the level you are operating at today. [Likely]
    - What part of that project would not have happened without you?
    - Who else could have done what you did on that project?
    - If I called your manager, what would they say was your scope?
2. Tell me about the largest-scope science project you have owned end to end, from problem framing to production. [Likely]

    - Where did the problem come from: did you find it or were you handed it?
    - What was the decision you made that most changed the outcome?
    - How many people or teams depended on your output?
3. Describe your exact individual contribution to that project, without saying "we". [Reported] [21][22]
    - Which design document or analysis did you write yourself?
    - What did you delegate, and why?
4. Tell me about a time your work influenced a team other than your own. [Likely]
    - How did they adopt it: mechanism, code, paper or persuasion?
    - What would have happened if they had said no?
5. What is the hardest technical problem you have solved in the last two years? [Likely]
    - Why was it hard for you specifically, and not just for the field?
    - Explain it so someone outside ML understands why it mattered.

B. Conflict, backbone and disagree-and-commit

6. Tell me about a time you had a conflict with a coworker or manager, and how you handled it. [Reported] [19][21][22]
    - What did you do once the decision went against you?
    - Looking back, who was right?
    - How is your relationship with that person now?
7. Tell me about a time you had a major conflict with a Principal Engineer (or a more senior scientist). How did you resolve it? [Reported] [24]
    - What data did you bring to the conversation?
    - Did you escalate? Why or why not?
8. Tell me about a time you disagreed with a product manager's launch decision based on your model's results. [Likely]
    - What metric were they looking at that you thought was wrong?
    - What would have convinced you to commit to their plan?
9. Tell me about a time you pushed back on a stakeholder's metric or launch decision using data. [Likely]
    - What happened after launch, and were you proven right?
10. Tell me about a time you committed to a decision you disagreed with. [Likely]
    - How did you make sure your doubts did not slow the team down?
    - Did you ever revisit the decision? What triggered that?
11. Tell me about a time an engineer said your model could not ship in its current form. [Likely]
    - What did you give up to get it into production?
    - Who made the final call?
12. Tell me about a time you had to tell a senior leader that their idea would not work scientifically. [Likely]
    - How did you frame it?
    - What did they do next?

C. Failure, being wrong and learning

13. Tell me about a project failure and how you managed the aftermath. [Reported] [24]
    - What would you do differently? [Reported] [20]

    - What mechanism did you put in place so it does not happen again?
14. Tell me about a time your model underperformed in production and you had to diagnose the issue. [Reported] [23]
    - What did you learn, and how did you pivot? [Reported] [25]
    - How long did it take you to notice, and why that long?
15. Tell me about a time you were wrong about a technical call. How did you find out? [Likely]
    - Who told you, or did you find it yourself?
    - What did it cost the team?
16. Tell me about a time you got critical feedback that surprised you. [Likely]
    - What did you change afterwards, and how would someone see that change?
17. Tell me about an experiment or research direction you killed. [Likely]
    - How did you decide it was dead rather than under-tuned?
    - How much effort had gone in, and how did you tell the team?
18. Tell me about a time you missed a deadline. [Likely]
    - When did you first know you would miss it, and when did you tell people?
19. How do you use AI tools in your work, and when did one give you a wrong answer? How did you catch it? [Reported] [28]
    - What do you now check by hand that you used to trust?

D. Decisions under ambiguity (Bias for Action, Are Right a Lot)

20. Tell me about a time you made a critical scientific or technical decision without complete data. [Reported] [25]
    - What data would you have wanted, and why couldn't you wait for it?
    - How did you limit the downside if you were wrong?
21. Tell me about a time you had to decide quickly with incomplete information and the decision turned out to be wrong. [Likely]
    - Was it a one-way or two-way door?
22. Tell me about a time you took initiative without waiting for permission. [Likely]
    - Whose permission would you normally have needed?
    - What did your manager say afterwards?
23. Tell me about a time you had to choose between a quick heuristic and a principled model. [Likely]
    - What did you ship first, and did the principled model ever replace it?
24. Tell me about a time two offline metrics disagreed and you had to make a ship call. [Likely]
    - Which metric did you trust, and why?
25. Tell me about a time the online A/B result contradicted your offline evaluation. [Likely]
    - What was the root cause?
    - Did you change your offline evaluation afterwards?

E. Ownership and Deliver Results (the core L5 signal)

26. Tell me about a time you took ownership of something outside your formal scope. [Likely]
    - Why didn't the actual owner handle it?

    - What did you stop doing to make room?
27. Tell me about a time you delivered a model under a hard deadline. [Likely]
    - What did you descope, and what did that cost?
    - What corners did you refuse to cut?
28. Tell me about the most significant business result your science has driven. [Likely]
    - How was that measured? [Reported] [21][22]
    - How do you know it was the model and not seasonality, a UI change or a marketing push?
    - What was the baseline?
29. Tell me about a time a project was about to fail and you turned it around. [Likely]
    - What was the first thing you did differently?
30. Tell me about a time you had to keep a model working long after launch (monitoring, retraining, on-call). [Likely]
    - Who owned it when it broke at 2am?
31. Tell me about a time you were blocked by another team. [Likely]
    - What did you do in the first week?
    - Did you find a way around them, or did you wait?

F. Customer Obsession and Working Backwards, for a scientist

32. Tell me about a time you started from the customer and worked backwards to the science problem. [Likely]
    - Who was the customer, and how did you get their signal?
    - Was the customer internal or external, and did that change how you worked?
33. Tell me about a time you chose a simpler model because it was better for the customer. [Likely]
    - What accuracy did you give up, and how did you justify it?
34. Tell me about a time a model you built hurt some customers even though the average metric improved. [Likely]
    - How did you find that segment?
    - What did you do about it?
35. Tell me about a time you used customer anecdotes to overturn what the data said. [Likely]
    - How many anecdotes was enough?
36. If you wrote a press release for your best project, what would the headline and customer quote be? [Likely]
    - What would the FAQ's hardest question be?

G. Dive Deep and scientific rigor (the technical side of the BR round)

37. Tell me about a time you dove into data or code to find a root cause that others missed. [Reported] [18][27]
    - What was the first anomaly you noticed?
    - How did you rule out the other explanations?
38. Tell me about a time you found a data-quality or label-leakage problem in your own work. [Likely]
    - How long had it been in production?
    - Which numbers did you have to retract?

39. Pick a model from your resume. Explain why you chose that architecture over the obvious alternative. [Likely]
    - What experiment told you the alternative was worse?
    - Would you make the same choice today?
40. Walk me through how you validated that your evaluation set reflected production traffic. [Likely]
    - What distribution shift did you see after launch?
41. Tell me about a time a result looked too good to be true. [Likely]
    - What did you check first?
42. Tell me about a time you had to defend a statistical result to a skeptical audience. [Likely]
    - What was the effect size, and was it practically significant?
43. You mentioned an improvement of X%. What was the confidence interval? [Likely]
    - How many samples, and how long did the test run?
44. Tell me about a time you reproduced a paper's results and could not match them. [Likely]
    - What did you conclude, and what did you ship instead?

H. Technical checks from a non-science Bar Raiser (TPM, SDE or business BR)

45. How do you set ETAs, map dependencies and communicate next steps on a science project? [Reported] [10]
    - When did a science timeline slip, and how did you tell stakeholders?
    - How do you estimate a project when you do not know if the idea will work?
46. Interpret a p-value or a confidence interval for me. [Reported] [10]
    - What would you tell a business leader who reads p = 0.06 as "no effect"?
47. Explain your most recent model to me as if I were a business leader with no ML background. [Likely]
    - What decision does it let someone make that they could not make before?
48. How do you decide when a research prototype is ready to hand to engineering? [Likely]
    - What does the handoff document contain?
49. Your science project needs six months but the business wants results in six weeks. What do you do? [Likely]
    - What would a six-week version look like?
50. How do you know a model is still working a year after launch? [Likely]
    - Who gets paged, and what is the rollback plan?
51. Tell me about a time you had to explain a model's mistake to a customer-facing or operations team. [Likely]

I. Invent and Simplify, Think Big, Frugality

52. Tell me about a time you simplified a complex scientific approach so it could ship. [Likely]
    - What did the complex version get right that the simple one missed?
53. Tell me about the most ambitious science bet you have proposed. [Likely]
    - Who funded it, and what did you have to prove first?

    - If it had been fully funded, what would the three-year version be?
54. Tell me about a time you did more with less: compute, labels or headcount. [Likely]
    - What was the cost of the cheaper approach in model quality?
55. Tell me about something you invented, such as a method, tool or mechanism, that others now use. [Likely]
    - How did you get people to adopt it?
56. Tell me about a time you used an LLM or foundation model where a custom model would have been the default. [Likely]
    - How did you weigh inference cost against development time?

J. Earn Trust, Hire and Develop the Best, influence without authority

57. Tell me about a time you earned the trust of an engineering or product partner who doubted your science. [Likely]
    - What was the turning point?
58. Tell me about a time you mentored someone or raised the bar of your team's science. [Likely]
    - How do you know they improved?
59. Tell me about a time you gave hard feedback to a peer. [Likely]
    - How did they react, and what changed?
60. Tell me about a time you had to admit a mistake publicly. [Likely]
61. Tell me about a time you influenced a decision without having authority over the people involved. [Likely]
    - Who was the hardest person to convince?
62. Tell me about a time you reviewed someone else's experiment and found a serious flaw. [Likely]
    - How did you raise it?

K. Insist on the Highest Standards, Broad Responsibility, Earth's Best Employer

63. Tell me about a time you raised the quality bar on a deliverable even though it delayed you. [Likely]
    - Who disagreed with the delay?
64. Tell me about a time you refused to launch something that was not ready. [Likely]
    - What was the minimum bar, and who set it?
65. Tell me about a time you considered the downstream harm or fairness impact of a model. [Likely]
    - How did you measure it?
    - What trade-off did you accept?
66. Tell me about a time you handled sensitive customer data in a modeling project. [Likely]
    - What would you not do even if it improved the model?
67. Tell me about a time you made your team's work environment better. [Likely]
68. Tell me about a time you learned a new area fast to unblock a project. [Likely]
    - How did you know you had learned enough?

L. Deliberate pushback and stress probes (BR-specific)

69. I think your approach was wrong. Convince me otherwise. [Reported] [21][22]
    - What data would make you concede?
    - What would the alternative have cost?
70. That sounds like a team result. What specifically did you do? [Reported] [21][22]
71. How was that measured? [Reported] [21][22]
    - What was the counterfactual?
72. That example is from three years ago. Do you have a more recent one? [Likely]
73. You used that story already. Give me a different example for the same principle. [Likely]
74. Your manager made that call. Where was your judgment in that? [Likely]
75. If you had not been on the project, what would have been different? [Likely]
76. What was the worst decision you made on that project? [Likely]
77. What would your harshest critic on that team say about how you handled it? [Likely]
78. You said the result was a success. Who would disagree? [Likely]
79. Stop the story there. What exactly did you say in that meeting? [Likely]

M. Leveling probes: L4 vs L5 vs L6

80. Tell me about a time you set the technical direction for a project, not just executed it. [Likely]
    - Who was in the room when the direction was decided?
81. Tell me about a time you identified a science problem nobody had asked you to solve. [Likely]
    - How did you get time and resources for it?
82. Tell me about a mechanism (process, tool, review) you created that outlived your involvement. [Likely]
    - Who runs it now?
83. Tell me about a time you had to coordinate science work across several people or teams. [Likely]
    - How did you divide the work, and what broke?
84. What is the biggest ambiguity you have resolved, from vague goal to concrete plan? [Likely]
    - What was the first artifact you produced?
85. How would your current manager describe the gap between you and the next level? [Likely]
    - What are you doing about it?

N. Mixed technical-behavioral prompts (BR who is a scientist)

86. Pick one modeling decision from your resume and walk me through the alternatives, the evidence and who you had to convince. [Likely]
    - What experiment would have changed your mind?
87. Tell me about a time a technical trade-off (latency, cost, accuracy) forced a business compromise. [Likely]
    - Who signed off on the compromise?
88. Tell me about a time you moved a model from research to production at scale. [Likely]
    - What failed first in production that never failed offline?
89. Tell me about a time you had to choose between publishing and shipping. [Likely]

- What did the business get out of the choice?

90. Tell me about a GenAI or LLM feature you built or evaluated. How did you decide it was safe to launch? [Likely]
    - How did you measure hallucination, and what threshold did you accept?
91. Tell me about a time an evaluation metric you chose turned out to be the wrong proxy. [Likely]
    - How did you detect the mismatch?

O. Post-loop and split-debrief scenarios

92. (Extra round after a split debrief) Let's revisit coding with a follow-up problem. [Reported] [17]
93. (Downlevel conversation) We think your scope maps to L4. Would you consider that level? [Likely] (BR-driven downleveling reported in [11][15])
    - What evidence from your past work would change our assessment?
94. Why Amazon, and why this team, over other offers you have? [Likely]
    - Which LP do you find hardest to live by?
95. What questions do you have for me about how Amazon hires and develops scientists? [Likely]

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11. Amazon Science Team Context: What Each Hiring Org Asks, and the Recent Amazon Science Work to Know

> Scope: the team-specific part of the L5 Applied Scientist (Applied Scientist II) onsite. That means the hiring-manager / Science Application round, the "why this team" probes, and the "have you seen our work?" questions. It covers the orgs that most often hire L5 AS: Search (Stores Foundational AI, formerly M5), Ads, Alexa AI / AGI (Nova), AWS AI, Rufus, Personalization and recommendations, SCOT (supply chain), Pricing, and Prime Video. It also lists recent Amazon Science publications and open-source releases, framed as questions an interviewer might ask. This chapter is questions only, with no answers.
>
> Research note (please read): this run's shared web-search budget was used up before the chapter started, and the environment's egress proxy blocked direct fetches to Glassdoor, Blind, Medium, LeetCode, 1point3acres, amazon.jobs, amazon.science, aboutamazon.com, arXiv and Wikipedia. Only github.com was reachable. The evidence comes from two places:
>
> - Candidate-report findings already gathered and cited in chapters 01-09 of this guide. Those were built from search-result extracts of the cited pages, not full reads. Each [Reported] tag here cites the original page, and I tag a question [Reported] only when one of those chapters attributes it to an Amazon science interview.
> - Pages fetched and read directly for this chapter: the amazon-science GitHub organization and 15 of its repositories [33]-[49], and three GitHub interview aggregators [19][20][21].
>
> **Team-specific first-person reports for Prime Video and Pricing were not found, so every question in those sections is [Likely].** Before your loop, ask your recruiter for the team's name and re-check its newest postings and papers. A follow-up session with a fresh search budget could add Blind and 1point3acres threads for each org.

Format notes for this round

- **Where team context shows up.** Every L5 loop includes a team-tailored round, usually called "Science Application" or "ML System Design (team-specific)", and it is often run by the hiring manager or a senior scientist from the hiring team. One 2025-2026 AS II loop was reported as "1 Science ML Breadth, 1 Science Depth, 1 Sys Design (Team-specific), 1 coding round and 1 Bar Raiser" [2]. Amazon's own prep page describes about four 55-minute interviews with members of the science community. It says each interviewer is assigned competencies and asks 2-3 behavioral

(Leadership Principle) questions [1]. Blind commenters say L5 loops usually include the application/design round and L4 loops may not [53].

- **Duration and interviewer.** Rounds run 55-60 minutes. Expect about 40 minutes on the team problem and 10-15 minutes on LPs [1][52]. The HM also judges team fit and, with the bar raiser, makes the final call [53].
- **What the L5 bar looks like here.** You take a vague problem from the team's domain and frame it end to end: data and labels, model, offline and online metrics, launch, and monitoring. You make the trade-off calls yourself [17][52]. Depth on the team's own stack, and awareness of what the team has published, separate a "strong hire" from a "hire" in this round. That point is an inference, not a published rubric.
- **The team can change after the loop.** Some candidates pass the loop and are then team-matched or "recycled" to another org's opening [54]. Prepare a 2-minute pitch on how your work maps to at least two orgs.
- **GenAI everywhere.** Even on non-LLM teams, 2025-2026 loops treat transformers and LLMs as core breadth. An Ads candidate reported being rejected after not preparing for them [5][50].
- **Org churn (treat as unverified).** One GitHub aggregator reports that Amazon's AGI org was reorganized in 2025-2026 and that Nova 2 Lite, Nova 2 Sonic, Nova Forge and Nova Act remain active products [21]. An earlier chapter's search extract says Rufus was renamed "Alexa for Shopping" in May 2026 (see [27]). Confirm the current org and product names with your recruiter.

Tag legend: **[Reported]** means a cited source attributes the question, or a close paraphrase, to an Amazon science or ML interview. Items from aggregators are marked "(aggregator)", and items from Research Scientist or MLE loops are marked "(RS loop)" or "(MLE loop)". **[Likely]** means a grounded inference from the team's products, job postings or Amazon Science publications. It has not been seen verbatim.

A. Cross-team "why this team" and team-fit questions

1. Why Amazon, why Applied Science, and why this particular team? [Likely]
    - Which of our team's products have you used as a customer, and what would you change first?
    - Which of our team's papers or launches have you read?
2. How would you apply your research or project to a problem at Amazon? [Reported] [55]
    - Which team's metric would your work move, and by how much, roughly?
3. Here is a high-level ML problem from our team. Propose data sources, an annotation strategy, a modeling technique, evaluation metrics, and the trade-offs and risks. [Reported] [17]
    - What is the version you would ship in one quarter, and what would v2 add?
4. Answer an ML application question about one of our Amazon products. [Reported] [18]
5. Pick an agentic product we could build. Who is the customer, and how do you measure success? [Reported] (Seattle, 2026) [2]
    - Write the press-release headline (Working Backwards).

6. If the loop goes well but this team's headcount closes, which other Amazon science org would you want, and why? [Likely] [54]
7. What science problem in our org do you think is unsolved, and how would you start on it? [Likely]
    - How would you write the 6-page document proposing it?

B. Search (Stores Foundational AI / Search Science, formerly M5)

8. You have a large collection of user click data for many search queries and product documents. Is click data alone sufficient? How would you train a retrieval or IR model with the data at hand? [Reported] [13]
    - What position, presentation and trust biases contaminate the clicks?
    - How would you estimate relevance from click logs while accounting for query intent? [Likely] (Amazon Science work on intent-based relevance from click logs [29])
9. Walk through how you would design, train, validate and deploy a ranking model for product search. [Reported] [56]
    - Pointwise, pairwise or listwise learning-to-rank: which, and why?
10. Answer a search-relevance application question for Amazon search. [Reported] [3]
11. Compare bi-encoders and cross-encoders. Where does each sit in a product-search stack? [Reported] [3]
    - How would you distill the cross-encoder into something that meets a retrieval latency budget?
12. You have 10k product listings in 100 categories, each with a title, description and photo. How would you categorize them? [Reported] [13]
    - How do you handle a long tail of categories with very few examples?
13. Design an end-to-end retrieval and ranking pipeline with candidate generation, serving, offline metrics versus online guardrails, A/B testing and position bias. [Reported] (Amazon search/recs/ads MLE loop, aggregator) [20]
14. Design a multi-turn search agent that handles long context. [Reported] (RS loop, aggregator) [19]
    - When would you summarize earlier turns, and when would you retrieve them instead?
15. The ESCI Shopping Queries dataset labels query-product pairs as Exact, Substitute, Complement or Irrelevant. Why is a 4-way label more useful than binary relevance for Amazon search? [Likely] [34]
    - How would you use Substitute and Complement labels differently in ranking and in recommendations?
    - The dataset covers English, Japanese and Spanish. How would you transfer a relevance model to a new marketplace?
16. How would you rank with several objectives at once (relevance, conversion, margin, delivery speed)? [Likely]
    - Would you use multi-task learning-to-rank, and how would you weight the tasks (for example uncertainty weighting, PCGrad or Nash-MTL)? [Likely] (Amazon's DeepMTL2R library implements about 22 such methods [45])

17. (Search/IR postings list learning-to-rank and recommender work [26].) How would you select a set of results that is both relevant and diverse from millions of candidates, faster than MMR? [Likely] (MUSS, UAI 2026 [39])
    - When does diversity hurt a high-intent query like a model number?
18. How would you use an LLM as a relevance judge for product search, and how would you validate it against human annotators? [Likely]
19. How would you detect and fix wrong attribute values (size, color, material) in catalog listings that hurt search? [Likely]
    - SynthAVE labels (attribute, value) pairs as Correct, Incorrect or Unknown, using a 21-configuration LLM arena with humans adjudicating disagreements. When would you trust an LLM arena as a labeler? [Likely] (EMNLP 2026 Industry [38])

C. Advertising (Sponsored Products, DSP, Ads Trust, Ads measurement)

20. Explain how transformers and attention work. How does a decoder-only LLM generate text at inference time? (An Ads candidate reported being rejected after not preparing this.) [Reported] [5]
21. Design a system that recommends trustworthy ads to Amazon users while maintaining brand safety and compliance. [Reported] (Advertising Trust) [8]
    - Walk through the full data pipeline from collection to deployment. [Reported] [8]
    - How do you get labels for rare policy violations?
22. Science Depth on an Ads loop: your resume, plus GenAI fundamentals, plus the business impact of your work. [Reported] [8]
23. Design a click counter over a sliding k-minute window, then an advertiser-metrics service with campaign and event entities, windowed pre-aggregation and idempotency for duplicate events. [Reported] (Ads MLE loop, aggregator) [20]
24. Design click-through-rate or conversion-rate prediction for Sponsored Products. [Likely]
    - Why does calibration matter for auction pricing?
    - How do you handle delayed conversions?
25. How would you learn a bidding policy offline from logged auction data? [Likely] (AuctionGym, KDD 2023 ADS track [37])
    - Why is offline evaluation of learning-to-bid hard (unobserved confounding, costly experiments)?
    - How does optimal bidding differ between second-price and first-price auctions, and what is bid shading?
26. How would you measure the incremental lift of an ad campaign for an advertiser? [Likely]
    - Ghost ads, geo experiments or observational causal methods: what are the trade-offs?
27. How would you blend sponsored products into a generative shopping answer without eroding trust, and how would you measure the effect? [Likely]
28. Design a budget-pacing algorithm for advertisers. How would you frame it as constrained optimization or as a bandit? [Likely]

D. Alexa AI, AGI and Amazon Nova (foundation models)

29. Expect several "modern LLM training technique" questions, with follow-ups on each stage (pretraining, SFT, preference tuning). [Reported] (AGI-team role) [6][7] (see chapter 07)
30. You are handed a Transformer implementation that trains but performs badly. Debug it. [Reported] (Amazon AGI, aggregator) [21]
    - Which bugs in masking, scaling, positional encoding or normalization produce silent quality loss?
31. Build and train a small classifier live. [Reported] (Amazon AGI, aggregator) [21]
32. Reinforcement-learning fundamentals plus coding. [Reported] (Amazon AGI, aggregator) [21]
    - Explain PPO's clipped objective and how GRPO removes the value model. [Reported] (RS loop, aggregator: PPO clipping, GRPO) [19]
33. Explain RoPE and ALiBi, KV caching and pipeline parallelism. [Reported] (RS loop, aggregator) [19]
34. Discuss LLM decoding and training choices in the context of an Amazon use case. [Reported] (AS II, LLM team, Santa Clara) [12]
    - How does beam search work, and when does it hurt open-ended generation? [Reported] [12]
35. How would you improve Alexa speech recognition for non-native speakers and users with strong regional accents? [Reported] (Alexa International Tech) [10]
    - Where would you get diverse accent data, and how would you handle phonetic variation? [Reported] [10]
36. The Alexa International loop is breadth, depth, science application, coding and a bar raiser, with LPs throughout. What would you expect the science-application problem to look like for a multilingual assistant? [Reported for the format] [9]; the follow-up is [Likely]
37. How would you align a multilingual assistant with SFT, DPO, RLHF or RLAIF when preference data is scarce in some locales? [Likely] (the Alexa International posting lists these methods [23])
38. Instruction-tuned LLMs are trained on text preferences, but Alexa speaks its answers. What makes a response "speechworthy", and how would you collect preferences for it? [Likely] (Speechworthy Instruction-tuned LMs, EMNLP 2024 [42])
39. AlexaTM 20B was a multilingual seq2seq model pretrained with denoising plus causal LM. When would an encoder-decoder beat a decoder-only model for an Alexa task? [Likely] [40]
40. Training LLMs with MXFP4: why can the backward pass run in 4-bit with stochastic rounding and a random Hadamard transform and still give unbiased gradients? [Likely] (AISTATS 2025 [41])
    - What breaks if rounding is biased?
41. Moving from an ASR, then LLM, then TTS cascade to an end-to-end speech model (in the spirit of Nova Sonic): what changes in latency, evaluation and failure modes? [Likely] [21]
42. Design a browser-operating agent (in the spirit of Nova Act) that completes multi-step web tasks reliably. [Likely] [21]

    - How do you build its evaluation set and measure step-level versus task-level success?
43. An Alexa+ style assistant calls real device and service APIs (timers, smart home, ordering). How do you make tool calls reliable and safe? [Likely]
44. How would you make an on-device wake-word or NLU model 10x smaller with minimal quality loss? [Likely]

E. AWS AI (Bedrock, SageMaker, Q / coding agents, AI research)

45. Explain transformers, then explain LoRA. [Reported] (AWS AS loop) [4]
    - Where are the LoRA adapters inserted, and what does the rank control?
46. Design an enterprise RAG service. [Reported] [51]
    - How do you enforce document-level access control in retrieval?
47. Design a GitHub Copilot-style coding agent. [Reported] (Senior AS, Boston) [3]
    - SWE-PolyBench scores agents on repository-level tasks in Java, JavaScript, TypeScript and Python with pass rate and file- and node-retrieval metrics. Why measure retrieval separately from pass rate? [Likely] (arXiv 2025 [46])
48. Evaluate an LLM agent with metrics beyond accuracy: tool use, hallucination and calibration. [Reported] [3]
49. RAGChecker separates retriever metrics (claim recall, context precision) from generator metrics (faithfulness, noise sensitivity, hallucination, context utilization). How would you use that split to debug a customer's RAG system? [Likely] (NeurIPS 2024 Datasets and Benchmarks [35])
50. RefChecker checks LLM outputs at the level of (subject, predicate, object) claim triplets in zero, noisy and accurate context settings. Why triplets rather than sentences? [Likely] (2024 [36])
    - Where does triplet extraction itself fail?
51. A customer's Bedrock-hosted workload costs too much. How would you cut inference cost sharply without unacceptable quality loss? [Likely] [21] (see chapter 07)
52. Chronos turns a time series into tokens by scaling and quantization, then trains a language model on them. Why does that work, and what does Chronos-Bolt change (patches, direct multi-step quantile output)? [Likely] (TMLR 2024; Chronos-Bolt Nov 2024 [33])
    - Chronos-2 (Oct 2025) adds multivariate and covariate-informed zero-shot forecasting. How would you evaluate it fairly against a tuned, dataset-specific model?
53. Synthesize realistic tabular data, or detect anomalies in tabular data, with generative models. What are the evaluation pitfalls? [Likely] (amazon-science TabSyn and AnoLLM repositories [49])

F. Rufus (generative shopping assistant)

54. Design Rufus: answer "What do I need for cold weather golf?" using the catalog, reviews and community Q&A. [Likely] [22][28]
    - How do you ground answers in product facts and avoid hallucinated specs?

    - How do you evaluate offline before an A/B test, and what is the online success metric?
55. Design an agentic recommendation system for Amazon, covering components, model choices, and quality and safety evaluation. [Reported] [16]
56. Rufus runs on several LLMs, including Amazon Nova, on Bedrock. How would you route each query between a small and a large model? [Likely] [27]
57. How would you personalize answers with a customer's purchase and browse history while respecting privacy? [Likely] [27]
58. You have one latency budget for an assistant answer (say, first token in under one second). How do you split it across retrieval, reranking, generation and safety filters? [Likely] [22]
59. How would you summarize thousands of reviews into a short "Customers say" blurb, and measure faithfulness and coverage of minority opinions? [Likely]
60. Personalized autocomplete for a chat interface: how would you rank candidate completions, and would you report saved-keystrokes@k or acceptance-rate@k? [Likely] (amazon-science personalized chat autocomplete experiments [49])
61. How would you red-team a shopping assistant for unsafe, biased or policy-violating answers? [Likely]

G. Personalization and recommendations (homepage, Amazon Music, explanations)

62. Design a pipeline to improve personalized product recommendations on the Amazon homepage. [Reported] [16]
    - What is the north-star metric, and what are the guardrails?
63. A user goes from viewing to clicking to purchasing. Model the probability of each action given a user and a product. [Reported] [14]
    - Three independent models, or a joint model? [Reported] [14]
    - How would you build the dataset and down-sample (for example with reservoir sampling)? [Reported] [14]
64. Design a recommender system end to end. [Reported] (reported as the design round in an AS loop) [57]
    - How do you handle cold start for new users and new ASINs?
65. You have logged rankings from the Amazon Music home page with logging-policy propensities. How would you estimate a new ranking policy's value offline? [Likely] (Music off-policy evaluation benchmark [44])
    - Compare IPS, self-normalized IPS and doubly-robust estimators for slate rankings.
    - How do you check whether your OPE estimate matches the online A/B result?
66. You have a pool of LLM-generated explanations for a recommendation. Would you choose among them with RL fine-tuning (PPO, GRPO, DPO) or with a pairwise ranker? [Likely] (RecSys 2026 finding that LambdaRank beat single-action RL [47])
67. How would you avoid feedback loops that recommend only already-popular items? [Likely]
68. How would you personalize email or push notifications as uplift modeling rather than response modeling? [Likely]

H. SCOT (Supply Chain Optimization Technologies), forecasting and inventory

69. What is the newsvendor problem, and how would you use it to set inventory levels? [Reported] (Amazon supply-chain interviews) [15]
    - How does a quantile forecast map to the critical ratio?
70. Model an optimization problem from your own work: objective, decision variables and constraints. [Reported] [52]
    - When would you use a heuristic instead of an exact solver?
71. How would you forecast demand for millions of products, including new ones with no history? [Likely] [30]
    - Why do buying decisions need probabilistic or quantile forecasts rather than point forecasts?
    - Would you use a global model or per-item models?
72. Design a system that decides how much of each product to buy, from which vendor, and when. [Likely] (SCOT Optimal Sourcing / Buying Science posting [24])
73. Design inventory placement across fulfillment centers. How do forecast errors propagate into transportation cost? [Likely] [31]
74. How do you forecast through Prime Day when history is sparse and demand was censored by stock-outs? [Likely]
75. Could a pretrained foundation model like Chronos replace your team's per-domain forecaster? What evidence would convince you? [Likely] [33]
76. Design a delivery-promise prediction (the date shown on the detail page). What loss would you use if late costs more than early? [Likely] (Last Mile candidate context [11])
77. How would you run an experiment on a buying policy when products share warehouses and trucks (interference)? [Likely]

I. Pricing and causal inference

78. Design a system that recommends competitive prices for millions of products. [Likely]
    - How do you estimate price elasticity when your own algorithm set the prices (endogeneity)?
    - What guardrails prevent price spirals and loss of customer trust?
79. How would you measure the effect of a pricing algorithm when prices interact across products? [Likely]
    - Switchback, cluster-randomized or synthetic-control designs: when would you use each?
80. How would you validate a causal model before trusting it? [Likely] (Amazon's causal-validation package simulates data with a known effect and supports placebo tests [48])
    - Design a placebo test for a synthetic-control estimate of a price change.
81. How would you design an A/B test for a model launch, and what would you do if it is underpowered? [Reported] [58]
82. How would you decide whether to discount a product for a deal event, accounting for cannibalization and pull-forward demand? [Likely]

83. When would you use double ML, instrumental variables or difference-in-differences for a promotion's effect? [Likely]

J. Prime Video (and other Amazon content and devices teams)

No first-person Prime Video AS reports were found in this run. Everything below is [Likely].

84. Design recommendations for Prime Video, where viewing is sequential and feedback is implicit. [Likely]
    - Is a show watched for 3 minutes a positive or a negative label?
85. How would you choose artwork or thumbnails per user, and evaluate the policy offline? [Likely]
86. How would you detect video-quality problems (buffering, artifacts, audio-video sync) at scale from playback telemetry? [Likely]
87. Design ad insertion for ad-supported streaming: where to place breaks, and how to measure viewer drop-off. [Likely]
88. How would you forecast live-sports streaming demand for capacity planning? [Likely]
89. How would you detect and skip intros and recaps, or generate scene-level metadata, with video models? [Likely]

K. Payments, trust, fraud and abuse

90. Design a model to predict payment-method failure and pick the best fallback for the customer. [Likely] (Amazon Payment Products L5 posting [25])
91. Train and evaluate a fraud model under extreme class imbalance, delayed chargeback labels and adaptive adversaries. [Likely]
    - Amazon's Fraud Dataset Benchmark groups public datasets by fraud type (card-not-present, bots, malicious traffic, content moderation, credit risk). How would you check that a method generalizes across them? [Likely] [43]
92. Report precision, recall and AUC for imbalanced fraud data, and say which you would optimize. [Reported] (RS loop phone screen, aggregator) [19]
93. Use computer vision to detect defective products in fulfillment-center warehouses. [Reported] [18]

L. "Have you read our work?" Recent Amazon Science publications as interview questions

These are [Likely] prompts. HMs often ask "What do you think of X?" about their own team's work. Read the abstract and the evaluation section of anything in your target org.

94. Chronos (TMLR 2024), Chronos-Bolt (Nov 2024) and Chronos-2 (arXiv, Oct 2025; released to SageMaker JumpStart in Dec 2025 and AutoGluon-Cloud in Jun 2026): what is the weakest part of the "language of time series" argument? [Likely] [33]
95. Shopping Queries Dataset (ESCI, arXiv 2022, used in a KDD Cup): what does ESCI miss that real Amazon search needs (personalization, freshness, session context)?

[Likely] [34]

96. RAGChecker (NeurIPS 2024 D&B) and RefChecker (2024): how would you meta-evaluate an automatic hallucination checker against human judgment? [Likely] [35] [36]
97. AuctionGym (KDD 2023 ADS; AdKDD 2022 best paper): what does a simulator buy you that logged data cannot, and what does it cost you? [Likely] [37]
98. SynthAVE (EMNLP 2026 Industry): what are the risks of evaluating on synthetically generated products with anonymized brands? [Likely] [38]
99. MUSS (UAI 2026): MMR is a special case of MUSS with one partition. What does the constant-factor approximation guarantee buy you in practice? [Likely] [39]
100. Training LLMs with MXFP4 (AISTATS 2025): why target the backward pass first? [Likely] [41]
101. Speechworthy instruction-tuned LMs (EMNLP 2024): how would you extend a listening-based preference dataset to multi-turn voice? [Likely] [42]
102. Multimodal chain-of-thought (mm-cot) and Auto-CoT: are these still relevant once reasoning models are trained with RL? [Likely] [49]
103. Pairwise ranking beats single-action RL for offline explanation selection (RecSys 2026): when would you expect that result to flip? [Likely] [47]
104. Amazon Science's 2026 open-source releases include agent self-improvement recipes (Ratchet, Double Ratchet), RL-training kernels (DualKV shared-prompt FlashAttention), multi-agent vLLM kernel optimization (ammo) and long-context hallucination benchmarks (ACL 2026). Pick one relevant to this team and critique it. [Likely] [49]
105. Earlier LLM distillation work, "Learning with less: knowledge distillation from LLMs via unlabeled data": when is distilling an LLM into a small classifier the right call for a production team? [Likely] [32]

M. L5-bar probes that recur across all teams

106. Of everything you proposed for our problem, what is the riskiest assumption, and how would you de-risk it in two weeks? [Likely]
107. Your design improves our team metric but hurts a sister team's metric (for example, conversion up and returns up). How do you decide? [Likely]
108. Which parts would you build, and which would you reuse from existing Amazon services or other science teams? [Likely]
109. How would you hand this system to engineering for long-term ownership, with monitoring and retraining triggers? [Likely]
110. Tell me about a time you influenced a team outside your own with your science. How would that look here? [Likely] (L5 scope: influence on peers and sister teams; see chapter 01)

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